```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

```
PUBLIC_ENDPOINTS = {"auth.login", "static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
    ("用户与操作日志", "/users", "users"),
]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
```

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            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
      badges = {}
      if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
      def build(links):
            return [{"label": label, "href": href, "icon": icon,
                     "active": hit(href), "badge": badges.get(href, 0)}
                    for label, href, icon in links]
      groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
      ]
      #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
      if path == "/":
            page_title = "工作台"
      elif path.startswith("/profile"):
            page_title = "个人设置"
      elif path.startswith(("/users", "/logs")):
            page_title = "用户与操作日志"
      elif path.startswith("/search"):
            page_title = "搜索结果"
      else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                  if hit(href):
                        page_title = label
                        break
      #  层级引导用：项目是归档单位，图斑才是评价对象。
      #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
      from app.models import Project, MinePatch
      projects = Project.query.order_by(Project.id).all()
      return {
            "nav_groups": groups,
            "home_active": path == "/",
            "page_title": page_title,
            "today": date.today().strftime("%Y-%m-%d"),
            "search_q": request.args.get("q", ""),
            "role_labels": ROLE_LABELS,
            "project_count": len(projects),
            "patch_count": MinePatch.query.count(),
            "projects_without_patch": [p for p in projects
                                       if p.patches.count() == 0],
      }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
```

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    @app.after_request                                                  201
    def audit_writes(response):                                         202
        if request.method in ("POST", "PUT", "PATCH",                   203
    "DELETE"):                                                          204
            audit.record(request.endpoint,                              205
    response.status_code)                                               206
        return response                                                 207
    # 错误页：给出可读提示，而不是默认的英文错误页                      208
    from flask import render_template                                   209
    @app.errorhandler(403)                                              210
    def forbidden(e):                                                   211
        return render_template("error.html", code=403,                  212
                               message="没有权限执行该操作"), 403       213
    @app.errorhandler(404)                                              214
    def not_found(e):                                                   215
        return render_template("error.html", code=404,                  216
                               message="请求的页面不存在"), 404         217
    return app                                                          218
# ==== app/config.py ====                                             219
# -*- coding: utf-8 -*-                                               220
"""应用配置。"""                                                        221
import os                                                               222
from datetime import timedelta                                          223
BASE_DIR = os.path.abspath(os.path.dirname(__file__))                   224
class Config:                                                           225
    SECRET_KEY = os.environ.get("SECRET_KEY",                           226
    "mine-restoration-dev-key")                                         227
    # 单机部署，SQLite 数据库文件放在应用目录下                         228
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                            229
    os.path.join(BASE_DIR, "app.db")                                    230
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              231
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项                  232
    WTF_CSRF_ENABLED = True                                             233
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录 234
    REMEMBER_COOKIE_DURATION = timedelta(days=30)                       235
    REMEMBER_COOKIE_HTTPONLY = True                                     236
    REMEMBER_COOKIE_SAMESITE = "Lax"                                    237
    # 影像与数据文件上传（遥感影像挂接、批量导入）                     238
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",                    239
    "uploads")                                                          240
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限       241
    32 MB                                                               242
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",                   243
    "webp", "bmp", "tif", "tiff"}                                       244
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}                           245
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积 246
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",                  247
    "webp"}                                                             248
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024             # 4 MB              249
# ==== app/models.py ====                                             250
```

```
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」中自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 显示名称（姓名）
    title = db.Column(db.String(64), nullable=True)
    # 职务 / 岗位
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def shown_name(self):
        """界面上显示的名字：优先用显示名称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字母占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
```

```
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
```

```
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)   # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)   # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)     # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
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    __tablename__ = "standard_rules"                                    401
    id = db.Column(db.Integer, primary_key=True)                        402
    standard_id = db.Column(db.Integer,                                 403
    db.ForeignKey("standards.id"), nullable=False)                      404
    item = db.Column(db.String(128), nullable=False)                    405
    # 判定项                                                            406
    operator = db.Column(db.String(8), nullable=False)                  407
    # 比较符：> >= < <= ==                                              408
    threshold = db.Column(db.String(64), nullable=True)                 409
    # 阈值                                                              410
    condition_type = db.Column(db.String(32),                           411
    default="必要条件")  # 必要条件 / 充分条件                          412
    region = db.Column(db.String(64), nullable=True)                    413
    # 适用分区（宁夏分区差异化）                                       414
    note = db.Column(db.Text, nullable=True)                            415
class ConstraintCheck(db.Model):                                        416
    """限制性判定记录。"""                                              417
    __tablename__ = "constraint_checks"                                 418
    id = db.Column(db.Integer, primary_key=True)                        419
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            420
    ches.id"), nullable=False)                                          421
    policy_pass = db.Column(db.Boolean, nullable=True)                  422
    # 政策约束                                                          423
    location_pass = db.Column(db.Boolean, nullable=True)                424
    # 区位约束                                                          425
    pollution_pass = db.Column(db.Boolean, nullable=True)               426
    # 土壤污染                                                          427
    geo_pass = db.Column(db.Boolean, nullable=True)                     428
    # 地质灾害                                                          429
    farmland_pass = db.Column(db.Boolean, nullable=True)                430
    # 耕地属性                                                          431
    passed = db.Column(db.Boolean, nullable=True)                       432
    # 总体是否通过                                                      433
    reason = db.Column(db.Text, nullable=True)                          434
    # 不通过原因                                                        435
    checked_at = db.Column(db.DateTime,                                 436
    default=datetime.utcnow)                                            437
class MicroUnit(db.Model):                                              438
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                      439
    __tablename__ = "micro_units"                                       440
    id = db.Column(db.Integer, primary_key=True)                        441
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            442
    ches.id"), nullable=False)                                          443
    unit_type = db.Column(db.String(32), nullable=False)                444
    # 单元类型                                                          445
    area = db.Column(db.Float, nullable=True)                           446
    # 面积                                                              447
    ratio = db.Column(db.Float, nullable=True)                          448
    # 面积占比（自动计算）                                              449
    description = db.Column(db.Text, nullable=True)                     450
```

```
    #  现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)   #  指标编码
    name = db.Column(db.String(64), nullable=False)
    #  指标名称
    level = db.Column(db.Integer, nullable=False)
    #  1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    #  父指标编码
    #  角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    #  单位
    ideal_value = db.Column(db.String(64), nullable=True)
    #  理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    #  benefit/cost/mid/discrete
    #  启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    #  取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    #  所属单元类型
    name = db.Column(db.String(64), nullable=True)
    #  方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)         #  1=一级指标  2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    #  是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
```

```
    nullable=False)        #  指标编码                                    501
    weight = db.Column(db.Float, nullable=False)                          502
    #  权重值                                                             503
class ExpertMatrix(db.Model):                                             504
    """专家判断矩阵：多专家打分的原始矩阵。"""                              505
    __tablename__ = "expert_matrices"                                     506
    id = db.Column(db.Integer, primary_key=True)                          507
    scheme_id = db.Column(db.Integer,                                     508
    db.ForeignKey("weight_schemes.id"), nullable=True)                    509
    #  直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             510
    unit_type = db.Column(db.String(32), nullable=True)                   511
    #  微地质单元类型                                                     512
    l1_name = db.Column(db.String(64), nullable=True)                     513
    #  所属一级指标（二级矩阵用）                                         514
    expert_name = db.Column(db.String(64), nullable=True)                 515
    #  专家姓名                                                           516
    level = db.Column(db.Integer, nullable=False,                         517
    default=1)          #  矩阵对应指标层级                               518
    matrix_json = db.Column(db.Text, nullable=False)                      519
    #  判断矩阵（ＪＳＯＮ 二维数组）                                      520
    cr = db.Column(db.Float, nullable=True)                               521
    #  该专家矩阵的一致性比例                                             522
    created_at = db.Column(db.DateTime,                                   523
    default=datetime.utcnow)                                              524
class IndicatorData(db.Model):                                            525
    """指标实测数据：按微地质单元组织。"""                                  526
    __tablename__ = "indicator_data"                                      527
    id = db.Column(db.Integer, primary_key=True)                          528
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni              529
    ts.id"), nullable=False, unique=True)                                 530
    soil_ratio = db.Column(db.Float, nullable=True)                       531
    #  土石比                                                             532
    thickness = db.Column(db.Float, nullable=True)                        533
    #  土层厚度  cm                                                       534
    ph = db.Column(db.Float, nullable=True)                               535
    organic_matter = db.Column(db.Float, nullable=True)                   536
    #  有机质  g/kg                                                       537
    #  重金属含量，ＪＳＯＮ 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null 538
    heavy_metals = db.Column(db.Text, nullable=True)                      539
    coverage = db.Column(db.Float, nullable=True)                         540
    #  植被覆盖度  %                                                      541
    shannon = db.Column(db.Float, nullable=True)                          542
    #  Shannon-Wiener 多样性指数                                          543
    slope = db.Column(db.Float, nullable=True)                            544
    #  坡度  °                                                            545
    aspect = db.Column(db.String(16), nullable=True)                      546
    #  坡向：阳坡/阴阳坡/阴坡                                             547
    def heavy_metals_dict(self):                                          548
        return json.loads(self.heavy_metals) if                           549
    self.heavy_metals else {}                                             550
```

```
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator，
```

```
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
class ProfileForm(FlaskForm):
    """个人资料：显示名称、职务与联系方式。字段都可留空。"""
    display_name = StringField("显示名称",
    validators=[Optional(), Length(max=64)])
    title = StringField("职务", validators=[Optional(),
    Length(max=64)])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
```

```
    old_password = PasswordField("当前密码",                                    651
    validators=[DataRequired()])                                                652
    new_password = PasswordField("新密码",                                      653
                                        validators=[DataRequired(),             654
    Length(min=6, max=64,                                                       655
                                                                                656
    message="新密码至少 6 位")])                                                657
    confirm = PasswordField("确认新密码",                                       658
                                    validators=[DataRequired(),                 659
                                                                                660
    EqualTo("new_password",                                                     661
                                                                                662
    message="两次输入的新密码不一致")])                                         663
    submit_pwd = SubmitField("修改密码")                                        664
# ==== app/utils/constants.py ====                                             665
# -*- coding: utf-8 -*-                                                        666
"""评价指标常量表。                                                            667
全部数值取自论文，作为系统计算的唯一基准：                                     668
- 论文 4.2.2 节：指标标准化理想极值与取值依据                                  669
- 论文 4.2.3 节 / 表 4：适宜性等级划分                                         670
- 论文表 3：按微地质单元分别确定的一级、二级指标权重                           671
- 论文 4.2.1 节：平均随机一致性指标 RI                                         672
"""                                                                            673
# ------------------------------------------------------------------------     674
    ----------------                                                           675
# 一、微地质单元类型                                                           676
# ------------------------------------------------------------------------     677
    ----------------                                                           678
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡           679
# 三种地质微单元类型"                                                          680
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")                                  681
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）  682
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"                         683
UNIT_ENABLED_INDICATORS = {                                                    684
    "岩质边坡": ("植被指标", "地形指标"),              # 不设土壤指标            685
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),                          686
    "平台": ("土壤指标", "植被指标"),                  # 不设地形指标            687
}                                                                              688
# ------------------------------------------------------------------------     689
    ----------------                                                           690
# 二、指标标准化理想极值（论文 4.2.2 节）                                      691
# ------------------------------------------------------------------------     692
    ----------------                                                           693
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值                                 694
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）                        695
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）                    696
IDEAL_ORGANIC = 30.0            # 有机质理想值                                 697
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值                     698
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）                 699
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统        700
```

```
#  重金属极值 （mg/kg），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 Ⅳ 级最高值
#  论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
#  污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
#  坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
#  坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
#  pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
#  ------------------------------------------------------------------------
    ------------------
#  三、适宜性等级划分（论文表 4）
#  ------------------------------------------------------------------------
    ------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    #  score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
#  ------------------------------------------------------------------------
    ------------------
#  四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2，3，4，5，其余为常用扩展值）
```

```
#  ----------------------------------------------------------------------  751
      ----------------                                                      752
RI_TABLE  =  {1:  0.0,  2:  0.0,  3:  0.58,  4:  0.90,  5:  1.12,  6:      753
      1.24,                                                                 754
                  7:  1.32,  8:  1.41,  9:  1.45,  10:  1.49}               755
#  一致性比例阈值（论文：当  CR<0.1  时，判断矩阵满足一致性要求）           756
CR_THRESHOLD  =  0.1                                                        757
#  ----------------------------------------------------------------------  758
      ----------------                                                      759
#  五、默认权重方案（论文表  3）                                           760
#  ----------------------------------------------------------------------  761
      ----------------                                                      762
#  论文  4.2.2  节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  763
#  分别计算评价指标权重"（论文摘要、图  2  流程与表  5  均作"碎石边坡"，      764
#  系统统一采用"碎石边坡"）。权重由  10  位专家打分后经  AHP  方法确定。      765
#  下表为论文表  3  所载的归一化权重，作为系统默认方案。                    766
DEFAULT_WEIGHTS  =  {                                                       767
      "岩质边坡":  {                                                        768
            "l1":  {"植被指标":  0.45,  "地形指标":  0.55},                 769
            "l2":  {                                                        770
                  "植被指标":  {"植被覆盖度":  0.67,  "生物多样性":  0.33},  771
                  "地形指标":  {"坡度":  0.64,  "坡向":  0.36},             772
            },                                                              773
      },                                                                    774
      "碎石边坡":  {                                                        775
            "l1":  {"土壤指标":  0.44,  "植被指标":  0.31,  "地形指标":  0.25},  776
            "l2":  {                                                        777
                  "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,  778
                              "有机质":  0.17,  "污染指标":  0.17},         779
                  "植被指标":  {"植被覆盖度":  0.64,  "生物多样性":  0.36},  780
                  "地形指标":  {"坡度":  0.67,  "坡向":  0.33},             781
            },                                                              782
      },                                                                    783
      "平台":  {                                                            784
            "l1":  {"土壤指标":  0.54,  "植被指标":  0.46},                 785
            "l2":  {                                                        786
                  "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,  787
                              "有机质":  0.17,  "污染指标":  0.17},         788
                  "植被指标":  {"植被覆盖度":  0.61,  "生物多样性":  0.39},  789
            },                                                              790
      },                                                                    791
}                                                                           792
#  ----------------------------------------------------------------------  793
      ----------------                                                      794
#  六、限制性指标（论文表  2  +  4.1  节流程）                             795
#  ----------------------------------------------------------------------  796
      ----------------                                                      797
#  论文摘要与  4.1  节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  798
CONSTRAINT_ITEMS  =  ("政策",  "区位约束",  "土壤污染",  "地质灾害",  "耕地属性")  799
#  ----------------------------------------------------------------------  800
```

```
    ------------------                                                  801
# 七、用户角色                                                          802
# ------------------------------------------------------------------- 803
    ------------------                                                  804
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",                 805
    "viewer": "查看者"}                                                 806
# ==== app/utils/schema.py ====                                       807
# -*- coding: utf-8 -*-                                               808
"""轻量结构迁移。                                                       809
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。            810
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，             811
使旧数据库无需重建即可用上新字段。                                     812
"""                                                                     813
from sqlalchemy import text                                             814
# (表名, 列名, 列定义)                                                  815
REQUIRED_COLUMNS = [                                                    816
    # 遥感影像挂接                                                      817
    ("mine_patches", "image_path", "VARCHAR(256)"),                     818
    ("mine_patches", "image_note", "VARCHAR(256)"),                     819
    ("mine_patches", "image_uploaded_at", "DATETIME"),                  820
    # 指标维护                                                          821
    ("indicators", "enabled_units", "VARCHAR(128)"),                    822
    ("indicators", "description", "TEXT"),                              823
    # 多专家矩阵                                                        824
    ("expert_matrices", "unit_type", "VARCHAR(32)"),                    825
    ("expert_matrices", "l1_name", "VARCHAR(64)"),                      826
    ("expert_matrices", "cr", "FLOAT"),                                 827
    # 用户个人资料（个人设置页）                                        828
    ("users", "display_name", "VARCHAR(64)"),                           829
    ("users", "title", "VARCHAR(64)"),                                  830
    ("users", "email", "VARCHAR(128)"),                                 831
    ("users", "phone", "VARCHAR(32)"),                                  832
    ("users", "avatar", "VARCHAR(256)"),                                833
]                                                                       834
def ensure_schema(db):                                                  835
    """为已有表补齐缺失列。返回补列清单。"""                           836
    added = []                                                          837
    try:                                                                838
        with db.engine.connect() as conn:                               839
            for table, column, ddl in REQUIRED_COLUMNS:                 840
                rows = conn.execute(text("PRAGMA                        841
    table_info(%s)" % table)).fetchall()                                842
                if not rows:                                            843
                    continue                                #           844
    表还不存在，create_all 会建                                         845
                existing = {r[1] for r in rows}                         846
                if column in existing:                                  847
                    continue                                            848
                conn.execute(text(                                      849
                    "ALTER TABLE %s ADD COLUMN %s %s" %                 850
```

```
        (table, column, ddl))                                                  851
                    conn.commit()                                              852
                    added.append("%s.%s" % (table, column))                    853
        except Exception as e:                                  #              854
        迁移失败不应阻断启动                                                   855
            print("[schema] 结构检查跳过：%s" % e)                             856
        return added                                                           857
# ==== app/utils/uploads.py ====                                               858
# -*- coding: utf-8 -*-                                                        859
"""上传文件处理（遥感影像、批量导入）。"""                                     860
import os                                                                      861
import re                                                                      862
import uuid                                                                    863
from datetime import datetime                                                  864
from werkzeug.utils import secure_filename                                     865
def _safe_ext(filename):                                                       866
        return os.path.splitext(filename)[1].lower().lstrip(".")               867
def ext_allowed(filename, allowed):                                            868
        return _safe_ext(filename) in allowed                                  869
def save_upload(file_storage, subdir, allowed_ext,                             870
        upload_root):                                                          871
        """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。  872
        文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。        873
        """                                                                    874
        if file_storage is None or not file_storage.filename:                  875
            return None, "未选择文件"                                          876
        ext = _safe_ext(file_storage.filename)                                 877
        if ext not in allowed_ext:                                             878
            return None, "不支持的文件类型：.%s" % ext                         879
        target_dir = os.path.join(upload_root, subdir)                         880
        os.makedirs(target_dir, exist_ok=True)                                 881
        # 仅保留安全字符，并加时间戳与随机串防重名                             882
        base = secure_filename(os.path.splitext(file_storage.fil               883
        ename)[0]) or "file"                                                   884
        base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                    885
        "file"                                                                 886
        name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M               887
        %S"),                                                                  888
                             uuid.uuid4().hex[:6], ext)                        889
        abs_path = os.path.join(target_dir, name)                              890
        file_storage.save(abs_path)                                            891
        rel = os.path.join("uploads", subdir,                                  892
        name).replace("\\", "/")                                               893
        return rel, None                                                       894
def remove_upload(rel_path, upload_root):                                      895
        """删除已保存的上传文件（相对 static 的路径）。"""                     896
        if not rel_path:                                                       897
            return                                                             898
        abs_path = os.path.join(upload_root,                                   899
        rel_path.replace("uploads/", "", 1))                                   900
```

```
    try:                                                                    901
        if os.path.isfile(abs_path):                                        902
            os.remove(abs_path)                                             903
    except OSError:                                                         904
        pass                                                                905
# ==== app/utils/charts.py ====                                             906
# -*- coding: utf-8 -*-                                                     907
"""轻量图表。                                                               908
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                                 909
- 服务端渲染，页面与打印（PDF）中都能稳定显示                               910
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）                  911
- 单色方案，符合界面「一个视图一个强调色」的约定                            912
"""                                                                         913
from html import escape                                                     914
ACCENT = "#1f3a5f"                                                          915
ACCENT_SOFT = "#7f93ad"                                                     916
INK = "#23262b"                                                             917
MUTED = "#6b7280"                                                           918
LINE = "#dfe3e8"                                                            919
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                              920
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                        921
    "#bcc9da", "#dbe3ec"]                                                   922
def _fmt(v, digits=4):                                                      923
    try:                                                                    924
        return ("%%.%df" % digits) % float(v)                               925
    except (TypeError, ValueError):                                         926
        return str(v)                                                       927
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,                  928
    value_w=76):                                                            929
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                       930
    if not items:                                                           931
        return '<p class="muted">暂无数据</p>'                              932
    n = len(items)                                                          933
    height = n * (bar_h + gap) + 10                                         934
    max_v = max((abs(float(v)) for _, v, _ in items),                       935
    default=1) or 1                                                         936
    body_w = width - label_w - value_w                                      937
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                        938
    height="%d" '                                                           939
             'role="img" aria-label="条形图" '                              940
             'style="max-width:%dpx;font-family:inherit;">'                 941
    % (width, height, height, width)]                                       942
    for i, (label, value, disp) in enumerate(items):                        943
        y = i * (bar_h + gap) + 5                                           944
        frac = (abs(float(value)) / max_v) if max_v else 0                  945
        w = max(2.0, body_w * frac)                                         946
        fill = GRAYS[i % len(GRAYS)]                                        947
        parts.append(                                                       948
            '<text x="%d" y="%d" text-anchor="end"                          949
    font-size="12" fill="%s">%s</text>'                                     950
```

```
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, float(v), d) for l, v, d in items if
    float(v) > 0]
    if not vals:
        return '<p class="muted">暂无数据</p>'
    total = sum(v for _, v, _ in vals) or 1
    r = (size - thickness) / 2
    cx = cy = size / 2
    import math
    parts = ['<svg viewBox="0 0 %d %d" width="%d"
    height="%d" role="img" '
             'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
            % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
```

```
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
            % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
               100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"
    ,
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom
    e.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed
    ge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe
    ",
]
#  允许通过环境变量覆盖，便于在其他机器上指定浏览器
```

```
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
```

```
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
```

```
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
```

```
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式（8） 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
```

```
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
    IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
```

```
                                std_benefit(data.get("thickness"),    1301
    IDEAL_THICKNESS),                                                 1302
                                l2w["土壤指标"]["土层厚度"]),                1303
            "pH": (data.get("ph"),                                    1304
                   std_mid_ph(data.get("ph"),                         1305
    *IDEAL_PH_RANGE),                                                 1306
                   l2w["土壤指标"]["pH"]),                               1307
            "有机质": (data.get("organic_matter"),                     1308
                       std_benefit(data.get("organic_matter")         1309
    , IDEAL_ORGANIC),                                                 1310
                       l2w["土壤指标"]["有机质"]),                          1311
            "污染指标": (data.get("heavy_metals") or {},               1312
                         pollution_score(data.get("heavy_meta         1313
    ls") or {}),                                                      1314
                         l2w["土壤指标"]["污染指标"]),                        1315
        }                                                             1316
    # ---- 植被指标 ----                                              1317
    if "植被指标" in l1_enabled:                                        1318
        result["植被指标"] = {                                          1319
            "植被覆盖度": (data.get("coverage"),                        1320
                           std_benefit(data.get("coverage"),          1321
    IDEAL_COVERAGE),                                                  1322
                           l2w["植被指标"]["植被覆盖度"]),                    1323
            "生物多样性": (data.get("shannon"),                         1324
                           std_benefit(data.get("shannon"),           1325
    IDEAL_SHANNON),                                                   1326
                           l2w["植被指标"]["生物多样性"]),                    1327
        }                                                             1328
    # ---- 地形指标 ----                                              1329
    if "地形指标" in l1_enabled:                                        1330
        result["地形指标"] = {                                          1331
            "坡度": (data.get("slope"),                               1332
                     std_slope(data.get("slope")),                    1333
                     l2w["地形指标"]["坡度"]),                            1334
            "坡向": (data.get("aspect"),                              1335
                     aspect_score(data.get("aspect")),                1336
                     l2w["地形指标"]["坡向"]),                            1337
        }                                                             1338
    return result                                                     1339
# ==== app/services/weight_service.py ====                          1340
# -*- coding: utf-8 -*-                                             1341
"""权重方案的存取与生效权重解析。                                          1342
系统有两套权重来源：                                                    1343
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），       1344
   是**权威取值来源**，案例复算与报告结论以此为准；                          1345
2. 通过层次分析法在本系统中实际算出的权重，保存在                           1346
   weight_schemes / weight_items 两张表中，可标记为「生效」。              1347
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                      1348
以论文表 3 为底，用生效方案中已保存的条目覆盖。                             1349
没有任何生效方案时，即完全使用论文表 3 的权重。                             1350
```

```
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
# 方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights       嵌套权重结构，可直接交给标准化与评价引擎
        source_label  权重来源说明，用于界面与报告标注
        scheme        生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
```

```
    merged["l2"][l1_name]:                                          1401
                merged["l2"][l1_name][l2_name] = w                  1402
                applied += 1                                        1403
    if applied == 0:                                                1404
        return copy.deepcopy(base), "论文表 3 默认权重", None          1405
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme   1406
def save_weights(unit_type, code_weight_pairs, note=None):          1407
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。          1408
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）              1409
    分次录入而无需一次填完。                                              1410
    """                                                             1411
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1412
    e,                                                              1413
                                                                    1414
    name=ACTIVE_NAME).first()                                       1415
    if scheme is None:                                              1416
        scheme = WeightScheme(unit_type=unit_type,                  1417
    name=ACTIVE_NAME,                                               1418
                                  level=1, is_default=True)         1419
        db.session.add(scheme)                                      1420
        db.session.flush()                                          1421
    if note:                                                        1422
        scheme.level = scheme.level or 1                            1423
    if not scheme.is_default:                                       1424
        scheme.is_default = True                                    1425
    written = 0                                                     1426
    for code, w in code_weight_pairs:                               1427
        if w is None or w <= 0:                                     1428
            continue                                                1429
        item = WeightItem.query.filter_by(scheme_id=scheme.i        1430
    d,                                                              1431
                                                                    1432
    indicator_code=code).first()                                    1433
        if item is None:                                            1434
            item = WeightItem(scheme_id=scheme.id,                  1435
    indicator_code=code)                                            1436
            db.session.add(item)                                    1437
        item.weight = float(w)                                      1438
        written += 1                                                1439
    db.session.commit()                                             1440
    return scheme, written                                          1441
def reset_weights(unit_type):                                       1442
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                1443
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1444
    pe,                                                             1445
                                                                    1446
    name=ACTIVE_NAME).all()                                         1447
    n = 0                                                           1448
    for s in schemes:                                               1449
        WeightItem.query.filter_by(scheme_id=s.id).delete(          1450
```

```
                synchronize_session=False)
            db.session.delete(s)
            n += 1
    db.session.commit()
    return n
def scheme_items(unit_type):
    """返回该单元类型已保存的自定义权重条目 {指标编码: 权重}。"""
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return {}, None
    items = {i.indicator_code: i.weight
             for i in WeightItem.query.filter_by(scheme_id=scheme.id).all()}
    return items, scheme
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type)
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式(9) 实现：
- 单元级：F' = Σ(二级指标权重 × 该二级指标下三级指标标准化值)
- 矿山级：F = Σ(Si × F'i)，Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details: [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
```

```
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    # 一级指标权重按实际参与计算的指标归一化。
    # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
    # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型： 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
```

```
    返回                                                                    1551
    ----                                                                    1552
    dict:{ final_score, grade, grade_info, unit_results }                   1553
    """                                                                     1554
    weights_map = weights_map or {}                                         1555
    total = 0.0                                                             1556
    unit_results = []                                                       1557
    for u in units:                                                         1558
        r = score_unit(u["unit_type"], u["data"],                           1559
                       weights_map.get(u["unit_type"]))                     1560
        weighted = u["ratio"] * r["unit_score"]                             1561
        total += weighted                                                   1562
        unit_results.append({                                               1563
            "unit_type": u["unit_type"],                                    1564
            "ratio": u["ratio"],                                            1565
            "unit_score": r["unit_score"],                                  1566
            "weighted": weighted,                                           1567
            "l1_scores": r["l1_scores"],                                    1568
            "details": r["details"],                                        1569
        })                                                                  1570
    grade_info = grade_of(total)                                            1571
    return {                                                                1572
        "final_score": total,                                               1573
        "grade": grade_info["grade"],                                       1574
        "grade_info": grade_info,                                           1575
        "unit_results": unit_results,                                       1576
    }                                                                       1577
# ==== app/services/report_service.py ====                                  1578
# -*- coding: utf-8 -*-                                                     1579
"""报告生成：生成 Word 评价报告。                                           1580
依赖 python-docx。报告结构：                                                1581
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。   1582
"""                                                                         1583
import io                                                                   1584
from datetime import datetime                                               1585
from docx import Document                                                   1586
from docx.shared import Pt, Cm                                              1587
from docx.oxml.ns import qn                                                 1588
from docx.enum.text import WD_ALIGN_PARAGRAPH                               1589
def _set_run(run, text=None, size=12, bold=False):                          1590
    if text is not None:                                                    1591
        run.text = text                                                     1592
    run.font.name = "Times New Roman"                                       1593
    run.font.size = Pt(size)                                                1594
    run.font.bold = bold                                                    1595
    # 中文字体                                                              1596
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")                   1597
    return run                                                              1598
def _para(doc, text, size=12, bold=False, align=None,                       1599
    space_after=6):                                                         1600
```

```
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d
    %H：%M：%S"),
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,
    space_after=12)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
```

```
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
]
_table(doc, ["项目", "内容"], info)
#  评价结论
_heading(doc, "二、评价结论", 1)
gi = mine_result["grade_info"]
concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
]
_table(doc, ["项目", "结果"], concl)
#  各单元得分
_heading(doc, "三、各微地质单元得分", 1)
rows = []
for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]
* 100),
                     "%.4f" % u["unit_score"], "%.4f" %
u["weighted"]))
_table(doc, ["微地质单元", "面积占比", "单元评价因子 F′", "加权贡献"],
rows)
#  完整得分明细
_heading(doc, "四、得分明细", 1)
for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f"
% d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else
"%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std,
w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
"贡献"],
               detail_rows)
        #  一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
```

```
            _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,
        space_after=10)
        buf = io.BytesIO()
        doc.save(buf)
        buf.seek(0)
        return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
```

```
        #  ----  二级指标：植被  ----
        ("coverage", "植被覆盖度", 2, "veg", "%",
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
          "论文 4.2.2 节：参照生态系统取值 75%", False),
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
        "benefit", ALL_UNITS,
          "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
        #  ----  二级指标：地形  ----
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
          "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
        为陡坡重要分界", False),
        ("aspect", "坡向", 2, "terrain", "—",
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
          "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
#  污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
        TREE.append((
              "hm_" + _m, _m, 3, "pollution", "mg/kg",
        str(_limit), "cost",
              "碎石边坡,平台",
              "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
        _limit,
              False,
        ))
def seed_indicators():
        """写入/更新指标定义（幂等）。返回写入条数。"""
        code_map = {
              "benefit": "越大越优", "cost": "越小越优",
              "mid": "适中型", "discrete": "离散赋值",
        }
        n = 0
        for (code, name, level, parent, unit, ideal, itype,
        units, desc, is_con) in TREE:
              ind = Indicator.query.filter_by(code=code).first()
              if ind is None:
                    ind = Indicator(code=code)
                    db.session.add(ind)
              ind.name = name
              ind.level = level
              ind.parent_code = parent
              ind.unit = unit
              ind.ideal_value = ideal
              ind.ideal_type = itype
              ind.enabled_units = units
              ind.description = desc
              ind.is_constraint = is_con
              ind.is_evaluative = (level in (2, 3))
              n += 1
```

```
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg， As， Ni， Cu， Cd， Pb， Cr， Zn，
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
```

```
        "Pb": "6.50",
                    "植被覆盖度": "87.92", "Shannon": "0.60",
                    "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
```

```
    ----
    (parsed, errors)
       parsed : list[dict]，每项 {unit_type, fields{...},
    heavy_metals{}}
       errors : list[str]，逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    #  表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       #  第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
    "、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
    key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
```

```
                    s = str(val).strip() if val is not None
    else ""
                    if s == "":
                        fields["aspect"] = None
                    elif s in ASPECT_TYPES:
                        fields["aspect"] = s
                    else:
                        errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                      % (idx, s,
    "、".join(ASPECT_TYPES)))
                        ok = False
                    continue
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
# 指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
               "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
```

```
        return True                                                    2001
    return bool(ind.heavy_metals_dict())                               2002
def patch_progress(patch_id):                                          2003
    """检查某图斑的评价流程完成情况。                                  2004
    返回 dict：                                                        2005
        units          {done, count, ratio_sum, detail}                2006
        constraint     {done, passed, detail}                          2007
        data           {done, ready, missing, detail}                  2008
        blocking       未满足的前置步骤列表（按流程顺序）              2009
        ready          是否可进入定量评价                              2010
    """                                                                2011
    units = MicroUnit.query.filter_by(patch_id=patch_id).all2012
    ()                                                                 2013
    ratio_sum = sum(u.ratio or 0.0 for u in units)                     2014
    # ---- 步骤一：微地质单元 ----                                     2015
    units_step = {                                                     2016
        "done": bool(units),                                           2017
        "count": len(units),                                           2018
        "ratio_sum": ratio_sum,                                        2019
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"       2020
                   % (len(units), ratio_sum * 100))                    2021
                  if units else "尚未划分微地质单元",                  2022
    }                                                                  2023
    # ---- 步骤二：限制性条件判定 ----                                 2024
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)            2025
    \                                                                  2026
        .order_by(ConstraintCheck.id.desc()).first()                   2027
    if cc is None:                                                     2028
        constraint_step = {"done": False, "passed": False,             2029
                           "detail": "尚未执行限制性条件判定"}         2030
    elif cc.passed:                                                    2031
        constraint_step = {"done": True, "passed": True,               2032
                           "detail": "限制性条件判定已通过"}           2033
    else:                                                              2034
        constraint_step = {"done": True, "passed": False,              2035
                           "detail": "限制性条件判定未通过（%s）" %    2036
    (cc.reason or "")}                                                 2037
    # ---- 步骤三：指标数据 ----                                       2038
    missing = []                                                       2039
    for u in units:                                                    2040
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi2041
    rst()                                                              2042
        if not _has_data(ind):                                         2043
            missing.append(u.unit_type)                                2044
    if not units:                                                      2045
        data_step = {"done": False, "ready": False,                    2046
    "missing": [],                                                     2047
                     "detail": "尚无可录入数据的微地质单元"}           2048
    elif missing:                                                      2049
        data_step = {"done": False, "ready": False,                    2050
```

```
    "missing": missing,                                                    2051
                        "detail": "以下微地质单元尚未录入指标数据：%s"          2052
                                  % "、".join(missing)}                    2053
    else:                                                                  2054
        data_step = {"done": True, "ready": True,                          2055
    "missing": [],                                                         2056
                        "detail": "全部 %d 个微地质单元的指标数据已录入" %      2057
    len(units)}                                                            2058
    ratio_ok = ratio_sum > 0.999 if units else False                       2059
    if units and not ratio_ok:                                             2060
        units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"      2061
    # ---- 汇总未满足的前置条件（按论文 4.1 节的流程顺序：限制性判定优先） ----  2062
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——                2063
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。     2064
    blocking = []                                                          2065
    if not constraint_step["done"]:                                        2066
        blocking.append({                                                  2067
            "step": "constraint", "label": "限制性条件判定未执行",          2068
            "reason": "五项限制性条件尚未逐项判定。",                        2069
            "fix_url": "/constraint/%d" % patch_id,                        2070
    "fix_label": "去执行判定",                                             2071
        })                                                                 2072
    elif not constraint_step["passed"]:                                    2073
        blocking.append({                                                  2074
            "step": "constraint", "label": "限制性条件判定未通过",          2075
            "reason": cc.reason or "",                                     2076
            "fix_url": "/constraint/%d" % patch_id,                        2077
    "fix_label": "去复核判定结果",                                         2078
        })                                                                 2079
    if not units_step["done"]:                                             2080
        blocking.append({                                                  2081
            "step": "units", "label": "未划分微地质单元",                  2082
            "reason": "该图斑下没有任何微地质单元。",                        2083
            "fix_url": "/unit/%d" % patch_id, "fix_label":                 2084
    "去划分",                                                              2085
        })                                                                 2086
    elif not ratio_ok:                                                     2087
        blocking.append({                                                  2088
            "step": "units", "label": "面积占比合计不为 100%",             2089
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),             2090
            "fix_url": "/unit/%d" % patch_id, "fix_label":                 2091
    "去核对面积",                                                          2092
        })                                                                 2093
    if units_step["done"] and not data_step["ready"]:                      2094
        blocking.append({                                                  2095
            "step": "data", "label": "指标数据未录入完整",                 2096
            "reason": data_step["detail"],                                 2097
            "fix_url": "/indicator/", "fix_label": "去录入",               2098
        })                                                                 2099
    return {                                                               2100
```

```
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_id) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#     key            步骤标识
```

```
#     no            步骤编号（01~08）                                          2151
#     title         名称                                                       2152
#     icon          图标名（见 templates/_macros.html 的 ICONS）               2153
#     desc          一句话说明                                                 2154
#     href          入口地址                                                   2155
STEPS = [                                                                      2156
      dict(key="project", no="01", title="项目与矿山档案",                     2157
      icon="folder",                                                           2158
            href="/project/",                                                  2159
            desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),  2160
      dict(key="standard", no="02", title="政策与标准知识库",                  2161
      icon="book",                                                             2162
            href="/standard/",                                                 2163
            desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),    2164
      dict(key="constraint", no="03", title="限制性指标判定",                  2165
      icon="shield",                                                           2166
            href="/constraint/",                                               2167
            desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),    2168
      dict(key="unit", no="04", title="微地质单元", icon="layers",             2169
            href="/unit/",                                                     2170
            desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),      2171
      dict(key="ahp", no="05", title="指标体系与权重",                         2172
      icon="sliders",                                                          2173
            href="/ahp/",                                                      2174
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),2175
      dict(key="indicator", no="06", title="指标数据采集",                     2176
      icon="database",                                                         2177
            href="/indicator/",                                                2178
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),2179
      dict(key="evaluate", no="07", title="适宜性定量评价",                    2180
      icon="target",                                                           2181
            href="/evaluate/",                                                 2182
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),        2183
      dict(key="report", no="08", title="报告与结果输出",                      2184
      icon="file",                                                             2185
            href="/report/",                                                   2186
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),          2187
]                                                                              2188
def _has_data(ind):                                                            2189
      return wf._has_data(ind)                                                 2190
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的用户与权限管理）       2191
MODULES = [                                                                    2192
      ("01", "用户与权限管理", "登录注销、三种角色权限、写操作自动留痕"),      2193
      ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),2194
      ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),2195
      ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),2196
      ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),2197
      ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),2198
      ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),2199
      ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),2200
```

```
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
    l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
    passed_constraint),
```

```
        "unit": sum(1 for p in patches if not
    unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
    us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
        "constraint": "需先建立矿山图斑",
        "unit": "需先建立矿山图斑",
        "indicator": "需先划分微地质单元",
        "evaluate": "需前置步骤齐备的图斑",
        "report": "需先执行适宜性评价",
    }
    steps = []
    for s in STEPS:
        steps.append(dict(
            s,
            available=avail[s["key"]],
            pending=pending.get(s["key"], 0),
            note=None if avail[s["key"]] else
    notes.get(s["key"], ""),
        ))
    unit_types = len({u.unit_type for u in units})
    stats = {
        "projects": projects,
        "patches": len(patches),
        "unit_types": unit_types,
        "standards": Standard.query.count(),
    }
```

```
    return {
        "stats": stats,
        "steps": steps,
        "badges": {k: v for k, v in pending.items() if v},
        "progress": {
            "patches": len(patches),
            "units": len(units),
            "data_units": len(data_unit_ids),
            "ready": ready_patches,
            "evaluated": len(evaluated),
        },
    }
def nav_badges():
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
        "/unit/": snap["badges"].get("unit", 0),
        "/indicator/": snap["badges"].get("indicator", 0),
        "/evaluate/": snap["badges"].get("evaluate", 0),
    }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.logout": "退出登录",
    "auth.add_user": "新增用户",
    "auth.edit_user": "编辑用户",
    "auth.delete_user": "删除用户",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
```

```
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += ";结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
```

```
                target_type=(endpoint or "").split(".")[0] or
    None,
                target_id=extract_target_id(),
                detail=detail,
            ))
            db.session.commit()
    except Exception:
            # 审计失败不应影响业务响应
            db.session.rollback()
# ==== app/views/auth.py ====
# -*- coding: utf-8 -*-
"""用户与权限管理 —— 登录、注销、用户维护、操作日志。"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                       flash, abort, current_app)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import (LoginForm, EmptyForm, ProfileForm,
    PasswordForm,
                           email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
from app.utils.constants import ROLE_LABELS
bp = Blueprint("auth", __name__)
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "error")
                return render_template("auth/login.html",
```

```
    form=form,                                                    2451
                                                  **_intro_data())  2452
            login_user(user, remember=form.remember.data)         2453
            # 用 toast 类别：渲染为页面正中的浮层，淡入停留后淡出。  2454
            # 登录成功只是告知性反馈，不需要用户停下来阅读。          2455
            flash("登录成功", "toast")                             2456
            return redirect(url_for("auth.index"))                2457
        flash("用户名或密码错误", "error")                         2458
    return render_template("auth/login.html", form=form,          2459
    **_intro_data())                                              2460
def _intro_data():                                                2461
    """登录页下方的系统介绍所用的静态内容。"""                      2462
    from app.services import dashboard as dash                    2463
    return {"modules": dash.MODULES}                              2464
@bp.route("/logout")                                              2465
@login_required                                                   2466
def logout():                                                     2467
    logout_user()                                                 2468
    flash("已退出登录", "ok")                                      2469
    return redirect(url_for("auth.login"))                        2470
# ---------------------------- 个人设置 ----------------------------2471
    ----                                                          2472
# 与「用户管理」的区别：那里是管理员维护别人，这里是任何登录用户维护自己。  2473
@bp.route("/profile", methods=["GET", "POST"])                    2474
@login_required                                                   2475
def profile():                                                    2476
    # 两个表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名  2477
    info_form = ProfileForm(prefix="info", obj=current_user)      2478
    pwd_form = PasswordForm(prefix="pwd")                         2479
    if request.method == "POST":                                  2480
        action = request.form.get("action")                       2481
        if action == "info" and info_form.validate_on_submit      2482
    ():                                                           2483
            current_user.display_name =                           2484
    (info_form.display_name.data or "").strip() or None           2485
            current_user.title = (info_form.title.data or         2486
    "").strip() or None                                           2487
            current_user.email = (info_form.email.data or         2488
    "").strip() or None                                           2489
            current_user.phone = (info_form.phone.data or         2490
    "").strip() or None                                           2491
            db.session.commit()                                   2492
            flash("个人资料已保存", "ok")                          2493
            return redirect(url_for("auth.profile"))              2494
        if action == "password" and pwd_form.validate_on_sub      2495
    mit():                                                        2496
            if not check_password_hash(current_user.password      2497
    _hash,                                                        2498
                                                                  2499
    pwd_form.old_password.data):                                  2500
```

```
                pwd_form.old_password.errors.append("当前密码不正确
    ")
            else:
                current_user.password_hash =
    generate_password_hash(
                    pwd_form.new_password.data)
                db.session.commit()
                flash("密码已修改", "ok")
                return redirect(url_for("auth.profile"))
    return render_template("auth/profile.html",
    info_form=info_form,
                           pwd_form=pwd_form,
    avatar_form=EmptyForm(),
                           role_labels=ROLE_LABELS)
def _avatar_length_ok(file_storage):
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *
    1024 * 1024)
    stream = file_storage.stream
    pos = stream.tell()
    stream.seek(0, 2)
    size = stream.tell()
    stream.seek(pos)
    return size <= limit, size
@bp.route("/profile/avatar", methods=["POST"])
@login_required
def upload_avatar():
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("auth.profile"))
    file = request.files.get("avatar")
    if file is None or not file.filename:
        flash("请选择头像图片", "error")
        return redirect(url_for("auth.profile"))
    ok, size = _avatar_length_ok(file)
    if not ok:
        flash("头像不能超过 %.0f MB（当前 %.1f MB）"
              % (current_app.config["MAX_AVATAR_LENGTH"] /
    1024 / 1024,
                 size / 1024 / 1024), "error")
        return redirect(url_for("auth.profile"))
    rel, err = up.save_upload(file, "avatars",
                              current_app.config["ALLOWED_AV
    ATAR_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("auth.profile"))
```

```
    # 换新头像时删掉旧文件，避免残留
    up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
    current_user.avatar = rel
    db.session.commit()
    flash("头像已更新", "ok")
    return redirect(url_for("auth.profile"))
@bp.route("/profile/avatar/delete", methods=["POST"])
@login_required
def delete_avatar():
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
        current_user.avatar = None
        db.session.commit()
        flash("头像已移除", "ok")
    return redirect(url_for("auth.profile"))
# ---------------------------- 用户维护 ----------------------------
    ----
@bp.route("/users")
@login_required
def users():
    all_users = User.query.order_by(User.id).all()
    return render_template("auth/users.html",
    users=all_users,
                           role_labels=ROLE_LABELS,
    form=EmptyForm())
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", "viewer")
        if not username or not password:
            flash("用户名和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("用户名已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(username=username, role=role,
                            password_hash=generate_password_
    hash(password)))
        db.session.commit()
        flash("用户已添加", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/add_user.html")
@bp.route("/users/<int:user_id>/edit", methods=["GET",
```

```
    "POST"])                                                              2601
@login_required                                                           2602
def edit_user(user_id):                                                   2603
    require_admin()                                                       2604
    user = db.get_or_404(User, user_id)                                   2605
    if request.method == "POST":                                          2606
        role = request.form.get("role", user.role)                        2607
        active = request.form.get("active") == "1"                        2608
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统                2609
        if user.role == "admin" and (role != "admin" or not               2610
    active):                                                              2611
            others = User.query.filter(User.role == "admin",              2612
                                       User.id != user.id,                2613
                                                                          2614
    User.active.is_(True)).count()                                        2615
            if others == 0:                                               2616
                flash("系统至少需保留一名启用状态的管理员", "error")          2617
                return redirect(url_for("auth.edit_user",                 2618
    user_id=user.id))                                                     2619
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）          2620
        email = request.form.get("email", "").strip()                     2621
        if not email_ok(email):                                           2622
            flash("邮箱格式不正确", "error")                               2623
            return redirect(url_for("auth.edit_user",                     2624
    user_id=user.id))                                                     2625
        user.role = role                                                  2626
        user.active = active                                              2627
        user.display_name = request.form.get("display_name",              2628
    "").strip() or None                                                   2629
        user.title = request.form.get("title", "").strip()                2630
    or None                                                               2631
        user.email = email or None                                        2632
        user.phone = request.form.get("phone", "").strip()                2633
    or None                                                               2634
        new_password = request.form.get("password",                       2635
    "").strip()                                                           2636
        if new_password:                                                  2637
            user.password_hash = generate_password_hash(new_              2638
    password)                                                             2639
        db.session.commit()                                               2640
        flash("用户信息已保存", "ok")                                      2641
        return redirect(url_for("auth.users"))                            2642
    return render_template("auth/edit_user.html", user=user,              2643
                           role_labels=ROLE_LABELS,                       2644
    form=EmptyForm())                                                     2645
@bp.route("/users/<int:user_id>/delete", methods=["POST"])                2646
@login_required                                                           2647
def delete_user(user_id):                                                 2648
    require_admin()                                                       2649
    user = db.get_or_404(User, user_id)                                   2650
```

```
    if user.id == current_user.id:                                    2651
        flash("不能删除当前登录的账号", "error")                          2652
        return redirect(url_for("auth.users"))                        2653
    if user.role == "admin":                                          2654
        others = User.query.filter(User.role == "admin",              2655
                                   User.id != user.id,                2656
                                   User.active.is_(True)).co          2657
    unt()                                                             2658
        if others == 0:                                               2659
            flash("系统至少需保留一名启用状态的管理员", "error")                  2660
            return redirect(url_for("auth.users"))                    2661
    db.session.delete(user)                                           2662
    db.session.commit()                                               2663
    flash("用户「%s」已删除" % user.username, "ok")                       2664
    return redirect(url_for("auth.users"))                            2665
# ---------------------------- 操作日志 ----------------------------  2666
    ----                                                              2667
@bp.route("/logs")                                                    2668
@login_required                                                       2669
def logs():                                                           2670
    page = request.args.get("page", 1, type=int)                      2671
    per_page = 50                                                     2672
    q = OperationLog.query.order_by(OperationLog.id.desc())           2673
    total = q.count()                                                 2674
    records = q.offset((page - 1) * per_page).limit(per_page          2675
    ).all()                                                           2676
    users_map = {u.id: u.username for u in User.query.all()}          2677
    pages = max(1, (total + per_page - 1) // per_page)                2678
    return render_template("auth/logs.html",                          2679
    records=records, page=page,                                       2680
                                pages=pages, total=total,             2681
    users_map=users_map)                                              2682
# ==== app/views/project.py ====                                      2683
# -*- coding: utf-8 -*-                                               2684
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""                  2685
from datetime import datetime                                         2686
from flask import (Blueprint, render_template, request,               2687
    redirect, url_for,                                                2688
                      flash, current_app)                             2689
from flask_login import login_required, current_user                  2690
from app.forms import EmptyForm                                       2691
from app.models import db, Project, MinePatch,                        2692
    ReferenceEcosystem                                                2693
bp = Blueprint("project", __name__, url_prefix="/project")            2694
@bp.route("/")                                                        2695
@login_required                                                       2696
def index():                                                          2697
    projects = Project.query.order_by(Project.id.desc()).all          2698
    ()                                                                2699
    return render_template("project/index.html",                      2700
```

```
    projects=projects,                                                    2701
                                form=EmptyForm())                         2702
@bp.route("/add", methods=["GET", "POST"])                                2703
@login_required                                                           2704
def add():                                                                2705
    form = EmptyForm()                                                    2706
    if request.method == "POST" and form.validate_on_submit(              2707
    ):                                                                    2708
        p = Project(                                                      2709
            name=request.form.get("name", "").strip(),                    2710
            code=request.form.get("code", "").strip() or                  2711
    None,                                                                 2712
            description=request.form.get("description",                   2713
    "").strip() or None,                                                  2714
            created_by=current_user.id,                                   2715
        )                                                                 2716
        if not p.name:                                                    2717
            flash("项目名称不能为空", "error")                                2718
        else:                                                             2719
            db.session.add(p)                                             2720
            db.session.commit()                                           2721
            flash("项目已创建", "ok")                                         2722
            return redirect(url_for("project.index"))                     2723
    return render_template("project/edit.html", form=form,                2724
    project=None)                                                         2725
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])                     2726
@login_required                                                           2727
def edit(pid):                                                            2728
    p = db.get_or_404(Project, pid)                                       2729
    form = EmptyForm()                                                    2730
    if request.method == "POST" and form.validate_on_submit(              2731
    ):                                                                    2732
        p.name = request.form.get("name", "").strip()                     2733
        p.code = request.form.get("code", "").strip() or                  2734
    None                                                                  2735
        p.description = request.form.get("description",                   2736
    "").strip() or None                                                   2737
        p.status = request.form.get("status", "进行中")                     2738
        db.session.commit()                                               2739
        flash("项目已保存", "ok")                                           2740
        return redirect(url_for("project.index"))                         2741
    return render_template("project/edit.html", form=form,                2742
    project=p)                                                            2743
@bp.route("/<int:pid>/delete", methods=["POST"])                          2744
@login_required                                                           2745
def delete(pid):                                                          2746
    p = db.get_or_404(Project, pid)                                       2747
    db.session.delete(p)                                                  2748
    db.session.commit()                                                   2749
    flash("项目已删除", "ok")                                               2750
```

```
    return redirect(url_for("project.index"))                    2751
# ---------------------------- 矿山图斑 ---------------------------2752
    ----                                                         2753
@bp.route("/<int:pid>/patches")                                  2754
@login_required                                                  2755
def patches(pid):                                                2756
    p = db.get_or_404(Project, pid)                              2757
    return render_template("project/patches.html",               2758
    project=p,                                                   2759
                                    form=EmptyForm())            2760
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])      2761
@login_required                                                  2762
def patch_delete(patch_id):                                      2763
    """删除矿山图斑，及其微地质单元与指标数据。"""                  2764
    from app.models import MicroUnit, IndicatorData,             2765
    EvaluationResult, \                                          2766
        ScoreDetail, ConstraintCheck, ReferenceEcosystem         2767
    patch = db.get_or_404(MinePatch, patch_id)                   2768
    pid = patch.project_id                                       2769
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc2770
    h_id=patch.id).all()]                                        2771
    if unit_ids:                                                 2772
        IndicatorData.query.filter(IndicatorData.unit_id.in_2773
    (unit_ids)).delete(                                          2774
            synchronize_session=False)                           2775
    for r in EvaluationResult.query.filter_by(patch_id=patch2776
    .id).all():                                                  2777
        ScoreDetail.query.filter_by(result_id=r.id).delete(      2778
            synchronize_session=False)                           2779
    EvaluationResult.query.filter_by(patch_id=patch.id).dele2780
    te(                                                          2781
        synchronize_session=False)                               2782
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet2783
    e(                                                           2784
        synchronize_session=False)                               2785
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync2786
    hronize_session=False)                                       2787
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de2788
    lete(                                                        2789
        synchronize_session=False)                               2790
    db.session.delete(patch)                                     2791
    db.session.commit()                                          2792
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")             2793
    return redirect(url_for("project.patches", pid=pid))         2794
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])       2795
@login_required                                                  2796
def patch_add(pid):                                              2797
    p = db.get_or_404(Project, pid)                              2798
    form = EmptyForm()                                           2799
    if request.method == "POST" and form.validate_on_submit(2800
```

```
    ):                                                                  2801
        def fv(name, cast=str):                                         2802
            v = request.form.get(name, "").strip()                      2803
            return cast(v) if v != "" else None                         2804
        patch = MinePatch(                                              2805
            project_id=p.id,                                            2806
            name=request.form.get("name", "").strip(),                  2807
            admin_region=fv("admin_region"),                            2808
            area_ha=fv("area_ha", float),                               2809
            mineral_type=fv("mineral_type"),                            2810
            mining_method=fv("mining_method"),                          2811
            stop_years=fv("stop_years", float),                         2812
            responsible_entity_missing=bool(request.form.get            2813
    ("responsible_entity_missing")),                                    2814
        )                                                               2815
        if not patch.name:                                              2816
            flash("图斑名称不能为空", "error")                              2817
        else:                                                           2818
            db.session.add(patch)                                       2819
            db.session.flush()                                          2820
            # 参照生态系统默认取论文值                                      2821
            db.session.add(ReferenceEcosystem(                          2822
                patch_id=patch.id, coverage_benchmark=75.0,             2823
    shannon_benchmark=0.7))                                             2824
            db.session.commit()                                         2825
            flash("图斑已创建", "ok")                                      2826
            return redirect(url_for("project.patches",                  2827
    pid=p.id))                                                          2828
    return render_template("project/patch_edit.html",                   2829
    form=form, project=p, patch=None)                                   2830
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",                 2831
    "POST"])                                                            2832
@login_required                                                         2833
def patch_edit(patch_id):                                               2834
    patch = db.get_or_404(MinePatch, patch_id)                          2835
    form = EmptyForm()                                                  2836
    if request.method == "POST" and form.validate_on_submit(            2837
    ):                                                                  2838
        def fv(name, cast=str):                                         2839
            v = request.form.get(name, "").strip()                      2840
            return cast(v) if v != "" else None                         2841
        patch.name = request.form.get("name", "").strip()               2842
        patch.admin_region = fv("admin_region")                         2843
        patch.area_ha = fv("area_ha", float)                            2844
        patch.mineral_type = fv("mineral_type")                         2845
        patch.mining_method = fv("mining_method")                       2846
        patch.stop_years = fv("stop_years", float)                      2847
        patch.responsible_entity_missing =                              2848
    bool(request.form.get("responsible_entity_missing"))                2849
        # 参照生态系统                                                  2850
```

```
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
        return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                           project=patch.project,
    patch=patch,
                           reference=ReferenceEcosystem.quer
    y.filter_by(
                               patch_id=patch.id).first())
# ---------------------------- 遥感影像挂接
    ----------------------------
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])
@login_required
def upload_image(patch_id):
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    rel, err = up.save_upload(request.files.get("image"),
    "patches",
                              current_app.config["ALLOWED_IM
    AGE_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    # 替换旧影像时清理原文件，避免残留
    if patch.image_path:
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
    patch.image_path = rel
```

```
    patch.image_uploaded_at = datetime.utcnow()
    note = request.form.get("image_note", "").strip()
    if note:
        patch.image_note = note
    db.session.commit()
    flash("遥感影像已挂接", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
@bp.route("/patch/<int:patch_id>/image/delete",
    methods=["POST"])
@login_required
def delete_image(patch_id):
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
        patch.image_path = None
        patch.image_uploaded_at = None
        db.session.commit()
        flash("遥感影像已移除", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
```

```
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
```

```
            db.session.add(u)                                                3001
            db.session.flush()                                               3002
            recalc_ratio(patch.id)                                           3003
            db.session.commit()                                              3004
            flash("微地质单元已添加", "ok")                                    3005
        return redirect(url_for("unit.index",                                3006
        patch_id=patch_id))                                                  3007
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])                    3008
@login_required                                                              3009
def edit(unit_id):                                                           3010
        """单个微地质单元的编辑页。"""                                          3011
        u = db.get_or_404(MicroUnit, unit_id)                                3012
        form = EmptyForm()                                                   3013
        if request.method == "POST" and form.validate_on_submit(             3014
        ):                                                                   3015
            new_type = request.form.get("unit_type",                         3016
        u.unit_type)                                                         3017
            if new_type not in UNIT_TYPES:                                   3018
                flash("微地质单元类型不正确", "error")                          3019
                return redirect(url_for("unit.edit",                         3020
        unit_id=u.id))                                                       3021
            #  改类型时同样不允许与本图斑已有的其他单元重复                      3022
            dup = MicroUnit.query.filter_by(patch_id=u.patch_id,             3023
                                                                             3024
        unit_type=new_type).first()                                          3025
            if dup is not None and dup.id != u.id:                           3026
                flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,        3027
        "error")                                                             3028
                return redirect(url_for("unit.edit",                         3029
        unit_id=u.id))                                                       3030
            u.unit_type = new_type                                           3031
            area = request.form.get("area", "").strip()                      3032
            u.area = float(area) if area else None                           3033
            u.description = request.form.get("description",                  3034
        "").strip() or None                                                  3035
            recalc_ratio(u.patch_id)                                         3036
            db.session.commit()                                              3037
            flash("微地质单元已保存", "ok")                                    3038
            return redirect(url_for("unit.index",                            3039
        patch_id=u.patch_id))                                                3040
        return render_template("unit/edit.html", unit=u,                     3041
        patch=u.patch,                                                       3042
                                    unit_types=UNIT_TYPES, form=form)        3043
@bp.route("/delete/<int:unit_id>", methods=["POST"])                         3044
@login_required                                                              3045
def delete(unit_id):                                                         3046
        u = db.get_or_404(MicroUnit, unit_id)                                3047
        pid = u.patch_id                                                     3048
        name = u.unit_type                                                   3049
        #  一并删除该单元的指标数据，避免残留                                  3050
```

```
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch
    ronize_session=False)
    db.session.delete(u)
    db.session.flush()
    recalc_ratio(pid)
    db.session.commit()
    flash("微地质单元「%s」已删除" % name, "ok")
    return redirect(url_for("unit.index", patch_id=pid))
# ==== app/views/standard.py ====
# -*- coding: utf-8 -*-
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, Standard, StandardRule
bp = Blueprint("standard", __name__, url_prefix="/standard")
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
```

```
                db.session.add(s)                                       3101
                db.session.commit()                                     3102
                flash("标准已添加", "ok")                                3103
                return redirect(url_for("standard.index"))              3104
        return render_template("standard/edit.html", form=form,         3105
        standard=None)                                                  3106
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])                   3107
@login_required                                                         3108
def edit(sid):                                                          3109
        s = db.get_or_404(Standard, sid)                                3110
        form = EmptyForm()                                              3111
        if request.method == "POST" and form.validate_on_submit(        3112
        ):                                                              3113
                s.province = request.form.get("province",               3114
        "").strip()                                                     3115
                s.name = request.form.get("name", "").strip()           3116
                s.doc_number = request.form.get("doc_number",           3117
        "").strip() or None                                             3118
                s.publish_date = request.form.get("publish_date",       3119
        "").strip() or None                                             3120
                s.implement_date = request.form.get("implement_date"    3121
        , "").strip() or None                                           3122
                s.category = request.form.get("category", "地方标准")    3123
                s.summary = request.form.get("summary", "").strip()     3124
        or None                                                         3125
                db.session.commit()                                     3126
                flash("已保存", "ok")                                    3127
                return redirect(url_for("standard.rules", sid=s.id))    3128
        return render_template("standard/edit.html", form=form,         3129
        standard=s)                                                     3130
@bp.route("/<int:sid>/delete", methods=["POST"])                        3131
@login_required                                                         3132
def delete(sid):                                                        3133
        s = db.get_or_404(Standard, sid)                                3134
        StandardRule.query.filter_by(standard_id=sid).delete()          3135
        db.session.delete(s)                                            3136
        db.session.commit()                                             3137
        flash("已删除", "ok")                                            3138
        return redirect(url_for("standard.index"))                      3139
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])                  3140
@login_required                                                         3141
def rules(sid):                                                         3142
        s = db.get_or_404(Standard, sid)                                3143
        form = EmptyForm()                                              3144
        if request.method == "POST" and form.validate_on_submit(        3145
        ):                                                              3146
                r = StandardRule(                                       3147
                        standard_id=s.id,                               3148
                        item=request.form.get("item", "").strip(),      3149
                        operator=request.form.get("operator", ">="),    3150
```

```
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
            flash("判定项不能为空", "error")
        else:
            db.session.add(r)
            db.session.commit()
            flash("规则已添加", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    rs = StandardRule.query.filter_by(standard_id=s.id).all(
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
```

```
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch
    _id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_
    id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit(
    ):
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field)
    == "1")
        record.passed = all(getattr(record, f) for f, _ in
    FIELDS)
        failed = [name for f, name in FIELDS if not
    getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check",
    patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
```

```
    field_states = [(f, name, None if record is None else
    getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
```

```
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
```

```
                                        level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                                   weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
    range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
    j), "1"))
                     for i in range(n) for j in range(i +
    1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
    "error")
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
```

```
                    flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                          "请调整后再保存" % result["cr"], "error")
                else:
                    pairs = to_code_pairs(unit_type, level, l1,
    names,
                                          result["weights"])
                    scheme, cnt = ws.save_weights(unit_type,
    pairs)
                    flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                          "此后该单元类型的评价将以此权重计算。"
                          % (unit_type, cnt, result["cr"]),
    "ok")
                    return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                            level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
    it_type)
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
        source_label=source_label, custom_scheme=custom_sche
    me,
    )
@bp.route("/weights/reset", methods=["POST"])
def weights_reset():
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""
    unit_type = request.form.get("unit_type")
    form = EmptyForm()
    if form.validate_on_submit():
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
        else:
            n = ws.reset_weights(unit_type)
            if n:
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,
    "ok")
            else:
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,
    "ok")
    return redirect(url_for("ahp.index"))
# ---------------------------- 指标维护 ----------------------------
    ----
@bp.route("/indicators")
def indicators():
```

```
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""
    if Indicator.query.count() == 0:
        ind_svc.seed_indicators()
    all_ind = Indicator.query.order_by(Indicator.level,
    Indicator.id).all()
    by_parent = {}
    for i in all_ind:
        by_parent.setdefault(i.parent_code, []).append(i)
    return render_template("ahp/indicators.html",
    by_parent=by_parent,
                           all_ind=all_ind,
    unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el,
                           form=EmptyForm())
@bp.route("/indicators/<code>/edit", methods=["GET",
    "POST"])
def indicator_edit(code):
    ind = Indicator.query.filter_by(code=code).first_or_404(
    )
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        name = request.form.get("name", "").strip()
        if not name:
            flash("指标名称不能为空", "error")
            return redirect(url_for("ahp.indicator_edit",
    code=code))
        ind.name = name
        ind.unit = request.form.get("unit", "").strip() or
    None
        ind.ideal_value = request.form.get("ideal_value",
    "").strip() or None
        ind.ideal_type = request.form.get("ideal_type") or
    ind.ideal_type
        units = request.form.getlist("enabled_units")
        ind.enabled_units = ",".join(units) if units else
    None
        ind.is_constraint = request.form.get("is_constraint"
    ) == "1"
        ind.description = request.form.get("description",
    "").strip() or None
        db.session.commit()
        flash("指标「%s」已保存" % ind.name, "ok")
        return redirect(url_for("ahp.indicators"))
    return render_template("ahp/indicator_edit.html",
    ind=ind,
                           unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el, form=form)
```

```
@bp.route("/indicators/reset", methods=["POST"])
def indicators_reset():
    """按论文取值重置指标定义。"""
    form = EmptyForm()
    if form.validate_on_submit():
        n = ind_svc.seed_indicators()
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")
    return redirect(url_for("ahp.indicators"))
# ---------------------------- 多专家矩阵 ----------------------------
-----
def expert_scope():
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""
    unit_type = request.values.get("unit_type") or
    UNIT_TYPES[0]
    if unit_type not in DEFAULT_WEIGHTS:
        unit_type = UNIT_TYPES[0]
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    return unit_type, level, l1
def expert_query(unit_type, level, l1):
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,
    level=level)
    if level == 2:
        q = q.filter_by(l1_name=l1)
    else:
        q = q.filter(ExpertMatrix.l1_name.is_(None))
    return q.order_by(ExpertMatrix.id)
def default_scheme(unit_type, level, l1=None):
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""
    name = "%s · %s%s" % (unit_type,
                          "一级指标" if level == 1 else "二级指标",
                          "(%s)" % l1 if l1 else "")
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,
                                          level=level,
    name=name).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    level=level, name=name,
                              is_default=False)
        db.session.add(scheme)
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
```

```
    return redirect(url_for("ahp.index"))
names, vec = indicator_list(unit_type, level, l1)
n = len(names)
form = EmptyForm()
if request.method == "POST" and form.validate_on_submit(
):
    action = request.form.get("action")
    if action == "add":
        expert_name = request.form.get("expert_name",
"").strip() or "未具名专家"
        try:
            upper = [float(request.form.get("m_%d_%d" %
(i, j), "1"))
                     for i in range(n) for j in range(i
+ 1, n)]
        except ValueError:
            flash("矩阵输入有误，请填入正数", "error")
            return redirect(url_for("ahp.experts",
unit_type=unit_type,
                                    level=level, l1=l1))
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        scheme = default_scheme(unit_type, level, l1)
        db.session.add(ExpertMatrix(
            scheme_id=scheme.id,
            unit_type=unit_type, l1_name=(l1 if level
== 2 else None),
            expert_name=expert_name, level=level,
            matrix_json=json.dumps([list(map(float,
row)) for row in A]),
            cr=round(r["cr"], 6)))
        db.session.commit()
        flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
r["cr"]), "ok")
    elif action == "save_agg":
        # 把汇总权重落库并生效，使多专家结果真正参与评价计算
        weights = _aggregate(unit_type, level, l1)
        if weights is None:
            flash("暂无可汇总的专家矩阵", "error")
        elif not weights["consistent"]:
            flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
                  "请先检查或调整专家矩阵" % weights["cr"],
"error")
        else:
            names_, _ = indicator_list(unit_type,
level, l1)
            pairs = to_code_pairs(unit_type, level, l1,
names_,
                                  weights["weights"])
            _, cnt = ws.save_weights(unit_type, pairs)
```

```
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
    "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
def _aggregate(unit_type, level, l1):
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
    rows = expert_query(unit_type, level, l1).all()
    mats = []
    for r in rows:
        try:
            mats.append([list(map(float, line))
                         for line in json.loads(r.matrix_jso
    n)])
        except Exception:
            continue
    if not mats:
        return None
    agg = ahp.aggregate_matrices(mats)
    res = ahp.weights_and_consistency(agg)
    return {
        "count": len(mats),
        "weights": res["weights"],
        "lambda_max": res["lambda_max"],
        "ci": res["ci"], "cr": res["cr"],
        "consistent": res["consistent"],
        "most_inconsistent": ahp.find_most_inconsistent(agg)
    ,
```

```
            "expert_crs": [r.cr for r in rows if r.cr is not
      None],
      }
@bp.route("/experts/aggregate")
def experts_aggregate():
      """按几何平均汇总多专家矩阵，并做一致性检验。"""
      unit_type, level, l1 = expert_scope()
      if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
            return redirect(url_for("ahp.index"))
      names, vec = indicator_list(unit_type, level, l1)
      rows = expert_query(unit_type, level, l1).all()
      result = _aggregate(unit_type, level, l1)
      if result is not None:
            result["weights"] = [round(v, 4) for v in
      result["weights"]]
            result["lambda_max"] = round(result["lambda_max"],
      4)
      return render_template("ahp/experts_aggregate.html",
      unit_type=unit_type,
                                   level=level, l1=l1, names=names,
      rows=rows,
                                   result=result, form=EmptyForm())
#  ==== app/views/indicator.py ====
#  -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
      redirect, url_for,
                         flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
      IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
      UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
      url_prefix="/indicator")
#  指标字段定义：字段名 -> （中文名， 单位， 是否可选）
SOIL_FIELDS = [
      ("soil_ratio", "土石比", ""),
      ("thickness", "土层厚度", "cm"),
      ("ph", "pH", ""),
      ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
```

```
    ("coverage", "植被覆盖度", "%"),                                    3701
    ("shannon", "Shannon-Wiener 多样性指数", ""),                       3702
]                                                                       3703
TERRAIN_FIELDS = [                                                      3704
    ("slope", "坡度", "°"),                                             3705
    ("aspect", "坡向", ""),                                             3706
]                                                                       3707
def unit_field_config(unit_type):                                       3708
    """根据单元类型返回需录入的字段分组。"""                                3709
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())                3710
    groups = []                                                         3711
    if "土壤指标" in enabled:                                            3712
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True        3713
    = 含重金属                                                          3714
    if "植被指标" in enabled:                                            3715
        groups.append(("植被指标", VEG_FIELDS, False))                  3716
    if "地形指标" in enabled:                                            3717
        groups.append(("地形指标", TERRAIN_FIELDS, False))              3718
    return groups                                                       3719
@bp.route("/")                                                          3720
def index():                                                            3721
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            3722
    all()                                                               3723
    return render_template("indicator/index.html",                      3724
    patches=patches)                                                    3725
@bp.route("/<int:unit_id>", methods=["GET", "POST"])                    3726
def edit(unit_id):                                                      3727
    unit = db.get_or_404(MicroUnit, unit_id)                            3728
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir            3729
    st()                                                                3730
    form = EmptyForm()                                                  3731
    groups = unit_field_config(unit.unit_type)                          3732
    if request.method == "POST" and form.validate_on_submit(            3733
    ):                                                                  3734
        def f(name, cast=float):                                        3735
            v = request.form.get(name, "").strip()                      3736
            return cast(v) if v != "" else None                         3737
        heavy = {}                                                      3738
        for m in HEAVY_METALS:                                          3739
            v = f("hm_" + m)                                            3740
            if v is not None:                                           3741
                heavy[m] = v                                            3742
        if ind is None:                                                 3743
            ind = IndicatorData(unit_id=unit.id)                        3744
            db.session.add(ind)                                         3745
        ind.soil_ratio = f("soil_ratio")                                3746
        ind.thickness = f("thickness")                                  3747
        ind.ph = f("ph")                                                3748
        ind.organic_matter = f("organic_matter")                        3749
        ind.heavy_metals = json.dumps(heavy,                            3750
```

```
    ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ------------------------------ 批量导入 ------------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if
    patch_id else None
        if patch is None:
            flash("请选择矿山图斑", "error")
            return redirect(url_for("indicator.import_data")
    )
        file = request.files.get("datafile")
        if file is None or not file.filename:
            flash("请选择要导入的文件", "error")
            return redirect(url_for("indicator.import_data")
```

```
)
    ext = file.filename.rsplit(".", 1)[-1].lower()
    if ext not in ("csv", "xlsx", "xls"):
        flash("仅支持 CSV 或 Excel(.xlsx/.xls)文件", "error")
        return redirect(url_for("indicator.import_data"))
)
    units = MicroUnit.query.filter_by(patch_id=patch.id)
.all()
    unit_map = {}
    for u in units:
        unit_map.setdefault(u.unit_type, u)
    if not unit_map:
        flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
        return redirect(url_for("indicator.import_data"))
)
    try:
        parsed, errors = imp.parse_upload(file.filename,
file.stream, unit_map)
    except Exception as e:
        flash("文件解析失败：%s" % e, "error")
        return redirect(url_for("indicator.import_data"))
)
    saved = 0
    for rec in parsed:
        unit = unit_map[rec["unit_type"]]
        ind = IndicatorData.query.filter_by(unit_id=unit
.id).first()
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        f = rec["fields"]
        for key in ("soil_ratio", "thickness", "ph",
"organic_matter",
                    "coverage", "shannon", "slope",
"aspect"):
            if key in f:
                setattr(ind, key, f[key])
        if rec["heavy_metals"]:
            ind.heavy_metals = json.dumps(rec["heavy_met
als"],

ensure_ascii=False)
        saved += 1
    db.session.commit()
    report = {"patch": patch.name, "saved": saved,
"errors": errors,
              "total": len(parsed) + len(
                  [e for e in errors if e.startswith("第
")])}
    if saved:
```

```
                flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
            else:
                flash("没有可导入的数据行，请检查文件内容", "error")
        return render_template("indicator/import.html",
        patches=patches,
                               form=form, report=report,
        headers=imp.HEADERS)
@bp.route("/import/template")
def import_template():
    """下载导入模板（CSV，含一行示例）。"""
    csv_text = imp.build_template_csv()
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于
    Excel 正确识别中文
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="指标数据导入模板.csv",
                     mimetype="text/csv")
# ==== app/views/evaluate.py ====
# -*- coding: utf-8 -*-
"""适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
```

```
                "coverage": ind.coverage if ind else None,
                "shannon": ind.shannon if ind else None,
                "slope": ind.slope if ind else None,
                "aspect": ind.aspect if ind else None,
            }
            result.append({
                "unit_id": u.id,
                "unit_type": u.unit_type,
                "ratio": u.ratio if u.ratio is not None else
    0.0,
                "data": data,
            })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
```

```
                    "std_value": _round(d["std_value"]),
                    "weight": _round(d["weight"]),
                    "contribution": _round(d["contribution"]),
                } for d in u["details"]],
            })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        # 先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
```

```
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patch_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit():
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
```

```
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
```

```
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight)
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    buf = rs.build_report(patch, result)
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformat
    s-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
```

```
            generated_at=datetime.now().strftime("%Y-%m-%d
        %H:%M"))
        data, err = browser.html_to_pdf(html)
        if err:
            flash("PDF 导出失败：%s" % err, "error")
            return redirect(url_for("report.detail",
        patch_id=patch_id))
        return send_file(io.BytesIO(data), as_attachment=True,
                         download_name="自然恢复适宜性评价报告_%s.pdf" %
        patch.name,
                         mimetype="application/pdf")
# ==== app/views/search.py ====
# -*- coding: utf-8 -*-
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。
"""
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
            Standard.province.like(like),
            Standard.summary.like(like),
        )).order_by(Standard.id).limit(LIMIT).all()
    total = len(projects) + len(patches) + len(standards)
    return render_template("search.html", q=q,
    projects=projects,
                           patches=patches,
    standards=standards, total=total)
# ==== app/templates/base.html ====
```

```
{% import "_macros.html" as ui %}<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
    :root {
      --accent: #1f6b45;                    /* 主色：深绿，呼应自然恢复主题 */
      --accent-hover: #185738;
      --accent-soft: #e9f4ee;
      --side-1: #17503a;                    /* 侧边栏渐变起止色 */
      --side-2: #113c2b;
      --danger: #b3261e;
      --danger-soft: #fdecea;
      --ok: #1e6b3a;
      --ok-soft: #eaf6ef;
      --todo: #d98324;                      /* 待办角标 */
      --ink: #23262b;
      --ink-soft: #6b7280;
      --line: #e3e7ea;
      --surface: #ffffff;
      --canvas: #f4f7f5;
      --radius: 10px;
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);
      --header-h: 60px;
      --nav-w: 226px;
    }
    * { box-sizing: border-box; }
    html, body { margin: 0; padding: 0; }
    body {
      font-family: "Microsoft YaHei", "PingFang SC",
    "SimSun", sans-serif;
      font-size: 14px;
      line-height: 1.6;
      color: var(--ink);
      background: var(--canvas);
    }
    /* ---------- 跳到主要内容（键盘可达性） ---------- */
    .skip-link {
      position: absolute;
      left: 12px;
      top: -60px;
      z-index: 30;
      padding: 8px 14px;
```

```
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ---------- */
   固定为一屏高，页面滚动交给右侧内容区，
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */
.app-shell {
    display: flex;
    align-items: stretch;
    height: 100vh;
    height: 100dvh;
    overflow: hidden;
}
/* ---------- 侧边栏 ---------- */
.app-side {
    flex: 0 0 var(--nav-w);
    width: var(--nav-w);
    display: flex;
    flex-direction: column;
    background: linear-gradient(180deg, var(--side-1) 0%,
var(--side-2) 100%);
    color: #fff;
}
.app-brand {
    display: flex;
    align-items: center;
    gap: 11px;
    padding: 17px 16px 15px;
    border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
    flex: none;
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    border-radius: 10px;
    background: rgba(255, 255, 255, .15);
}
.app-brand__name {
    font-size: 14.5px;
    font-weight: 600;
    line-height: 1.35;
    letter-spacing: .2px;
}
.app-brand__sub {
```

```
  font-size: 11.5px;
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s ease;
}
.app-nav a:hover { background: rgba(255, 255, 255, .10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
  background: rgba(255, 255, 255, .17);
  color: #fff;
  font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
  margin-left: auto;
  min-width: 20px;
  padding: 0 6px;
  font-size: 11px;
  font-weight: 600;
  line-height: 18px;
  text-align: center;
  border-radius: 9px;
  background: var(--todo);
  color: #fff;
  font-variant-numeric: tabular-nums;
```

```
}
.app-user {
  display: flex;
  align-items: center;
  gap: 8px;
  padding: 10px 14px;
  border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
  display: flex;
  flex: 1;
  min-width: 0;
  align-items: center;
  gap: 10px;
  padding: 4px 6px;
  border-radius: 8px;
  color: inherit;
  text-decoration: none;
  transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255,
.10); }
.app-user__me:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
  object-fit: cover;
}
.app-user__name {
  display: block;
  font-size: 13px;
  line-height: 1.3;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
```

```
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
.app-user__out {
    flex: none;
    padding: 6px;
    border-radius: 6px;
    color: rgba(255, 255, 255, .68);
    text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
    flex: 1;
    min-width: 0;
    display: flex;
    flex-direction: column;
    overflow-y: auto;
    overflow-x: hidden;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
    position: sticky;
    top: 0;
    z-index: 20;
    display: flex;
    align-items: center;
    gap: 14px;
    height: var(--header-h);
    padding: 0 22px;
    background: var(--surface);
    border-bottom: 1px solid var(--line);
}
.crumb {
    display: flex;
    align-items: center;
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
```

```
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
    place-items: center;
    padding: 0;
    border: none;
    background: transparent;
    color: inherit;
    cursor: pointer;
}
.icon-btn {
    display: grid;
    place-items: center;
    width: 34px;
    height: 34px;
    border-radius: 9px;
    border: 1px solid transparent;
    color: var(--ink-soft);
```

```
    text-decoration: none;                                          4501
    transition: background-color .16s ease, color .16s              4502
ease,                                                               4503
                border-color .16s ease;                             4504
}                                                                   4505
.icon-btn:hover {                                                   4506
    color: var(--accent);                                           4507
    background: var(--accent-soft);                                 4508
    border-color: rgba(31, 107, 69, .18);                           4509
}                                                                   4510
.app-main {                                                         4511
    flex: 1;                                                        4512
    min-width: 0;                                                   4513
    padding: 22px 26px 26px;                                        4514
}                                                                   4515
.container { max-width: 1120px; }                                   4516
/* ---------- 页脚 ---------- */                                    4517
.app-foot {                                                         4518
    display: flex;                                                  4519
    align-items: center;                                            4520
    justify-content: space-between;                                 4521
    gap: 14px;                                                      4522
    padding: 0 26px 18px;                                           4523
    font-size: 12px;                                                4524
    color: var(--ink-soft);                                         4525
}                                                                   4526
/* ---------- 未登录时的裸页面（登录页） ---------- */              4527
.login-shell {                                                      4528
    display: flex;                                                  4529
    flex-direction: column;                                         4530
    min-height: 100dvh;                                             4531
}                                                                   4532
/* ---------- 卡片 ---------- */                                    4533
.card {                                                             4534
    background: var(--surface);                                     4535
    border: 1px solid var(--line);                                  4536
    border-radius: var(--radius);                                   4537
    box-shadow: var(--shadow);                                      4538
    padding: 22px 24px;                                             4539
    margin-bottom: 18px;                                            4540
}                                                                   4541
.card > h2 {                                                        4542
    margin: 0 0 14px;                                               4543
    font-size: 16px;                                                4544
    font-weight: 600;                                               4545
    color: var(--accent);                                           4546
    text-wrap: balance;                                             4547
}                                                                   4548
.card > h3 {                                                        4549
    margin: 20px 0 10px;                                            4550
```

```
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
  font-size: 12px;
  padding-bottom: 6px;
}
th, td {
  border: 1px solid var(--line);
  padding: 8px 12px;
  text-align: center;
  vertical-align: middle;
}
th {
  background: var(--accent-soft);
  font-weight: 600;
  white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
  color: var(--ink-soft);
  font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
```

```
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 14px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color
.16s ease,
              color .16s ease, box-shadow .16s ease,
transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow:
```

```
none; }
.btn:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px;
border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
  border-color: rgba(31, 107, 69, .30);
  box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107,
69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ----------
   取代原来「文字 | 文字 | 文字」的写法：
   圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
  align-items: center;
  padding: 4px 11px;
  font-family: inherit;
```

```
    font-size: 13px;
    line-height: 1.5;
    color: var(--accent);
    background: rgba(31, 107, 69, .07);
    border: 1px solid rgba(31, 107, 69, .20);
    border-radius: 7px;
    cursor: pointer;
    text-decoration: none;
    white-space: nowrap;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
    color: var(--danger);
    background: rgba(179, 38, 30, .07);
    border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
    padding: 10px 14px;
    margin-bottom: 14px;
    border-radius: var(--radius);
    border-left: 4px solid;
}
.flash.error {
    color: var(--danger);
    background: var(--danger-soft);
    border-left-color: var(--danger);
}
```

```
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 居中浮层提示（登录成功、各类操作完成） ---------- */
   外层 toast-stack 固定在页面正中心并纵向排列，
   内层每条各自淡入 → 停留 → 淡出，多条提示会依次排开而不重叠。
   半透明底 + 背景模糊做出毛玻璃质感，不做成实心块；
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 50%;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 10px;
  transform: translate(-50%, -50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(560px, 78vw);
  padding: 13px 28px;
  border-radius: 999px;
  background: rgba(31, 107, 69, .62);          /* 半透明，非实体 */
  color: #fff;
  font-size: 16px;
  font-weight: 600;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
  text-wrap: pretty;
  border: 1px solid rgba(255, 255, 255, .28);
  backdrop-filter: blur(9px) saturate(1.3);
  -webkit-backdrop-filter: blur(9px) saturate(1.3);
  box-shadow: 0 10px 34px rgba(19, 36, 58, .18);
  animation: toast-fade 2.6s ease-in-out forwards;
}
@keyframes toast-fade {
  0%          { opacity: 0; transform: scale(.90); }
  14%, 74%    { opacity: 1; transform: scale(1); }
  100%        { opacity: 0; transform: scale(1); }
}
/* 系统设置里要求「减少动态效果」时，去掉缩放，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s linear forwards; }
```

```
  @keyframes toast-fade-plain {                                    4801
    0%        { opacity: 0; }                                      4802
    14%, 74%  { opacity: 1; }                                      4803
    100%      { opacity: 0; }                                      4804
  }                                                                4805
}                                                                  4806
/* 不支持背景模糊时，加深底色保证文字仍然清楚 */                    4807
@supports not (backdrop-filter: blur(1px)) {                       4808
  .toast-center { background: rgba(31, 107, 69, .88); }            4809
}                                                                  4810
/* ---------- 文字工具类 ---------- */                            4811
.muted { color: var(--ink-soft); font-size: 12.5px; }              4812
.mono { font-family: Consolas, "Courier New",                      4813
monospace; font-variant-numeric: tabular-nums; }                   4814
.num { font-variant-numeric: tabular-nums; }                       4815
.warn { color: var(--danger); font-weight: 600; }                  4816
.ok { color: var(--ok); font-weight: 600; }                        4817
.nowrap { white-space: nowrap; }                                   4818
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，   4819
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */ 4820
.nowrap form { display: inline; margin: 0; }                       4821
.stack { display: flex; flex-direction: column; gap:               4822
14px; }                                                            4823
/* 状态标签：不依赖颜色单独表意 */                                 4824
.tag {                                                             4825
  display: inline-block;                                           4826
  padding: 1px 9px;                                                4827
  font-size: 12.5px;                                               4828
  border-radius: 10px;                                             4829
  border: 1px solid;                                               4830
  white-space: nowrap;                                             4831
}                                                                  4832
.tag--ok { color: var(--ok); background:                           4833
var(--ok-soft); border-color: #b8ddc6; }                           4834
.tag--warn { color: var(--danger); background:                     4835
var(--danger-soft); border-color: #eec4c0; }                       4836
.tag--muted { color: var(--ink-soft); background:                  4837
#f1f3f6; border-color: #dde1e6; }                                  4838
/* ---------- 登录页 ---------- */                                4839
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，   4840
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */  4841
.body--login {                                                     4842
  background:                                                      4843
    radial-gradient(1100px 640px at 50% 8%,                        4844
                    #edf6f1 0%, rgba(237, 246, 241, 0)             4845
72%),                                                              4846
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);             4847
}                                                                  4848
.body--login .app-main { padding: 0; }                             4849
.body--login .container { max-width: none; width: 100%;            4850
```

```
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,
                            rgba(11, 48, 33, .68) 50%,
                            rgba(11, 48, 33, .06) 100%),
    url("{{ url_for('static', filename='hero.svg') }}")
    center bottom / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
```

```
  margin-top: 18px;
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease;
}
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 12px;
  margin-bottom: 22px;
}
.stat-card {
  display: block;
  padding: 15px 17px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease;
}
.stat-card:hover {
```

```
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
    gap: 12px;
}
.step-card {
    position: relative;
    display: block;
```

```
    padding: 16px 16px 15px;                                    5001
    background: var(--surface);                                 5002
    border: 1px solid var(--line);                              5003
    border-radius: var(--radius);                               5004
    box-shadow: var(--shadow);                                  5005
    color: var(--ink);                                          5006
    text-decoration: none;                                      5007
    transition: border-color .16s ease, box-shadow .16s         5008
ease,                                                           5009
                transform .16s ease;                            5010
}                                                               5011
.step-card:hover {                                              5012
    border-color: rgba(31, 107, 69, .55);                       5013
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);               5014
    transform: translateY(-1px);                                5015
}                                                               5016
.step-card:focus-visible { outline: 2px solid                   5017
var(--accent); outline-offset: 2px; }                           5018
.step-card__no {                                                5019
    position: absolute;                                         5020
    top: 13px;                                                  5021
    right: 15px;                                                5022
    font-size: 17px;                                            5023
    font-weight: 700;                                           5024
    color: #d7e3db;                                             5025
    font-variant-numeric: tabular-nums;                         5026
}                                                               5027
.step-card__icon {                                              5028
    display: grid;                                              5029
    place-items: center;                                        5030
    width: 32px;                                                5031
    height: 32px;                                               5032
    margin-bottom: 10px;                                        5033
    border-radius: 9px;                                         5034
    background: var(--accent-soft);                             5035
    color: var(--accent);                                       5036
}                                                               5037
.step-card__title {                                             5038
    display: block;                                             5039
    margin-bottom: 5px;                                         5040
    font-size: 14px;                                            5041
    font-weight: 600;                                           5042
    color: var(--accent);                                       5043
}                                                               5044
.step-card__desc {                                              5045
    display: block;                                             5046
    font-size: 12.5px;                                          5047
    line-height: 1.65;                                          5048
    color: var(--ink-soft);                                     5049
    text-wrap: pretty;                                          5050
```

```
}
.step-card__badge {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #fdf1e2;
  color: #9a5a12;
  border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
  background: #fafbfa;
  border-color: #eceff0;
  box-shadow: none;
  cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
  background: #f1f3f2;
  color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #f1f3f2;
  color: #8b948f;
}
/* ---------- 工作台：整行入口 ---------- */
.row-card {
  display: flex;
  align-items: center;
  gap: 14px;
  padding: 15px 18px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease;
}
.row-card:hover {
```

```
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.row-card__icon {
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    flex: none;
    border-radius: 10px;
    background: var(--accent-soft);
    color: var(--accent);
}
.row-card__main { flex: 1; min-width: 0; }
.row-card__title {
    display: block;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.row-card__desc {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.row-card__arrow { color: #c3c9d0; flex: none; }
/* ---------- 结果摘要 ---------- */
.stat-row {
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
    display: flex; align-items: center; gap: 7px;
    padding: 8px 12px; font-size: 13px;
    background: #fff; border: 1px solid var(--line);
    border-radius: var(--radius);
}
.step__no {
    display: inline-flex; align-items: center;
justify-content: center;
    width: 20px; height: 20px; border-radius: 50%;
```

```
    background: #eef1f5; color: var(--ink-soft);
    font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
    padding: 14px 16px; margin: 0 0 16px;
    border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
    background: var(--danger-soft);
    border-left-color: var(--danger);
    color: #7a1c17;
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
    border: 1px solid var(--line);
    border-radius: var(--radius);
    background: var(--surface);
    padding: 14px 16px;
    margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
```

```
      margin: 0 0 12px;                                                    5201
      padding-bottom: 9px;                                                 5202
      border-bottom: 1px solid var(--line);                                5203
    }                                                                      5204
    .patch-panel__name {                                                   5205
      margin: 0;                                                           5206
      font-size: 15px;                                                     5207
      font-weight: 600;                                                    5208
    }                                                                      5209
    .patch-panel__meta {                                                   5210
      font-size: 12.5px;                                                   5211
      color: var(--ink-soft);                                              5212
    }                                                                      5213
    .patch-panel__meta a { color: var(--accent); }                         5214
    .patch-panel__tail { margin-left: auto; }                              5215
    .patch-panel .notice { margin: 0; }                                    5216
    .patch-panel table { margin: 0; }                                      5217
    /* ---------- 窄屏适配 ---------- */                                   5218
    @media (max-width: 960px) {                                            5219
      /* 上下布局时取消「一屏高 + 内容区内部滚动」，                       5220
         否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */               5221
      .app-shell {                                                         5222
        flex-direction: column;                                            5223
        height: auto;                                                      5224
        min-height: 100vh;                                                 5225
        min-height: 100dvh;                                                5226
        overflow: visible;                                                 5227
      }                                                                    5228
      .app-content { overflow: visible; }                                  5229
      .app-header { position: static; }                                    5230
      .app-side { width: auto; flex: none; }                               5231
      .app-brand { padding: 12px 14px; }                                   5232
      .app-nav { padding: 4px 10px 10px; }                                 5233
      .app-nav ul { display: flex; flex-wrap: wrap; gap:                   5234
    4px; }                                                                 5235
      .app-nav a { margin-bottom: 0; }                                     5236
      .app-nav__group { margin: 8px 0 4px; }                               5237
      .app-user { padding: 10px 14px; }                                    5238
      .app-header { padding: 0 14px; gap: 10px; }                          5239
      .search-box { width: 180px; }                                        5240
      .app-main { padding: 16px 14px 20px; }                               5241
      .app-foot { padding: 0 14px 14px; }                                  5242
    }                                                                      5243
    @media (max-width: 640px) {                                            5244
      .crumb { display: none; }                                            5245
      .search-box { width: auto; flex: 1; }                                5246
    }                                                                      5247
  </style>                                                                 5248
</head>                                                                    5249
<body class="{% if not current_user.is_authenticated                       5250
```

```
  %}body--login{% endif %}">
<a class="skip-link" href="#main">跳到主要内容</a>
{# 取一次即可（同一请求内重复调用返回相同内容）。
   类别为 toast 的渲染成居中浮层，其余仍是页面顶部的提示条 ——
   错误提示需要停下来细读，不适合做成会消失的浮层。 #}
{% set flash_msgs = get_flashed_messages(with_categories=t
  rue) %}
{# 类别为 toast / ok 的渲染成页面正中的浮层（操作完成的即时反馈）；
   其余（尤其 error）仍是页面顶部的提示条 ——
   错误提示需要停下来细读，不适合做成会消失的浮层。
   没有浮层时这个容器是空的，不占任何位置。 #}
<div class="toast-stack">
  {% for category, message in flash_msgs if category in
  ("toast", "ok") %}
  <div class="toast-center" role="status"
  aria-live="polite">{{ message }}</div>
  {% endfor %}
</div>
{% if current_user.is_authenticated %}
<div class="app-shell">
  <aside class="app-side">
    <div class="app-brand">
      <span class="app-brand__mark">{{
  ui.icon("mountain", 21) }}</span>
      <span>
        <span class="app-brand__name">矿山自然恢复评价</span><br>
        <span class="app-brand__sub">适宜性评价系统 Ｖ1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %}
  aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %}
  aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label
  }}</span>
            {% if item.badge %}<span
  class="app-nav__badge">{{ item.badge }}</span>{% endif
  %}
```

```
          </a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
    <div class="app-user">
      <a class="app-user__me" href="/profile"
title="个人设置">
        {% if current_user.avatar %}
        <img class="app-user__avatar"
             src="{{ url_for('static',
filename=current_user.avatar) }}" alt="">
        {% else %}
        <span class="app-user__avatar"
aria-hidden="true">{{ current_user.initial }}</span>
        {% endif %}
        <span style="min-width:0;">
          <span class="app-user__name">{{
current_user.shown_name }}</span>
          <span class="app-user__state">
            <span class="app-user__dot"
aria-hidden="true"></span>
            {{ role_labels.get(current_user.role,
current_user.role) }}
          </span>
        </span>
      </a>
      <a class="app-user__out" href="/logout" title="退出登录"
        aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>
    </div>
</aside>
<div class="app-content">
    <header class="app-header">
      <nav class="crumb" aria-label="当前位置">
        <a href="/">首页</a>
        {{ ui.icon("chevron", 13) }}
        <strong>{{ page_title }}</strong>
      </nav>
      <span class="app-header__spacer"></span>
      <form class="search-box" method="get"
action="/search" role="search">
        {{ ui.icon("search", 16) }}
        <label for="q" class="visually-hidden">搜索项目、图斑、标准</
/label>
        <input id="q" type="search" name="q"
autocomplete="off"
               placeholder="搜索项目、图斑、标准" value="{{
search_q }}">
        <button type="submit" aria-label="搜索">{{
```

```
      ui.icon("chevron", 15) }}</button>
          </form>
          <a class="icon-btn" href="/logs" title="操作日志"
      aria-label="操作日志">
            {{ ui.icon("bell", 18) }}
          </a>
        </header>
    {% else %}
    <div class="login-shell">
    {% endif %}
      <main id="main" class="app-main">
        <div class="container">
          {% for category, message in flash_msgs if category
      not in ("toast", "ok") %}
          <p class="flash {{ category }}" role="status">{{
      message }}</p>
          {% endfor %}
          {% block content %}{% endblock %}
        </div>
      </main>
    {% if current_user.is_authenticated %}
        <footer class="app-foot">
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0</span>
          <span>数据截止 {{ today }}</span>
        </footer>
      </div>
    </div>
    {% else %}
    </div>
    {% endif %}
    {% block scripts %}{% endblock %}
</body>
</html>
# ==== app/templates/_macros.html ====
{# 共用宏：图标、流程状态面板 #}
{# ============================================================
   ================
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）
   不引图标字体或外部图片，避免额外请求与资源依赖；
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。
   ============================================================
   ================ #}
{% set ICONS = {
  "dot": '<circle cx="12" cy="12" r="3"/>',
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4
```

```
      19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',
    "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
      8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
    "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
      d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
    "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
      21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
      17h5"/>',
    "database": '<ellipse cx="12" cy="5.5" rx="8"
      ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
      8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
      8-3.2"/>',
    "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
      cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
    "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
      2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
      2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
    "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
      20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
      3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
      d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
    "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
      21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
      4.4V21"/>',
    "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
    "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
    "plus": '<path d="M12 5v14M5 12h14"/>',
    "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
    "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
    "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
    "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
    "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
```

```
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
{% endmacro %}
{# 前置条件未满足时的提示块：只列缺什么，并给出处理入口 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<div class="notice notice--warn">
  <p class="notice__title">尚不能执行本步骤，需要先完成：</p>
  <ol class="notice__list">
    {% for b in progress.blocking %}
    <li>
```

```
        <strong>{{ b.label }}</strong>{% if b.reason %}：{{ b.reason }}{% endif %}
        <a class="link-btn" href="{{ b.fix_url }}">{{ b.fix_label }}</a>
      </li>
      {% endfor %}
    </ol>
  </div>
  {% endif %}
  {% endmacro %}
  {# 一行式状态摘要，用于列表页 #}
  {% macro workflow_badge(progress) %}
  {% if progress.ready %}
  <span class="tag tag--ok">可执行评价</span>
  {% elif progress.blocking %}
  <span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
  <span class="muted">{{ progress.blocking[0].reason }}</span>
  {% endif %}
  {% endmacro %}
  {# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
     下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就直接列出并给出入口。 #}
  {% macro patch_guide(projects_without_patch) %}
  {% if projects_without_patch %}
  <div class="notice notice--warn">
    <p class="notice__title">
      以下 {{ projects_without_patch|length }}
      个项目下还没有图斑，本页不会出现它们：
    </p>
    <ul class="notice__list">
      {% for p in projects_without_patch %}
      <li>
        {{ p.name }}
        <a class="link-btn" href="/project/{{ p.id }}/patch/add">新建图斑</a>
      </li>
      {% endfor %}
    </ul>
  </div>
  {% endif %}
  {% endmacro %}
  # ==== app/templates/_login_badge.svg ====
  <svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg"
       viewBox="0 0 240 240" role="img"
       aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
    <title>系统徽标</title>
    <g class="lp-ring" style="--k: 2.7">
      <circle cx="162.00" cy="120.00" r="2.60"
      fill="#c4f5ab" />
      <circle cx="157.84" cy="138.22" r="2.97"
```

```
    fill="#b8f5ab"/>                                                    5551
    <circle cx="146.19" cy="152.84" r="3.06"                            5552
    fill="#acf5ab"/>                                                    5553
    <circle cx="129.35" cy="160.95" r="2.80"                            5554
    fill="#abf5b7"/>                                                    5555
    <circle cx="110.65" cy="160.95" r="2.40"                            5556
    fill="#abf5c2"/>                                                    5557
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>            5558
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>            5559
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>            5560
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>            5561
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>             5562
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>            5563
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>            5564
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>            5565
    <circle cx="157.84" cy="101.78" r="2.23"                            5566
    fill="#b8f5ab"/>                                                    5567
  </g>                                                                  5568
  <g class="lp-ring" style="--k: 2.15">                                 5569
    <circle cx="177.29" cy="129.07" r="3.92"                            5570
    fill="#a4e28d"/>                                                    5571
    <circle cx="171.68" cy="146.33" r="4.01"                            5572
    fill="#9be28d"/>                                                    5573
    <circle cx="161.01" cy="161.01" r="3.87"                            5574
    fill="#91e28d"/>                                                    5575
    <circle cx="146.33" cy="171.68" r="3.55"                            5576
    fill="#8de292"/>                                                    5577
    <circle cx="129.07" cy="177.29" r="3.17"                            5578
    fill="#8de29c"/>                                                    5579
    <circle cx="110.93" cy="177.29" r="2.88"                            5580
    fill="#8de2a6"/>                                                    5581
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af"/>            5582
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9"/>            5583
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2"/>            5584
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc"/>            5585
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc"/>            5586
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2"/>             5587
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9"/>             5588
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af"/>             5589
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6"/>            5590
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>            5591
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292"/>            5592
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d"/>            5593
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d"/>            5594
    <circle cx="177.29" cy="110.93" r="3.63"                            5595
    fill="#a4e28d"/>                                                    5596
  </g>                                                                  5597
  <g class="lp-ring" style="--k: 1.85">                                 5598
    <circle cx="194.00" cy="120.00" r="4.94"                            5599
    fill="#91cf72"/>                                                    5600
```

```
    <circle cx="191.85" cy="137.71" r="4.92"                    5601
    fill="#88cf72"/>                                            5602
    <circle cx="185.52" cy="154.39" r="4.73"                    5603
    fill="#80cf72"/>                                            5604
    <circle cx="175.39" cy="169.07" r="4.42"                    5605
    fill="#78cf72"/>                                            5606
    <circle cx="162.04" cy="180.90" r="4.06"                    5607
    fill="#72cf73"/>                                            5608
    <circle cx="146.24" cy="189.19" r="3.73"                    5609
    fill="#72cf7b"/>                                            5610
    <circle cx="128.92" cy="193.46" r="3.51"                    5611
    fill="#72cf83"/>                                            5612
    <circle cx="111.08" cy="193.46" r="3.45"                    5613
    fill="#72cf8b"/>                                            5614
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>    5615
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>    5616
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>    5617
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>    5618
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>    5619
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>    5620
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>    5621
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>     5622
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>     5623
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>     5624
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>     5625
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>    5626
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>    5627
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>    5628
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>    5629
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>    5630
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>    5631
    <circle cx="191.85" cy="102.29" r="4.80"                    5632
    fill="#88cf72"/>                                            5633
  </g>                                                          5634
  <g class="lp-ring" style="--k: 1.62">                         5635
    <circle cx="209.57" cy="128.82" r="5.68"                    5636
    fill="#76bb59"/>                                            5637
    <circle cx="206.12" cy="146.13" r="5.40"                    5638
    fill="#6fbb59"/>                                            5639
    <circle cx="199.37" cy="162.43" r="5.06"                    5640
    fill="#68bb59"/>                                            5641
    <circle cx="189.57" cy="177.10" r="4.71"                    5642
    fill="#61bb59"/>                                            5643
    <circle cx="177.10" cy="189.57" r="4.41"                    5644
    fill="#5abb59"/>                                            5645
    <circle cx="162.43" cy="199.37" r="4.19"                    5646
    fill="#59bb5f"/>                                            5647
    <circle cx="146.13" cy="206.12" r="4.10"                    5648
    fill="#59bb66"/>                                            5649
    <circle cx="128.82" cy="209.57" r="4.15"                    5650
```

```
    fill="#59bb6d"/>                                                    5651
    <circle cx="111.18" cy="209.57" r="4.32"                            5652
    fill="#59bb73"/>                                                    5653
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a"/>            5654
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81"/>            5655
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88"/>            5656
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f"/>            5657
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96"/>            5658
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d"/>            5659
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4"/>            5660
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4"/>            5661
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d"/>             5662
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96"/>             5663
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f"/>             5664
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88"/>             5665
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81"/>             5666
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a"/>             5667
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73"/>            5668
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d"/>            5669
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66"/>            5670
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f"/>            5671
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59"/>            5672
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59"/>            5673
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59"/>            5674
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59"/>            5675
    <circle cx="209.57" cy="111.18" r="5.85"                            5676
    fill="#76bb59"/>                                                    5677
  </g>                                                                  5678
  <g class="lp-ring" style="--k: 1.46">                                 5679
    <circle cx="226.00" cy="120.00" r="5.94"                            5680
    fill="#65a843"/>                                                    5681
    <circle cx="224.55" cy="137.45" r="5.62"                            5682
    fill="#5fa843"/>                                                    5683
    <circle cx="220.26" cy="154.42" r="5.29"                            5684
    fill="#59a843"/>                                                    5685
    <circle cx="213.22" cy="170.45" r="5.00"                            5686
    fill="#53a843"/>                                                    5687
    <circle cx="203.65" cy="185.11" r="4.77"                            5688
    fill="#4da843"/>                                                    5689
    <circle cx="191.79" cy="197.99" r="4.63"                            5690
    fill="#47a843"/>                                                    5691
    <circle cx="177.98" cy="208.74" r="4.59"                            5692
    fill="#43a846"/>                                                    5693
    <circle cx="162.58" cy="217.07" r="4.67"                            5694
    fill="#43a84c"/>                                                    5695
    <circle cx="146.02" cy="222.76" r="4.85"                            5696
    fill="#43a852"/>                                                    5697
    <circle cx="128.75" cy="225.64" r="5.10"                            5698
    fill="#43a858"/>                                                    5699
    <circle cx="111.25" cy="225.64" r="5.42"                            5700
```

```
      fill="#43a85e"/>                                                    5701
      <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>            5702
      <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>            5703
      <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>            5704
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>            5705
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>            5706
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>            5707
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>            5708
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>            5709
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>            5710
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>            5711
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>             5712
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>             5713
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>             5714
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>             5715
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>             5716
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>             5717
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>             5718
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>            5719
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>            5720
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>            5721
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>            5722
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>            5723
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>            5724
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>            5725
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>            5726
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>            5727
      <circle cx="224.55" cy="102.55" r="6.23"                            5728
      fill="#5fa843"/>                                                    5729
    </g>                                                                  5730
    <g class="lp-badge__mark">                                            5731
      <circle cx="120" cy="120" r="27" fill="#14432f"/>                   5732
      <g transform="translate(120 120) scale(1.5)                         5733
      translate(-12 -12)" fill="none" stroke="#ffffff"                    5734
      stroke-width="1.9" stroke-linecap="round"                           5735
      stroke-linejoin="round">                                            5736
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>                       5737
        <circle cx="17" cy="6" r="2.2"/>                                  5738
      </g>                                                                5739
    </g>                                                                  5740
</svg>                                                                    5741
# ==== app/templates/error.html ====                                      5742
{% extends "base.html" %}                                                 5743
{% block title %}{{ code }} · {{ message }}{% endblock %}                 5744
{% block content %}                                                       5745
<div class="card" style="max-width:560px;">                               5746
  <h2>{{ code }} · {{ message }}</h2>                                     5747
  <p class="muted">                                                       5748
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。                                5749
  </p>                                                                    5750
```

```
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ================== */
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp { color: #1d1d1f; }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
    opacity: calc(1 - var(--p) * .94);
    filter: blur(calc(var(--p) * 2.6px));
  }
  .lp-ring {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
  }
  .lp-badge__mark {
```

```
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + .18 * var(--p)));
  }
  .lp-title {
    margin: 0 0 18px;
    font-size: clamp(28px, 4.2vw, 42px);
    font-weight: 700;
    line-height: 1.3;
    letter-spacing: -.4px;
    animation: lp-in .7s ease .06s both;
  }
  .lp-cta {
    padding: 13px 52px;
    font-family: inherit;
    font-size: 17px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    animation: lp-in .7s ease .18s both;
    transition: background-color .2s ease, box-shadow .2s ease,
                transform .2s ease;
  }
  .lp-cta:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 26px rgba(31, 107, 69, .26);
  }
  .lp-cta:active { transform: translateY(1px); box-shadow: none; }
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107, 69, .40); outline-offset: 3px; }
  .lp-more {
    margin-top: 38px;
    font-size: 13px;
    color: #86868b;
    text-decoration: none;
    animation: lp-in .7s ease .24s both;
  }
  .lp-more:hover { color: var(--accent); }
  @keyframes lp-in {
    from { opacity: 0; transform: translateY(12px); }
    to   { opacity: 1; transform: none; }
  }
  /* ---------- 滚动进场 ---------- */
     进入视口时按方向弹入，离开视口时弹回，可反复。
     交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
```

```
.lp-reveal {
  opacity: 0;
  transition: opacity .62s ease,
              transform .62s cubic-bezier(.16, .84, .34,
  1),
              border-color .62s ease, background-color
  .62s ease;
  transition-delay: 0s;
  will-change: opacity, transform;
}
.lp-reveal[data-dir="left"]  { transform:
  translate3d(-48px, 0, 0); }
.lp-reveal[data-dir="right"] { transform:
  translate3d(48px, 0, 0); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0,
  36px, 0); }
.lp-reveal.is-in {
  opacity: 1;
  transform: none;
  transition-delay: calc(var(--i, 0) * 85ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:
  color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: 980px;
  margin: 0 auto;
  padding: 20px 24px 72px;
}
.lp-intro__title {
  margin: 0 0 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
.lp-points {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px,
  1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 24px 22px;
```

```
    background: #f5f5f7;
    border-radius: 18px;
  }
  .lp-point__icon {
    display: grid;
    place-items: center;
    width: 38px;
    height: 38px;
    margin-bottom: 14px;
    border-radius: 11px;
    background: #fff;
    color: var(--accent);
  }
  .lp-point h3 { margin: 0 0 8px; font-size: 16px; }
  .lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
  }
  .lp-sub-title { margin: 0 0 20px; font-size: 17px; }
  .lp-mods {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(258px,
    1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
  }
  /* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
  .lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
  }
  .lp-mod:hover,
  .lp-mod:focus-within,
  .lp-mod:focus {
    border-color: rgba(31, 107, 69, .42);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
```

```
    font-size: 13px;
    font-weight: 700;
    color: #c8d6cd;
    font-variant-numeric: tabular-nums;
  }
  .lp-mod__name {
    display: block;
    font-size: 14px;
    font-weight: 600;
  }
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
  .lp-mod__body {
    display: grid;
    grid-template-rows: 0fr;
    transition: grid-template-rows .34s ease, margin-top
    .34s ease;
  }
  .lp-mod:hover .lp-mod__body,
  .lp-mod:focus-within .lp-mod__body {
    grid-template-rows: 1fr;
    margin-top: 7px;
  }
  .lp-mod__desc {
    overflow: hidden;
    margin: 0;
    font-size: 12.5px;
    line-height: 1.7;
    color: #6e6e73;
    opacity: 0;
    transition: opacity .3s ease;
    text-wrap: pretty;
  }
  .lp-mod:hover .lp-mod__desc,
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }
  /* 触摸设备没有悬停，直接把说明摊开 */
  @media (hover: none) {
    .lp-mod__body { grid-template-rows: 1fr; margin-top:
    7px; }
    .lp-mod__desc { opacity: 1; }
  }
  .lp-foot {
    margin: 0;
    padding-top: 26px;
    border-top: 1px solid #e6e6e8;
    text-align: center;
    font-size: 12.5px;
    color: #86868b;
  }
  /* ---------- 登录面板：点击后渐变浮现 ---------- */
  .lp-auth {
```

```
    position: fixed;                                                    6001
    inset: 0;                                                           6002
    z-index: 60;                                                        6003
    display: grid;                                                      6004
    place-items: center;                                                6005
    padding: 24px;                                                      6006
    opacity: 0;                                                         6007
    visibility: hidden;                                                 6008
    transition: opacity .42s ease, visibility 0s linear                 6009
    .42s;                                                               6010
}                                                                       6011
.lp-auth.is-open {                                                      6012
    opacity: 1;                                                         6013
    visibility: visible;                                                6014
    transition: opacity .42s ease, visibility 0s;                       6015
}                                                                       6016
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                                  6017
.lp-auth__scrim {                                                       6018
    position: absolute;                                                 6019
    inset: 0;                                                           6020
    background:                                                         6021
      radial-gradient(1000px 640px at 50% 0%,                           6022
                      rgba(31, 107, 69, .22), rgba(31, 107,             6023
    69, 0) 70%),                                                        6024
      linear-gradient(180deg, rgba(255, 255, 255, .84),                 6025
                                rgba(246, 250, 247, .95));              6026
    backdrop-filter: blur(16px) saturate(1.25);                         6027
    -webkit-backdrop-filter: blur(16px) saturate(1.25);                 6028
}                                                                       6029
.lp-auth__panel {                                                       6030
    position: relative;                                                 6031
    width: 100%;                                                        6032
    max-width: 396px;                                                   6033
    padding: 36px 36px 26px;                                            6034
    background: #fff;                                                   6035
    border-radius: 20px;                                                6036
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                      6037
    opacity: 0;                                                         6038
    transform: translateY(18px) scale(.975);                            6039
    transition: transform .55s cubic-bezier(.16, .84, .34,              6040
    1) .05s,                                                            6041
                opacity .55s ease .05s;                                 6042
}                                                                       6043
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:               6044
    none; }                                                             6045
.lp-auth__close {                                                       6046
    position: absolute;                                                 6047
    top: 14px;                                                          6048
    right: 14px;                                                        6049
    display: grid;                                                      6050
```

```
    place-items: center;                                                    6051
    width: 30px;                                                            6052
    height: 30px;                                                           6053
    color: #86868b;                                                         6054
    background: transparent;                                                6055
    border: none;                                                           6056
    border-radius: 50%;                                                     6057
    cursor: pointer;                                                        6058
    transition: background-color .18s ease, color .18s ease;                6059
}                                                                           6060
.lp-auth__close:hover { background: #f5f5f7; color:                         6061
    #1d1d1f; }                                                              6062
.lp-auth__title {                                                           6063
    margin: 0 0 6px;                                                        6064
    text-align: center;                                                     6065
    font-size: 24px;                                                        6066
    font-weight: 700;                                                       6067
    letter-spacing: -.2px;                                                  6068
}                                                                           6069
.lp-auth__sub {                                                             6070
    margin: 0 0 26px;                                                       6071
    text-align: center;                                                     6072
    font-size: 13px;                                                        6073
    color: #86868b;                                                         6074
}                                                                           6075
.lp-field { margin-bottom: 15px; }                                          6076
.lp-field label {                                                           6077
    display: block;                                                         6078
    margin-bottom: 7px;                                                     6079
    font-size: 13px;                                                        6080
    color: #6e6e73;                                                         6081
}                                                                           6082
.lp-field input {                                                           6083
    width: 100%;                                                            6084
    padding: 13px 16px;                                                     6085
    font-family: inherit;                                                   6086
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    6087
    color: #1d1d1f;                                                         6088
    background: #f5f5f7;                                                    6089
    border: 1px solid transparent;                                          6090
    border-radius: 12px;                                                    6091
    outline: none;                                                          6092
    transition: background-color .18s ease, border-color                    6093
    .18s ease,                                                              6094
                box-shadow .18s ease;                                       6095
}                                                                           6096
.lp-field input::placeholder { color: #a1a1a6; }                            6097
.lp-field input:focus {                                                     6098
    background: #fff;                                                       6099
    border-color: rgba(31, 107, 69, .55);                                   6100
```

```
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.lp-remember {
    display: flex;
    align-items: center;
    gap: 8px;
    margin: 4px 0 20px;
    font-size: 13px;
    color: #6e6e73;
    cursor: pointer;
}
.lp-remember input {
    width: 16px;
    height: 16px;
    margin: 0;
    accent-color: var(--accent);
    cursor: pointer;
}
.lp-submit {
    width: 100%;
    padding: 13px 20px;
    font-family: inherit;
    font-size: 16px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
}
.lp-submit:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);
}
.lp-submit:active { transform: translateY(1px);
    box-shadow: none; }
.lp-submit:focus-visible { outline: 3px solid rgba(31,
    107, 69, .40); outline-offset: 3px; }
.lp-auth__hint {
    margin: 18px 0 0;
    text-align: center;
    font-size: 12.5px;
    color: #a1a1a6;
}
/* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
@media (prefers-reduced-motion: reduce) {
    .lp-badge, .lp-title, .lp-cta, .lp-more { animation:
```

```
      none; }                                                                    6151
      .lp-badge__svg { opacity: 1; filter: none; }                               6152
      .lp-ring, .lp-badge__mark { transform: none; }                             6153
      .lp-reveal { opacity: 1; transform: none !important; }                     6154
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name                     6155
      { color: inherit; }                                                        6156
      .lp-reveal.is-in h2 { color: #1d1d1f; }                                    6157
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {                      6158
      color: var(--accent); }                                                    6159
      .lp-auth, .lp-auth__panel { transition: opacity .2s                        6160
      linear; }                                                                  6161
      .lp-auth__panel { transform: none; }                                       6162
    }                                                                            6163
    @media (max-width: 560px) {                                                  6164
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }                    6165
      .lp-badge { width: 160px; height: 160px; margin-bottom:                    6166
      26px; }                                                                    6167
      .lp-auth__panel { padding: 30px 24px 22px; }                               6168
    }                                                                            6169
</style>                                                                         6170
<div class="lp">                                                                 6171
  {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口                        6172
     ---------------- #}                                                        6173
  <section class="lp-hero">                                                      6174
    <div class="lp-badge" id="lp-badge" aria-hidden="true">                      6175
      {% include "_login_badge.svg" %}                                           6176
    </div>                                                                       6177
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>          6178
    <button type="button" class="lp-cta" id="lp-open"                            6179
            aria-expanded="false" aria-controls="lp-auth">登录                   6180
    </button>                                                                    6181
    <a class="lp-more" href="#intro">                                            6182
      向下滑动，了解系统 <span aria-hidden="true">↓</span>                       6183
    </a>                                                                         6184
  </section>                                                                     6185
  {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}               6186
  <section class="lp-intro" id="intro">                                          6187
    <h2 class="lp-intro__title lp-reveal"                                        6188
    data-dir="up">一套把评价流程固定下来的工具</h2>                              6189
    <div class="lp-points">                                                      6190
      {% set pdirs = ['left', 'up', 'right'] %}                                  6191
      {% set points = [                                                          6192
        ('sliders', '方法可溯',                                                  6193
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回 6194
    看。'),                                                                      6195
        ('shield', '流程可控',                                                   6196
         '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重 6197
    建或人工辅助的结论。'),                                                     6198
        ('file', '结果可复算',                                                   6199
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')  6200
```

```
    ] %}                                                                6201
    {% for ic, title, text in points %}                                 6202
    <div class="lp-point lp-reveal" data-dir="{{                        6203
  pdirs[loop.index0 % 3] }}"                                            6204
        style="--i: {{ loop.index0 }}">                                 6205
      <span class="lp-point__icon">{{ icon(ic, 20)                      6206
  }}</span>                                                             6207
      <h3>{{ title }}</h3>                                              6208
      <p>{{ text }}</p>                                                 6209
    </div>                                                              6210
    {% endfor %}                                                        6211
  </div>                                                                6212
  <h3 class="lp-sub-title lp-reveal"                                    6213
  data-dir="up">九个功能模块</h3>                                       6214
  <div class="lp-mods">                                                 6215
    {% set mdirs = ['left', 'up', 'right'] %}                           6216
    {% for no, name, desc in modules %}                                 6217
    <div class="lp-mod lp-reveal" data-dir="{{                          6218
  mdirs[loop.index0 % 3] }}"                                            6219
        style="--i: {{ loop.index0 % 3 }}" tabindex="0">                6220
      <span class="lp-mod__no">{{ no }}</span>                          6221
      <span class="lp-mod__name">{{ name }}</span>                      6222
      <div class="lp-mod__body"><p class="lp-mod__desc">{{              6223
  desc }}</p></div>                                                     6224
    </div>                                                              6225
    {% endfor %}                                                        6226
  </div>                                                                6227
  <p class="lp-foot lp-reveal" data-dir="up">                           6228
    历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０ ·                   6229
    运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库                    6230
  </p>                                                                  6231
</section>                                                              6232
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}  6233
<div class="lp-auth" id="lp-auth" aria-hidden="true">                   6234
  <div class="lp-auth__scrim" data-lp-close></div>                      6235
  <div class="lp-auth__panel" role="dialog"                             6236
  aria-modal="true"                                                     6237
       aria-labelledby="lp-auth-title">                                 6238
    <button type="button" class="lp-auth__close"                        6239
  data-lp-close                                                         6240
            aria-label="关闭登录框">                                    6241
      <svg width="16" height="16" viewBox="0 0 24 24"                   6242
  fill="none"                                                           6243
           stroke="currentColor" stroke-width="2"                       6244
  stroke-linecap="round"                                                6245
           aria-hidden="true"><path d="M6 6l12 12M18 6L6                6246
  18"/></svg>                                                           6247
    </button>                                                           6248
    <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>             6249
    <p class="lp-auth__sub">使用系统账号登录</p>                        6250
```

```
        <form method="post" novalidate>                                    6251
          {{ form.hidden_tag() }}                                          6252
          <div class="lp-field">                                           6253
            <label for="{{ form.username.id }}">用户名</label>             6254
            {{ form.username(id=form.username.id,                          6255
    autocomplete="username",                                               6256
                             placeholder="请输入用户名") }}                6257
          </div>                                                           6258
          <div class="lp-field">                                           6259
            <label for="{{ form.password.id }}">密码</label>               6260
            {{ form.password(id=form.password.id,                          6261
    autocomplete="current-password",                                       6262
                             placeholder="请输入密码") }}                  6263
          </div>                                                           6264
          <label class="lp-remember" for="{{ form.remember.id              6265
    }}">                                                                   6266
            {{ form.remember(id=form.remember.id) }}                       6267
            <span>记住我（下次自动登录，一个月内有效）</span>              6268
          </label>                                                         6269
          {{ form.submit(class_="lp-submit") }}                            6270
        </form>                                                            6271
        <p class="lp-auth__hint">演示账号：admin 密码：admin123</p>        6272
      </div>                                                               6273
    </div>                                                                 6274
</div>                                                                     6275
{% endblock %}                                                             6276
{% block scripts %}                                                        6277
<script>                                                                   6278
(function () {                                                             6279
  var reduce = window.matchMedia &&                                        6280
    window.matchMedia('(prefers-reduced-motion:                            6281
    reduce)').matches;                                                     6282
  /* ---------- 登录面板 ---------- */                                     6283
  var btn = document.getElementById('lp-open');                            6284
  var auth = document.getElementById('lp-auth');                           6285
  var lastFocus = null;                                                    6286
  function openPanel() {                                                   6287
    lastFocus = document.activeElement;                                    6288
    auth.classList.add('is-open');                                         6289
    auth.setAttribute('aria-hidden', 'false');                             6290
    btn.setAttribute('aria-expanded', 'true');                             6291
    document.body.style.overflow = 'hidden';                               6292
    // 记到地址栏，直接访问 /login#login 也能展开登录框                   6293
    if (location.hash !== '#login') {                                      6294
      history.replaceState(null, '', location.pathname +                   6295
    '#login');                                                             6296
    }                                                                      6297
    var first = auth.querySelector('input');                               6298
    if (first) window.setTimeout(function () {                             6299
    first.focus(); }, 300);                                                6300
```

```
}
function closePanel() {
  auth.classList.remove('is-open');
  auth.setAttribute('aria-hidden', 'true');
  btn.setAttribute('aria-expanded', 'false');
  document.body.style.overflow = '';
  if (location.hash === '#login') {
    history.replaceState(null, '', location.pathname);
  }
  if (lastFocus && lastFocus.focus) lastFocus.focus();
}
if (btn && auth) {
  btn.addEventListener('click', openPanel);
  Array.prototype.forEach.call(
    auth.querySelectorAll('[data-lp-close]'),
    function (el) { el.addEventListener('click',
  closePanel); }
  );
  document.addEventListener('keydown', function (e) {
    if (e.key === 'Escape' && auth.classList.contains('is-
  open')) closePanel();
  });
  // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框
  if (location.hash === '#login' ||
  document.querySelector('.flash.error')) {
    openPanel();
  }
}
/* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
   原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
   内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
   与徽标用同一次滚动回调，行为可预期、也便于排查。 */
var items = document.querySelectorAll('.lp-reveal');
var badge = document.getElementById('lp-badge');
var ticking = false;
function applyScroll() {
  ticking = false;
  var vh = window.innerHeight || 800;
  /* 徽标：滚动前约三分之一屏内完成散开淡出 */
  if (badge) {
    var range = vh * 0.34;
    var p = range > 0 ? window.scrollY / range : 0;
    badge.style.setProperty('--p', Math.min(1,
  Math.max(0, p)).toFixed(3));
  }
  /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */
  if (reduce || !items.length) return;
  var box = [];
  for (var i = 0; i < items.length; i++) {
    box.push(items[i].getBoundingClientRect());
```

```
      }                                                                                                  6351
      for (var j = 0; j < items.length; j++) {                                                           6352
        var r = box[j];                                                                                  6353
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内                                          6354
        var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;                                           6355
        if (shown) items[j].classList.add('is-in');                                                      6356
        else items[j].classList.remove('is-in');                                                         6357
      }                                                                                                  6358
    }                                                                                                    6359
    function onScroll() {                                                                                6360
      if (!ticking) {                                                                                    6361
        ticking = true;                                                                                  6362
        window.requestAnimationFrame(applyScroll);                                                       6363
      }                                                                                                  6364
    }                                                                                                    6365
    // 减少动态效果时不做滚动特效，内容直接显示                                                          6366
    if (reduce) {                                                                                        6367
      Array.prototype.forEach.call(items, function (el) {                                                6368
        el.classList.add('is-in');                                                                       6369
      });                                                                                                6370
      if (badge) badge.style.setProperty('--p', '0');                                                    6371
    } else {                                                                                             6372
      window.addEventListener('scroll', onScroll, { passive:                                             6373
      true });                                                                                           6374
      window.addEventListener('resize', onScroll);                                                       6375
      applyScroll();                                                                                     6376
    }                                                                                                    6377
})();                                                                                                    6378
</script>                                                                                                6379
{% endblock %}                                                                                           6380
# ==== app/templates/auth/home.html ====                                                                 6381
{% extends "base.html" %}                                                                                6382
{% from "_macros.html" import icon %}                                                                    6383
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}                           6384
{% block content %}                                                                                      6385
{# ---------- 页首横幅 ---------- #}                                                                    6386
<section class="hero">                                                                                   6387
  <div class="hero__body">                                                                               6388
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>                                  6389
    <p class="hero__desc">                                                                               6390
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》                               6391
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，                                  6392
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。                                   6393
    </p>                                                                                                 6394
    <div class="hero__actions">                                                                          6395
      <a class="hero-btn hero-btn--ghost"                                                                6396
    href="/indicator/import">                                                                            6397
        {{ icon("upload", 16) }}批量导入                                                                 6398
      </a>                                                                                               6399
      <a class="hero-btn hero-btn--solid"                                                                6400
```

```
      href="/project/add">                                                    6401
          {{ icon("plus", 16) }}新建评价项目                                  6402
        </a>                                                                  6403
      </div>                                                                  6404
    </div>                                                                    6405
</section>                                                                    6406
{# ---------- 数据概览 ---------- #}                                          6407
<section class="stat-cards">                                                  6408
    <a class="stat-card" href="/project/">                                    6409
      <span class="stat-card__label">评价项目</span>                          6410
      <span class="stat-card__value">{{ dash.stats.projects                   6411
      }}<span class="stat-card__unit">个</span></span>                        6412
      <span class="stat-card__note">                                          6413
        {%- if projects_without_patch -%}                                     6414
          其中 {{ projects_without_patch|length }} 个项目尚未建立图斑         6415
        {%- elif dash.progress.patches -%}                                    6416
          已建立 {{ dash.progress.patches }} 个矿山图斑                       6417
        {%- else -%}                                                          6418
          尚未建立矿山图斑                                                    6419
        {%- endif -%}                                                         6420
      </span>                                                                 6421
    </a>                                                                      6422
    <a class="stat-card" href="/project/">                                    6423
      <span class="stat-card__label">矿山图斑</span>                          6424
      <span class="stat-card__value">{{ dash.stats.patches                    6425
      }}<span class="stat-card__unit">个</span></span>                        6426
      <span class="stat-card__note">                                          6427
        {% if dash.progress.ready %}其中 {{ dash.progress.ready               6428
      }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}           6429
      </span>                                                                 6430
    </a>                                                                      6431
    <a class="stat-card" href="/unit/">                                       6432
      <span class="stat-card__label">微地质单元</span>                        6433
      <span class="stat-card__value">{{ dash.stats.unit_types                 6434
      }}<span class="stat-card__unit">类</span></span>                        6435
      <span class="stat-card__note">                                          6436
        {% if dash.progress.units %}共划分 {{                                 6437
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}          6438
      </span>                                                                 6439
    </a>                                                                      6440
    <a class="stat-card" href="/standard/">                                   6441
      <span class="stat-card__label">标准与政策</span>                        6442
      <span class="stat-card__value">{{ dash.stats.standards                  6443
      }}<span class="stat-card__unit">项</span></span>                        6444
      <span class="stat-card__note">地方标准与政策文件</span>                 6445
    </a>                                                                      6446
</section>                                                                    6447
{# ---------- 评价流程 ---------- #}                                          6448
<section class="section">                                                     6449
    <div class="section__head">                                               6450
```

```
      <h2 class="section__title">评价流程</h2>
      <p class="section__desc">
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
      </p>
    </div>
    <div class="step-grid">
      {% for s in dash.steps %}
      {% if s.available %}
      <a class="step-card" href="{{ s.href }}">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        {% if s.pending %}
        <span class="step-card__badge">待处理 {{ s.pending
      }}</span>
        {% endif %}
      </a>
      {% else %}
      <div class="step-card step-card--locked"
      aria-disabled="true">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        <span class="step-card__lock">{{ s.note }}</span>
      </div>
      {% endif %}
      {% endfor %}
    </div>
  </section>
  {# ---------- 系统管理 ---------- #}
  <section class="section">
    <div class="section__head">
      <h2 class="section__title">系统管理</h2>
    </div>
    <a class="row-card" href="/users">
      <span class="row-card__icon">{{ icon("users", 19)
      }}</span>
      <span class="row-card__main">
        <span class="row-card__title">用户与操作日志</span>
        <span class="row-card__desc">用户维护与角色权限；全部写操作自动留痕，可查询追溯
      </span>
      </span>
      <span class="row-card__arrow">{{ icon("chevron", 17)
      }}</span>
    </a>
  </section>
```

```
{% endblock %}
# ==== app/templates/auth/users.html ====
{% extends "base.html" %}
{% block title %}用户与权限管理{% endblock %}
{% block content %}
<div class="card">
  <h2>用户管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    {% if current_user.role == "admin" %}
    <a class="btn btn--sm" href="/users/add">新增用户</a>
    {% endif %}
    <a class="btn btn--sm btn--ghost"
    href="/logs">查看操作日志</a>
  </p>
  {% if current_user.role != "admin" %}
  <p class="muted">
    当前角色为「{{ role_labels.get(current_user.role,
    current_user.role) }}」，
    仅管理员可新增、编辑或删除用户。
  </p>
  {% endif %}
  <table>
    <caption class="muted">共 {{ users|length }}
    个用户</caption>
    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">用户名</th>
        <th scope="col">显示名称</th>
        <th scope="col">角色</th>
        <th scope="col">状态</th>
        {% if current_user.role == "admin" %}<th
    scope="col">操作</th>{% endif %}
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr>
        <td class="num">{{ u.id }}</td>
        <td>
          {{ u.username }}
          {% if u.id == current_user.id %}<span
    class="muted">（当前登录）</span>{% endif %}
        </td>
        <td style="text-align:left;">
          {% if u.display_name %}
          {{ u.display_name }}
          {% if u.title %}<span class="muted">· {{ u.title
    }}</span>{% endif %}
          {% else %}
```

```
            <span class="muted">—</span>
            {% endif %}
          </td>
          <td>{{ role_labels.get(u.role, u.role) }}</td>
          <td>
            {% if u.active %}<span class="tag
    tag--ok">启用</span>
            {% else %}<span class="tag tag--muted">停用</span>{%
    endif %}
          </td>
          {% if current_user.role == "admin" %}
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/users/{{ u.id
    }}/edit">编辑</a>
              <form method="post" action="/users/{{ u.id
    }}/delete"
                    onsubmit="return confirm('确定删除用户「{{
    u.username }}」吗？\n该操作无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
    link-btn--danger"
                        {% if u.id == current_user.id
    %}disabled
                        title="不能删除当前登录的账号"{% endif
    %}>删除</button>
              </form>
            </span>
          </td>
          {% endif %}
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endblock %}
# ==== app/templates/auth/add_user.html ====
{% extends "base.html" %}
{% block title %}新增用户{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增用户</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">用户名</label>
      <input id="username" type="text" name="username"
    required
             autocomplete="off" style="width:280px;">
    </div>
```

```
      <div class="field">
        <label class="field__label" for="password">密码</label>
        <input id="password" type="password" name="password"
      required
                autocomplete="new-password"
      style="width:280px;">
      </div>
      <div class="field">
        <label class="field__label" for="role">角色</label>
        <select id="role" name="role">
          <option value="viewer">查看者（仅可查阅）</option>
          <option value="evaluator">评价员（可建项目、录数据、执行评价）</option
      >
          <option value="admin">管理员（可维护用户与基础数据）</option>
        </select>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/users">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/edit_user.html ====
{% extends "base.html" %}
{% block title %}编辑用户 · {{ user.username }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>编辑用户 · {{ user.username }}</h2>
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"
    >
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="display_name">显示名称</l
    abel>
      <input id="display_name" type="text"
    name="display_name" maxlength="64"
                value="{{ user.display_name or '' }}"
                placeholder="留空则显示登录账号">
    </div>
    <div class="field">
      <label class="field__label" for="title">职务</label>
      <input id="title" type="text" name="title"
    maxlength="64"
                value="{{ user.title or '' }}"
    placeholder="如：评价员、项目负责人">
```

```
    </div>
    <div class="field">
      <label class="field__label" for="email">邮箱</label>
      <input id="email" type="text" name="email"
    maxlength="128"
             value="{{ user.email or '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="phone">手机号</label>
      <input id="phone" type="text" name="phone"
    maxlength="32"
             value="{{ user.phone or '' }}">
    </div>
    <hr style="margin:22px 0;border:none;border-top:1px
    solid var(--line);">
    <h3 style="margin:0 0 12px;font-size:15px;">账号与权限</h3>
    <div class="field">
      <label class="field__label" for="role">角色</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if user.role == code
    %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="checkbox" for="active">
        <input id="active" type="checkbox" name="active"
    value="1"
               {% if user.active %}checked{% endif %}>
        <span>启用该账号</span>
      </label>
      <p class="muted" style="margin:6px 0 0;">
        停用后该账号无法登录，但其历史操作记录会保留。
        系统至少需保留一名启用状态的管理员。
      </p>
    </div>
    <div class="field">
      <label class="field__label" for="password">重置密码（留空则不修改
    ）</label>
      <input id="password" type="password" name="password"
             autocomplete="new-password"
    style="width:280px;"
             placeholder="留空表示保持原密码">
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/users">返回</a>
    </p>
  </form>
```

```
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  .profile-grid {
    display: grid;
    grid-template-columns: 260px 1fr;
    gap: 18px;
    align-items: start;
  }
  @media (max-width: 820px) {
    .profile-grid { grid-template-columns: 1fr; }
  }
  /* 头像 */
  .avatar-box {
    display: flex;
    flex-direction: column;
    align-items: center;
    text-align: center;
  }
  .avatar-lg {
    display: grid;
    place-items: center;
    width: 132px;
    height: 132px;
    margin-bottom: 14px;
    border-radius: 50%;
    background: var(--accent-soft);
    color: var(--accent);
    font-size: 48px;
    font-weight: 600;
    object-fit: cover;
    border: 3px solid #fff;
    box-shadow: 0 4px 16px rgba(19, 60, 43, .12);
  }
  .avatar-lg--initial { user-select: none; }
  .avatar-name {
    margin: 0 0 2px;
    font-size: 15px;
    font-weight: 600;
  }
  .avatar-sub {
    margin: 0 0 16px;
    font-size: 12.5px;
    color: var(--ink-soft);
  }
  .avatar-form { width: 100%; }
```

```
  .avatar-form .avatar-pick { width: 100%; }
  .avatar-filename {
    margin: 8px 0 10px;
    font-size: 12px;
    color: var(--ink-soft);
    word-break: break-all;
  }
  /* 只读信息 */
  .kv {
    display: grid;
    grid-template-columns: 96px 1fr;
    gap: 6px 12px;
    margin: 0;
    font-size: 13.5px;
  }
  .kv dt { color: var(--ink-soft); }
  .kv dd { margin: 0; }
  .field__error {
    margin: 5px 0 0;
    font-size: 12.5px;
    color: var(--danger);
  }
</style>
<div class="profile-grid">
  {# ---------------- 左：头像 ---------------- #}
  <div class="card">
    <div class="avatar-box">
      {% if current_user.avatar %}
      <img class="avatar-lg"
           src="{{ url_for('static', filename=current_user.avatar) }}"
           alt="{{ current_user.shown_name }} 的头像">
      {% else %}
      <span class="avatar-lg avatar-lg--initial"
      aria-hidden="true">{{ current_user.initial }}</span>
      {% endif %}
      <p class="avatar-name">{{ current_user.shown_name
      }}</p>
      <p class="avatar-sub">
        {{ role_labels.get(current_user.role,
      current_user.role) }} ·
        {{ current_user.username }}
      </p>
      <form class="avatar-form" method="post"
      action="/profile/avatar"
            enctype="multipart/form-data">
        {{ avatar_form.hidden_tag() }}
        <label class="btn btn--sm btn--ghost avatar-pick"
      for="avatar-file">
          选择图片
```

```
        </label>                                                    6801
        <input id="avatar-file" type="file" name="avatar"           6802
    class="visually-hidden"                                         6803
                accept=".png,.jpg,.jpeg,.gif,.webp">                6804
        <p class="avatar-filename" id="avatar-filename">未选择任    6805
    何文件</p>                                                      6806
        <button type="submit" class="btn                            6807
    btn--sm">上传头像</button>                                      6808
      </form>                                                       6809
      {% if current_user.avatar %}                                  6810
      <form method="post" action="/profile/avatar/delete"           6811
    style="margin-top:8px;"                                         6812
            onsubmit="return confirm('确定移除当前头像吗？');">     6813
        {{ avatar_form.hidden_tag() }}                              6814
        <button type="submit" class="btn btn--sm                    6815
    btn--ghost">移除头像</button>                                   6816
      </form>                                                       6817
      {% endif %}                                                   6818
      <p class="muted" style="margin:12px 0                         6819
    0;font-size:12px;">                                             6820
        PNG / JPG / GIF / WebP，不超过 4 MB                         6821
      </p>                                                          6822
    </div>                                                          6823
  </div>                                                            6824
  {# ---------------- 右：资料 + 密码 ---------------- #}           6825
  <div>                                                             6826
    <div class="card">                                              6827
      <h2>个人资料</h2>                                             6828
      <form method="post">                                          6829
        {{ info_form.hidden_tag() }}                                6830
        <input type="hidden" name="action" value="info">            6831
        <div class="field">                                         6832
          <label class="field__label" for="{{                       6833
    info_form.display_name.id }}">                                  6834
            显示名称                                                6835
          </label>                                                  6836
          {{ info_form.display_name(id=info_form.display_nam        6837
    e.id,                                                           6838
                                      placeholder="留空则显示账号名")  6839
    }}                                                              6840
          {% for e in info_form.display_name.errors %}              6841
          <p class="field__error">{{ e }}</p>                       6842
          {% endfor %}                                              6843
        </div>                                                      6844
        <div class="field">                                         6845
          <label class="field__label" for="{{                       6846
    info_form.title.id }}">职务</label>                             6847
          {{ info_form.title(id=info_form.title.id,                 6848
    placeholder="如：评价员、项目负责人") }}                        6849
          {% for e in info_form.title.errors %}                     6850
```

```
        <p class="field__error">{{ e }}</p>
        {% endfor %}
      </div>
      <div class="field">
        <label class="field__label" for="{{
info_form.email.id }}">邮箱</label>
        {{ info_form.email(id=info_form.email.id) }}
        {% for e in info_form.email.errors %}
        <p class="field__error">{{ e }}</p>
        {% endfor %}
      </div>
      <div class="field">
        <label class="field__label" for="{{
info_form.phone.id }}">手机号</label>
        {{ info_form.phone(id=info_form.phone.id) }}
        {% for e in info_form.phone.errors %}
        <p class="field__error">{{ e }}</p>
        {% endfor %}
      </div>
      <p class="actions" style="margin-top:18px;">
        {{ info_form.submit_info(class_="btn") }}
      </p>
    </form>
    <hr style="margin:20px 0;border:none;border-top:1px
solid var(--line);">
    <h3 style="margin:0 0 10px;font-size:15px;">账号信息</h3>
    <dl class="kv">
      <dt>登录账号</dt><dd class="mono">{{
current_user.username }}</dd>
      <dt>角色</dt>
      <dd>{{ role_labels.get(current_user.role,
current_user.role) }}</dd>
      <dt>状态</dt>
      <dd>
        {% if current_user.active %}
        <span class="tag tag--ok">启用</span>
        {% else %}
        <span class="tag tag--warn">停用</span>
        {% endif %}
      </dd>
      <dt>创建时间</dt>
      <dd class="mono">
        {{ current_user.created_at.strftime("%Y-%m-%d
%H:%M") if current_user.created_at else "—" }}
      </dd>
    </dl>
  </div>
  <div class="card">
    <h2>修改密码</h2>
    <form method="post" style="max-width:420px;">
```

```
          {{ pwd_form.hidden_tag() }}                                    6901
          <input type="hidden" name="action" value="password">          6902
          <div class="field">                                            6903
            <label class="field__label" for="{{                          6904
      pwd_form.old_password.id }}">                                      6905
              当前密码                                                   6906
            </label>                                                     6907
            {{ pwd_form.old_password(id=pwd_form.old_password.           6908
      id,                                                                6909
                                       autocomplete="current-pas         6910
      sword") }}                                                         6911
            {% for e in pwd_form.old_password.errors %}                  6912
            <p class="field__error">{{ e }}</p>                          6913
            {% endfor %}                                                 6914
          </div>                                                         6915
          <div class="field">                                            6916
            <label class="field__label" for="{{                          6917
      pwd_form.new_password.id }}">                                      6918
              新密码                                                     6919
            </label>                                                     6920
            {{ pwd_form.new_password(id=pwd_form.new_password.           6921
      id,                                                                6922
                                       autocomplete="new-passwor         6923
      d") }}                                                             6924
            {% for e in pwd_form.new_password.errors %}                  6925
            <p class="field__error">{{ e }}</p>                          6926
            {% endfor %}                                                 6927
          </div>                                                         6928
          <div class="field">                                            6929
            <label class="field__label" for="{{                          6930
      pwd_form.confirm.id }}">                                           6931
              确认新密码                                                 6932
            </label>                                                     6933
            {{ pwd_form.confirm(id=pwd_form.confirm.id,                  6934
                                autocomplete="new-password")             6935
      }}                                                                 6936
            {% for e in pwd_form.confirm.errors %}                       6937
            <p class="field__error">{{ e }}</p>                          6938
            {% endfor %}                                                 6939
          </div>                                                         6940
          <p class="actions" style="margin-top:18px;">                   6941
            {{ pwd_form.submit_pwd(class_="btn") }}                      6942
          </p>                                                           6943
        </form>                                                          6944
      </div>                                                             6945
    </div>                                                               6946
</div>                                                                   6947
{% endblock %}                                                           6948
{% block scripts %}                                                      6949
<script>                                                                 6950
```

```
//  原生文件选择框样式难以统一，这里换成自定义按钮 + 显示所选文件名
(function () {
  var input = document.getElementById('avatar-file');
  var label = document.getElementById('avatar-filename');
  if (!input || !label) return;
  input.addEventListener('change', function () {
    label.textContent = input.files && input.files.length
      ? input.files[0].name : '未选择任何文件';
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/logs.html ====
{% extends "base.html" %}
{% block title %}操作日志{% endblock %}
{% block content %}
<div class="card">
  <h2>操作日志</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm btn--ghost"
    href="/users">返回用户管理</a>
  </p>
  {% if not records %}
  <p class="muted">暂无操作记录。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}
    / {{ pages }} 页</caption>
    <thead>
      <tr>
        <th scope="col">时间</th>
        <th scope="col">操作人</th>
        <th scope="col">操作</th>
        <th scope="col">对象</th>
        <th scope="col">说明</th>
      </tr>
    </thead>
    <tbody>
      {% for r in records %}
      <tr>
        <td class="mono nowrap">{{ r.created_at.strftime("%Y
    -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>
        <td>{{ users_map.get(r.user_id, "—") }}</td>
        <td style="text-align:left;">{{ r.action }}</td>
        <td>
          {% if r.target_type %}{{ r.target_type }}{% if
    r.target_id %}#{{ r.target_id }}{% endif %}
          {% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ r.detail or "—"
```

```
      }}</td>                                                                    7001
        </tr>                                                                    7002
        {% endfor %}                                                             7003
      </tbody>                                                                   7004
    </table>                                                                     7005
    {% if pages > 1 %}                                                           7006
    <nav class="actions" aria-label="日志分页">                                  7007
      {% if page > 1 %}                                                          7008
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      7009
      page - 1 }}">上一页</a>                                                    7010
      {% endif %}                                                                7011
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>                  7012
      {% if page < pages %}                                                      7013
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      7014
      page + 1 }}">下一页</a>                                                    7015
      {% endif %}                                                                7016
    </nav>                                                                       7017
    {% endif %}                                                                  7018
    {% endif %}                                                                  7019
</div>                                                                           7020
{% endblock %}                                                                   7021
# ==== app/templates/project/index.html ====                                     7022
{% extends "base.html" %}                                                        7023
{% block title %}项目与矿山档案{% endblock %}                                    7024
{% block content %}                                                              7025
<div class="card">                                                               7026
    <h2>项目与矿山档案</h2>                                                     7027
    <p class="actions" style="margin-bottom:12px;">                              7028
      <a class="btn btn--sm" href="/project/add">新建项目</a>                    7029
    </p>                                                                         7030
    {% if not projects %}                                                        7031
    <p class="muted">                                                            7032
      暂无项目。点击上方「新建项目」开始建立评价项目，                          7033
      或运行 <span class="mono">python seed_demo.py</span>                       7034
      载入论文案例。                                                             7035
    </p>                                                                         7036
    {% else %}                                                                   7037
    <table>                                                                      7038
      <caption class="muted">共 {{ projects|length }}                            7039
      个项目</caption>                                                           7040
      <thead>                                                                    7041
        <tr>                                                                     7042
          <th scope="col">项目名称</th>                                          7043
          <th scope="col">编号</th>                                              7044
          <th scope="col">状态</th>                                              7045
          <th scope="col">图斑数</th>                                            7046
          <th scope="col">操作</th>                                              7047
        </tr>                                                                    7048
      </thead>                                                                   7049
      <tbody>                                                                    7050
```

```
        {% for p in projects %}                                          7051
        <tr>                                                             7052
          <td style="text-align:left;">{{ p.name }}</td>                 7053
          <td class="mono">{{ p.code or "—" }}</td>                      7054
          <td><span class="tag tag--muted">{{ p.status                   7055
    }}</span></td>                                                       7056
          <td class="num">                                               7057
            {% if p.patches.count() %}                                   7058
              {{ p.patches.count() }}                                    7059
            {% else %}                                                   7060
              <span class="tag tag--warn">0</span>                       7061
            {% endif %}                                                  7062
          </td>                                                          7063
          <td class="nowrap">                                            7064
            <span class="row-actions">                                   7065
              <a class="link-btn" href="/project/{{ p.id                 7066
    }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+            7067
    新建图斑{% endif %}</a>                                              7068
              <a class="link-btn" href="/project/{{ p.id                 7069
    }}/edit">编辑</a>                                                    7070
              <form method="post" action="/project/{{ p.id               7071
    }}/delete"                                                           7072
                    onsubmit="return confirm('确定删除项目「{{           7073
    p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  7074
                {{ form.hidden_tag() }}                                  7075
                <button type="submit" class="link-btn                    7076
    link-btn--danger">删除</button>                                      7077
              </form>                                                    7078
            </span>                                                      7079
          </td>                                                          7080
        </tr>                                                            7081
        {% endfor %}                                                     7082
      </tbody>                                                           7083
    </table>                                                             7084
    {% endif %}                                                          7085
</div>                                                                   7086
{% endblock %}                                                           7087
# ==== app/templates/project/edit.html ====                              7088
{% extends "base.html" %}                                                7089
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%           7090
    endblock %}                                                          7091
{% block content %}                                                      7092
<div class="card" style="max-width:640px;">                              7093
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                   7094
  <form method="post">                                                   7095
    {{ form.hidden_tag() }}                                              7096
    <div class="field">                                                  7097
      <label class="field__label" for="name">项目名称</label>            7098
      <input id="name" type="text" name="name"                           7099
    style="width:100%;" required                                         7100
```

```
                value="{{ project.name if project else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="code">项目编号</label>
        <input id="code" type="text" name="code"
      style="width:240px;"
                value="{{ project.code or '' if project else
      '' }}">
      </div>
      {% if project %}
      <div class="field">
        <label class="field__label" for="status">状态</label>
        <select id="status" name="status">
          {% for s in ["进行中", "已完成", "已归档"] %}
          <option value="{{ s }}" {% if project.status == s
      %}selected{% endif %}>{{ s }}</option>
          {% endfor %}
        </select>
      </div>
      {% endif %}
      <div class="field">
        <label class="field__label" for="description">项目说明</la
      bel>
        <textarea id="description" name="description">{{
      project.description or '' if project else ''
      }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id
    }}/patch/add">新建图斑</a>
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
```

```
      <caption class="muted">共 {{ project.patches.count() }}
      个图斑</caption>
      <thead>
        <tr>
          <th scope="col">图斑名称</th>
          <th scope="col">行政区</th>
          <th scope="col">面积（公顷）</th>
          <th scope="col">开采矿种</th>
          <th scope="col">微地质单元</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in project.patches %}
        <tr>
          <td style="text-align:left;">{{ p.name }}</td>
          <td>{{ p.admin_region or "—" }}</td>
          <td class="num">{{ p.area_ha if p.area_ha is not
      none else "—" }}</td>
          <td>{{ p.mineral_type or "—" }}</td>
          <td class="num">{{ p.units.count() }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/unit/{{ p.id
      }}">微地质单元</a>
              <a class="link-btn" href="/constraint/{{ p.id
      }}">限制性判定</a>
              <a class="link-btn" href="/project/patch/{{
      p.id }}/edit">编辑</a>
              <form method="post" action="/project/patch/{{
      p.id }}/delete"
                    onsubmit="return confirm('确定删除图斑「{{
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
      link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/project/patch_edit.html ====
{% extends "base.html" %}
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%
      endblock %}
```

```
{% block content %}
<div class="card" style="max-width:720px;">
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-name">图斑名称</label>
      <input id="patch-name" type="text" name="name" style="width:100%;" required
             value="{{ patch.name if patch else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="admin-region">所在行政区</label>
      <input id="admin-region" type="text" name="admin_region" style="width:100%;"
             value="{{ patch.admin_region or '' if patch else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="area-ha">图斑面积（公顷）</label>
      <input id="area-ha" type="number" step="any" name="area_ha"
             value="{{ patch.area_ha if patch and patch.area_ha is not none else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="mineral-type">开采矿种</label>
      <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
             value="{{ patch.mineral_type or '' if patch else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="mining-method">开采方式</label>
      <input id="mining-method" type="text" name="mining_method" style="width:240px;"
             value="{{ patch.mining_method or '' if patch else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="stop-years">停产年限</label>
      <input id="stop-years" type="number" step="any" name="stop_years"
```

```
                value="{{ patch.stop_years if patch and
      patch.stop_years is not none else '' }}">
      </div>
      <div class="field">
        <label class="checkbox" for="responsible-missing">
          <input id="responsible-missing" type="checkbox"
      name="responsible_entity_missing" value="1"
                  {% if not patch or patch.responsible_entity_m
      issing %}checked{% endif %}>
          <span>责任主体缺失</span>
        </label>
      </div>
      <h3>参照生态系统（指标标准化基准）</h3>
      <p class="muted">
        论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。
      </p>
      <div class="field">
        <label class="field__label" for="coverage-benchmark">植
      被覆盖度基准（%）</label>
        <input id="coverage-benchmark" type="number"
      step="any" name="coverage_benchmark"
                value="{{ reference.coverage_benchmark if
      reference else 75 }}">
      </div>
      <div class="field">
        <label class="field__label" for="shannon-benchmark">Sh
      annon-Wiener 多样性指数基准</label>
        <input id="shannon-benchmark" type="number"
      step="any" name="shannon_benchmark"
                value="{{ reference.shannon_benchmark if
      reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种
      </label>
        <input id="dominant-species" type="text"
      name="dominant_species" style="width:100%;"
                value="{{ reference.dominant_species or '' if
      reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
      project.id }}/patches">返回</a>
      </p>
    </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
```

```
<h2>遥感影像挂接</h2>
<p class="muted">
  挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
  支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
</p>
{% if patch.image_path %}
<figure style="margin:0 0 16px;">
  <img src="{{ url_for('static', filename=patch.image_path) }}"
       alt="图斑「{{ patch.name }}」已挂接的遥感影像"
       style="max-width:100%;border:1px solid var(--line);border-radius:6px;">
  <figcaption class="muted" style="margin-top:6px;">
    {{ patch.image_note or "遥感影像" }}
    {% if patch.image_uploaded_at %}
    · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d %H:%M") }}
    {% endif %}
  </figcaption>
</figure>
<form method="post" action="/project/patch/{{ patch.id }}/image/delete"
      onsubmit="return confirm('确定移除该遥感影像吗？');">
  {{ form.hidden_tag() }}
  <button type="submit" class="btn btn--sm btn--danger">移除影像</button>
</form>
<hr style="margin:16px 0;border:none;border-top:1px solid var(--line);">
{% endif %}
<form method="post" action="/project/patch/{{ patch.id }}/image"
      enctype="multipart/form-data">
  {{ form.hidden_tag() }}
  <div class="field">
    <label class="field__label" for="image-file">
      {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
    </label>
    <input id="image-file" type="file" name="image"
           accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.tiff" required>
  </div>
  <div class="field">
    <label class="field__label" for="image-note">影像说明（如来源、时相）</label>
    <input id="image-note" type="text" name="image_note"
           style="width:100%;"
           placeholder="如：2025 年 6 月无人机正射影像"
           value="{{ patch.image_note or '' }}">
  </div>
```

```
      <p class="actions">                                                    7351
        <button type="submit" class="btn">{{ "替换影像" if                  7352
      patch.image_path else "上传并挂接" }}</button>                         7353
      </p>                                                                   7354
    </form>                                                                  7355
</div>                                                                       7356
{% endif %}                                                                  7357
{% endblock %}                                                               7358
# ==== app/templates/unit/list.html ====                                     7359
{% extends "base.html" %}                                                    7360
{% from "_macros.html" import patch_guide %}                                 7361
{% block title %}微地质单元{% endblock %}                                    7362
{% block content %}                                                          7363
<div class="card">                                                           7364
  <h2>微地质单元</h2>                                                        7365
  {{ patch_guide(projects_without_patch) }}                                  7366
  {% if not rows %}                                                          7367
  <p class="muted">                                                          7368
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。                        7369
  </p>                                                                       7370
  {% else %}                                                                 7371
  <table>                                                                    7372
    <thead>                                                                  7373
      <tr>                                                                   7374
        <th scope="col">所属项目</th>                                        7375
        <th scope="col">矿山图斑</th>                                        7376
        <th scope="col">已划分单元</th>                                      7377
        <th scope="col">占比合计</th>                                        7378
        <th scope="col">操作</th>                                            7379
      </tr>                                                                  7380
    </thead>                                                                 7381
    <tbody>                                                                  7382
      {% for r in rows %}                                                    7383
      <tr>                                                                   7384
        <td>{{ r.patch.project.name if r.patch.project else                  7385
    "—" }}</td>                                                              7386
        <td style="text-align:left;">{{ r.patch.name }}</td>                 7387
        <td class="num">{{ r.count }} 个</td>                                7388
        <td class="num">                                                     7389
          {% if r.count %}                                                   7390
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}                         7391
            {% if r.ratio_sum < 0.999 or r.ratio_sum >                       7392
    1.001 %}                                                                 7393
              <span class="warn">（不等于 100%）</span>                      7394
            {% endif %}                                                      7395
          {% else %}—{% endif %}                                             7396
        </td>                                                                7397
        <td class="nowrap">                                                  7398
          <span class="row-actions">                                         7399
            <a class="link-btn" href="/unit/{{ r.patch.id                    7400
```

```
      }}">划分与维护</a>                                                    7401
                <a class="link-btn" href="/indicator/">录入指标数据</a 7402
      >                                                                    7403
            </span>                                                        7404
          </td>                                                            7405
        </tr>                                                              7406
        {% endfor %}                                                       7407
      </tbody>                                                             7408
    </table>                                                               7409
    {% endif %}                                                            7410
</div>                                                                     7411
{% endblock %}                                                             7412
# ==== app/templates/unit/index.html ====                                  7413
{% extends "base.html" %}                                                  7414
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}               7415
{% block content %}                                                        7416
<div class="card">                                                         7417
  <h2>微地质单元 · {{ patch.name }}</h2>                                   7418
  {% if units %}                                                           7419
  <table>                                                                  7420
    <caption class="muted">                                                7421
      共 {{ units|length }} 个单元；                                       7422
      面积合计 <span class="num">{{ "%.4f"|format(total_area)              7423
    }}</span>；                                                            7424
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *            7425
    100) }}</span>                                                         7426
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >                7427
    1.001) %}                                                              7428
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>    7429
      {% endif %}                                                          7430
    </caption>                                                             7431
    <thead>                                                                7432
      <tr>                                                                 7433
        <th scope="col">微地质单元</th>                                    7434
        <th scope="col">面积</th>                                          7435
        <th scope="col">面积占比</th>                                      7436
        <th scope="col">说明</th>                                          7437
        <th scope="col">操作</th>                                          7438
      </tr>                                                                7439
    </thead>                                                               7440
    <tbody>                                                                7441
      {% for u in units %}                                                 7442
      <tr>                                                                 7443
        <td>{{ u.unit_type }}</td>                                         7444
        <td class="num">{{ u.area if u.area is not none                    7445
    else "—" }}</td>                                                       7446
        <td class="num">                                                   7447
          {% if u.ratio is not none %}{{                                   7448
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                7449
        </td>                                                              7450
```

```
        <td style="text-align:left;">{{ u.description or "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id }}">编辑</a>
            <form method="post" action="/unit/delete/{{ u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{ u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>
  {% endif %}
  <h3>添加微地质单元</h3>
  <form method="post" action="/unit/{{ patch.id }}/add">
    {{ form.hidden_tag() }}
    <div class="actions">
      <label class="field__label" for="add-unit-type" style="margin:0;">类型</label>
      <select id="add-unit-type" name="unit_type">
        {% for t in unit_types %}<option value="{{ t }}">{{ t }}</option>{% endfor %}
      </select>
      <label class="field__label" for="add-unit-area" style="margin:0;">面积</label>
      <input id="add-unit-area" type="number" step="any" name="area">
      <label class="field__label" for="add-unit-desc" style="margin:0;">说明</label>
      <input id="add-unit-desc" type="text" name="description" style="width:220px;">
      <button type="submit" class="btn btn--sm">添加</button>
    </div>
  </form>
  <p class="actions" style="margin-top:20px;">
    <a class="btn btn--sm btn--ghost" href="/project/{{ patch.project_id }}/patches">返回图斑列表</a>
    <a class="btn btn--sm btn--ghost" href="/indicator/">去录入指标数据</a>
  </p>
```

```
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>编辑微地质单元 · {{ patch.name }}</h2>
  <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type">
        {% for t in unit_types %}
        <option value="{{ t }}" {% if unit.unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="unit-area">面积</label>
      <input id="unit-area" type="number" step="any" name="area"
             value="{{ unit.area if unit.area is not none else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit-desc">说明</label>
      <input id="unit-desc" type="text" name="description" style="width:420px;"
             value="{{ unit.description or '' }}">
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/unit/{{ unit.patch_id }}">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
```

```
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python
    seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
```

```
      <thead>                                                                  7601
        <tr>                                                                   7602
          <th scope="col">省份</th>                                            7603
          <th scope="col">名称</th>                                            7604
          <th scope="col">文号</th>                                            7605
          <th scope="col">规则数</th>                                          7606
          <th scope="col">操作</th>                                            7607
        </tr>                                                                  7608
      </thead>                                                                 7609
      <tbody>                                                                  7610
        {% for s in policy %}                                                  7611
        <tr>                                                                   7612
          <td>{{ s.province }}</td>                                            7613
          <td style="text-align:left;">{{ s.name }}</td>                       7614
          <td class="mono">{{ s.doc_number or "—" }}</td>                      7615
          <td class="num">{{ s.rules|length }}</td>                            7616
          <td class="nowrap">                                                  7617
            <span class="row-actions">                                         7618
              <a class="link-btn" href="/standard/{{ s.id                      7619
      }}/rules">规则</a>                                                       7620
              <a class="link-btn" href="/standard/{{ s.id                      7621
      }}/edit">编辑</a>                                                        7622
              <form method="post" action="/standard/{{ s.id                    7623
      }}/delete"                                                               7624
                    onsubmit="return confirm('确定删除「{{                     7625
      s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length           7626
      }} 条规则将一并删除，且无法恢复。');">                                   7627
                {{ form.hidden_tag() }}                                        7628
                <button type="submit" class="link-btn                          7629
      link-btn--danger">删除</button>                                          7630
              </form>                                                          7631
            </span>                                                            7632
          </td>                                                                7633
        </tr>                                                                  7634
        {% endfor %}                                                           7635
      </tbody>                                                                 7636
    </table>                                                                   7637
    {% else %}                                                                 7638
    <p class="muted">暂无政策文件。</p>                                        7639
    {% endif %}                                                                7640
</div>                                                                         7641
{% endblock %}                                                                 7642
# ==== app/templates/standard/edit.html ====                                   7643
{% extends "base.html" %}                                                      7644
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%                 7645
    endblock %}                                                                7646
{% block content %}                                                            7647
<div class="card" style="max-width:800px;">                                    7648
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                        7649
  <form method="post">                                                         7650
```

```
{{ form.hidden_tag() }}
<div class="field">
  <label class="field__label" for="province">省份</label>
  <input id="province" type="text" name="province"
style="width:160px;" required
         value="{{ standard.province if standard else
'' }}">
</div>
<div class="field">
  <label class="field__label" for="std-name">名称</label>
  <input id="std-name" type="text" name="name"
style="width:100%;" required
         value="{{ standard.name if standard else ''
}}">
</div>
<div class="field">
  <label class="field__label" for="doc-number">标准编号 /
文号</label>
  <input id="doc-number" type="text" name="doc_number"
style="width:320px;"
         value="{{ standard.doc_number or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="category">类别</label>
  <select id="category" name="category">
    {% for c in ["地方标准", "政策文件"] %}
    <option value="{{ c }}" {% if standard and
standard.category == c %}selected{% endif %}>{{ c
}}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="publish-date">发布时间</l
abel>
  <input id="publish-date" type="text"
name="publish_date" style="width:200px;"
         placeholder="如 2023-03-30"
         value="{{ standard.publish_date or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="implement-date">实施时间<
/label>
  <input id="implement-date" type="text"
name="implement_date" style="width:200px;"
         placeholder="如 2023-04-30（未明确可留空）"
         value="{{ standard.implement_date or '' if
standard else '' }}">
```

```
      </div>
      <div class="field">
        <label class="field__label" for="summary">特点 /
      判定标准</label>
        <textarea id="summary" name="summary"
      style="min-height:96px;">{{ standard.summary or '' if
      standard else '' }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
      }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
      standard.province }} {{ standard.name
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
      btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
<div class="card">
    <h2>结构化规则 · {{ standard.province }} {{ standard.name
      }}</h2>
    {% if rules %}
    <table>
      <caption class="muted">共 {{ rules|length }}
      条规则</caption>
      <thead>
        <tr>
          <th scope="col">判定项</th>
          <th scope="col">条件</th>
          <th scope="col">条件类型</th>
          <th scope="col">适用分区</th>
          <th scope="col">备注</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in rules %}
```

```
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
        <td>
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗?');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item" placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="] %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="r-thr">阈值</label>
        <input id="r-thr" type="text" name="threshold" placeholder="如：50" style="width:110px;">
      </div>
      <div>
        <label class="field__label" for="r-cond">条件类型</label>
        <select id="r-cond" name="condition_type">
```

```
            <option value="必要条件">必要条件</option>                    7801
            <option value="充分条件">充分条件</option>                    7802
          </select>                                                       7803
        </div>                                                            7804
        <div>                                                             7805
          <label class="field__label" for="r-region">适用分区</lab        7806
    el>                                                                   7807
          <input id="r-region" type="text" name="region"                  7808
    placeholder="如宁夏分区" style="width:130px;">                        7809
        </div>                                                            7810
        <div>                                                             7811
          <label class="field__label" for="r-note">备注</label>           7812
          <input id="r-note" type="text" name="note"                      7813
    style="width:220px;">                                                 7814
        </div>                                                            7815
        <div>                                                             7816
          <button type="submit" class="btn">添加规则</button>             7817
        </div>                                                            7818
      </div>                                                              7819
    </form>                                                               7820
    <p class="actions" style="margin-top:20px;">                          7821
      <a class="btn btn--sm btn--ghost"                                   7822
      href="/standard/">返回标准列表</a>                                  7823
      <a class="btn btn--sm btn--ghost"                                   7824
      href="/standard/compare">省际口径比对</a>                           7825
    </p>                                                                  7826
</div>                                                                    7827
{% endblock %}                                                            7828
# ==== app/templates/standard/compare.html ====                           7829
{% extends "base.html" %}                                                 7830
{% block title %}省际口径比对{% endblock %}                               7831
{% block content %}                                                       7832
<div class="card">                                                        7833
    <h2>省际口径比对</h2>                                                 7834
    {% if not groups %}                                                   7835
    <p class="muted">                                                     7836
      暂无规则数据。请先在「标准知识库」中为标准录入规则，                7837
      或运行 <span class="mono">python seed_demo.py</span> 载入论文表     7838
      1 数据。                                                            7839
    </p>                                                                  7840
    {% else %}                                                            7841
    {% for item, rs in groups.items() %}                                  7842
    <h3>{{ item }}</h3>                                                   7843
    <table>                                                               7844
      <thead>                                                             7845
        <tr>                                                              7846
          <th scope="col">省份</th>                                       7847
          <th scope="col">标准 / 文件</th>                                7848
          <th scope="col">条件</th>                                       7849
          <th scope="col">条件类型</th>                                   7850
```

```
            <th scope="col">备注</th>
          </tr>
        </thead>
        <tbody>
          {% for r in rs %}
          <tr>
            <td>{{ r.standard.province }}</td>
            <td style="text-align:left;">{{ r.standard.name
        }}</td>
            <td class="mono nowrap">{{ r.operator }} {{
        r.threshold or "" }}</td>
            <td>{{ r.condition_type }}</td>
            <td style="text-align:left;">{{ r.note or "—"
        }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endfor %}
      {% endif %}
      <p class="actions" style="margin-top:16px;">
        <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
      </p>
    </div>
    {% endblock %}
    # ==== app/templates/constraint/index.html ====
    {% extends "base.html" %}
    {% from "_macros.html" import patch_guide %}
    {% block title %}限制性指标判定{% endblock %}
    {% block content %}
    <div class="card">
      <h2>限制性指标判定</h2>
      {{ patch_guide(projects_without_patch) }}
      {% if not patches %}
      <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
      {% else %}
      <table>
        <thead>
          <tr>
            <th scope="col">所属项目</th>
            <th scope="col">矿山图斑</th>
            <th scope="col">判定结果</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
          {% for p in patches %}
          {% set r = latest.get(p.id) %}
          <tr>
            <td>{{ p.project.name if p.project else "—" }}</td>
```

```
            <td style="text-align:left;">{{ p.name }}</td>
            <td style="text-align:left;">
              {% if r %}
                {% if r.passed %}
                <span class="tag tag--ok">通过</span>
                {% else %}
                <span class="tag tag--warn">不通过</span>
                {% endif %}
                <span class="muted">{{ r.reason }}</span>
              {% else %}
              <span class="tag tag--muted">未判定</span>
              {% endif %}
            </td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/constraint/{{ p.id
      }}">执行判定</a>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/constraint/check.html ====
{% extends "base.html" %}
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:760px;">
  <h2>限制性条件判定 · {{ patch.name }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for field, name, val in field_states %}
    <fieldset style="border:1px solid
    var(--line);border-radius:6px;padding:10px
    14px;margin:0 0 10px;">
      <legend style="padding:0 6px;font-weight:600;">{{
    name }}</legend>
      <div class="actions">
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="1"
                 {% if val is none or val %}checked{% endif
    %}>
          <span>通过</span>
        </label>
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="0"
                 {% if val is not none and not val
```

```
      %}checked{% endif %}>                                                    7951
            <span>不通过</span>                                                7952
          </label>                                                             7953
        </div>                                                                 7954
      </fieldset>                                                              7955
      {% endfor %}                                                             7956
      <p class="actions" style="margin-top:16px;">                             7957
        <button type="submit" class="btn">保存判定</button>                    7958
        <a class="btn btn--ghost" href="/constraint/">返回</a>                 7959
      </p>                                                                     7960
    </form>                                                                    7961
    {% if record %}                                                            7962
    <div style="margin-top:18px;padding:14px;border-radius:6px 7963
      ;border-left:4px solid;                                                  7964
                  {% if record.passed %}background:var(--ok-soft 7965
      );border-left-color:var(--ok);                                           7966
                  {% else %}background:var(--danger-soft);border 7967
      -left-color:var(--danger);{% endif %}">                                  7968
      <p style="margin:0 0 4px;font-weight:600;">                              7969
        判定结论：                                                             7970
        {% if record.passed %}                                                 7971
        <span class="ok">通过，可进入定量评价</span>                           7972
        {% else %}                                                             7973
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>             7974
        {% endif %}                                                            7975
      </p>                                                                     7976
      <p class="muted" style="margin:0;">{{ record.reason                      7977
      }}</p>                                                                   7978
    </div>                                                                     7979
    {% endif %}                                                                7980
</div>                                                                         7981
{% endblock %}                                                                 7982
# ==== app/templates/ahp/index.html ====                                       7983
{% extends "base.html" %}                                                      7984
{% block title %}AHP 权重计算{% endblock %}                                    7985
{% block content %}                                                            7986
<div class="card" style="max-width:720px;">                                    7987
  <h2>AHP 权重计算 · 选择指标层级</h2>                                        7988
  <form method="post">                                                         7989
    {{ form.hidden_tag() }}                                                    7990
    <div class="field">                                                        7991
      <label class="field__label" for="{{ form.unit_type.id                    7992
    }}">微地质单元类型</label>                                                 7993
      {{ form.unit_type(id=form.unit_type.id) }}                               7994
    </div>                                                                     7995
    <div class="field">                                                        7996
      <label class="field__label" for="{{ form.level.id                        7997
    }}">指标层级</label>                                                       7998
      {{ form.level(id=form.level.id) }}                                       7999
    </div>                                                                     8000
```

```
      <div class="field">
        <label class="field__label" for="{{ form.l1.id }}">
          一级指标（层级选「二级指标」时需选择）
        </label>
        {{ form.l1(id=form.l1.id) }}
      </div>
      <p class="actions" style="margin-top:16px;">
        {{ form.submit(class_="btn") }}
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维护</a>
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a>
      </p>
    </form>
</div>
<div class="card">
    <h2>各微地质单元当前生效权重</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">微地质单元</th>
          <th scope="col">当前权重来源</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in weight_rows %}
        <tr>
          <td style="text-align:left;">{{ r.unit_type }}</td>
          <td style="text-align:left;">
            {% if r.custom %}
            <span class="tag tag--warn">{{ r.label }}</span>
            {% else %}
            <span class="tag tag--muted">{{ r.label }}</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1">一级指标权重</a>
              <a class="link-btn" href="/ahp/matrix?unit_type={{ r.unit_type }}&level=1&calc=1">查看一致性</a>
              {% if r.custom %}
              <form method="post" action="/ahp/weights/reset"
                    onsubmit="return confirm('确定把「{{ r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                <input type="hidden" name="csrf_token" value="{{ csrf_token() }}">
                <input type="hidden" name="unit_type" value="{{ r.unit_type }}">
```

```
                    <button type="submit" class="link-btn
    link-btn--danger">恢复默认</button>
                </form>
                {% endif %}
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
    </table>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
    <h2>
      判断矩阵录入
      · {{ unit_type }}
      · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
      endif %}
    </h2>
    <p class="muted">
      采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
      2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
      矩阵已按论文表 3 的权重预填。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <table style="width:auto;">
        <thead>
          <tr>
            <th></th>
            {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
          </tr>
        </thead>
        <tbody>
          {% for i in range(names|length) %}
          <tr>
            <th scope="row">{{ names[i] }}</th>
            {% for j in range(names|length) %}
              {% if i == j %}
              <td class="muted num">1</td>
              {% elif i < j %}
              <td>
                <label class="muted" style="position:absolute;
    left:-9999px;"
                       for="m-{{ i }}-{{ j }}">{{ names[i] }}
```

```
    相对 {{ names[j] }} 的重要性</label>                                    8101
                <input id="m-{{ i }}-{{ j }}" type="number"                 8102
    step="0.01" min="0.01"                                                  8103
                        name="m_{{ i }}_{{ j }}" value="{{                  8104
    default[i][j] }}">                                                      8105
              </td>                                                         8106
              {% else %}                                                    8107
              <td class="muted num">{{ default[i][j] }}</td>                8108
              {% endif %}                                                   8109
            {% endfor %}                                                    8110
        </tr>                                                               8111
        {% endfor %}                                                        8112
      </tbody>                                                              8113
    </table>                                                                8114
    <p class="actions">                                                     8115
      <button type="submit" class="btn" name="action"                       8116
    value="calc">计算权重</button>                                          8117
      {% if result and result.consistent %}                                 8118
      <button type="submit" class="btn" name="action"                       8119
    value="save">保存并生效</button>                                        8120
      {% endif %}                                                           8121
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                   8122
    </p>                                                                    8123
</form>                                                                     8124
<p class="muted" style="margin-top:10px;">                                  8125
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label                8126
    }}</strong>。                                                           8127
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；    8128
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。         8129
</p>                                                                        8130
{% if result %}                                                             8131
<hr style="margin:22px 0;border:none;border-top:1px solid                   8132
    var(--line);">                                                          8133
<h3>计算结果</h3>                                                           8134
<table style="width:auto;">                                                 8135
    <thead>                                                                 8136
      <tr>                                                                  8137
        <th scope="col">指标</th>                                           8138
        {% for name in names %}<th scope="col">{{ name                      8139
    }}</th>{% endfor %}                                                     8140
      </tr>                                                                 8141
    </thead>                                                                8142
    <tbody>                                                                 8143
      <tr>                                                                  8144
        <th scope="row">权重</th>                                           8145
        {% for w in result.weights %}<td class="mono">{{ w                  8146
    }}</td>{% endfor %}                                                     8147
      </tr>                                                                 8148
    </tbody>                                                                8149
</table>                                                                    8150
```

```
  <div class="stat-row">
    <div>
      <div class="stat__label">最大特征值 λmax</div>
      <div class="stat__value mono">{{ result.lambda_max
    }}</div>
    </div>
    <div>
      <div class="stat__label">一致性指标 CI</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.ci) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性比例 CR</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </div>
    </div>
  </div>
  {% if not result.consistent and result.most_inconsistent
    %}
  <p style="margin-top:14px;padding:12px;border-radius:6px;b
    ackground:var(--danger-soft);
              border-left:4px solid var(--danger);">
    <strong class="warn">建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
    result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono num">{{ result.most_inconsistent.v
    alue|round(3) }}</span>，
    理想约 <span class="mono num">{{ result.most_inconsistent.e
    xpected|round(3) }}</span>）
  </p>
  {% endif %}
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicators.html ====
{% extends "base.html" %}
{% block title %}指标定义维护{% endblock %}
{% block content %}
<div class="card">
  <h2>指标体系维护</h2>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/ahp/">返回权重计算</a>
```

```
    <a class="btn btn--sm btn--ghost"                                    8201
    href="/ahp/experts">多专家矩阵</a>                                    8202
    <form method="post" action="/ahp/indicators/reset"                   8203
    style="display:inline;"                                              8204
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 8205
    确定继续？');">                                                       8206
      {{ form.hidden_tag() }}                                            8207
      <button type="submit" class="btn btn--sm                           8208
    btn--danger">按论文取值重置</button>                                  8209
    </form>                                                              8210
  </p>                                                                   8211
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,          8212
    "三级指标")] %}                                                      8213
  {% set level_items = all_ind|selectattr("level",                       8214
    "equalto", level)|list %}                                            8215
  {% if level_items %}                                                   8216
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>                    8217
  <table>                                                                8218
    <thead>                                                              8219
      <tr>                                                               8220
        <th scope="col">指标</th>                                        8221
        <th scope="col">上级指标</th>                                    8222
        <th scope="col">单位</th>                                        8223
        <th scope="col">标准化类型</th>                                  8224
        <th scope="col">理想取值</th>                                    8225
        <th scope="col">启用单元</th>                                    8226
        <th scope="col">限制性</th>                                      8227
        <th scope="col">操作</th>                                        8228
      </tr>                                                              8229
    </thead>                                                             8230
    <tbody>                                                              8231
      {% for i in level_items %}                                         8232
      <tr>                                                               8233
        <td style="text-align:left;">                                    8234
          <span class="mono muted">{{ i.code                             8235
    }}</span><br>{{ i.name }}                                            8236
        </td>                                                            8237
        <td>{{ (all_ind|selectattr("code", "equalto",                   8238
    i.parent_code)|list|first).name if i.parent_code else                8239
    "—" }}</td>                                                          8240
        <td>{{ i.unit or "—" }}</td>                                     8241
        <td>{{ type_label(i.ideal_type) }}</td>                          8242
        <td class="mono">{{ i.ideal_value or "—" }}</td>                 8243
        <td style="text-align:left;">{{ i.enabled_units or               8244
    "—" }}</td>                                                          8245
        <td>                                                             8246
          {% if i.is_constraint %}<span class="tag                       8247
    tag--warn">是</span>                                                 8248
          {% else %}<span class="tag tag--muted">否</span>{%             8249
    endif %}                                                             8250
```

```
          </td>                                                          8251
          <td class="nowrap">                                            8252
            <span class="row-actions">                                   8253
              <a class="link-btn" href="/ahp/indicators/{{               8254
    i.code }}/edit">编辑</a>                                             8255
            </span>                                                      8256
          </td>                                                          8257
        </tr>                                                            8258
        {% endfor %}                                                     8259
      </tbody>                                                           8260
    </table>                                                             8261
    {% endif %}                                                          8262
    {% endfor %}                                                         8263
</div>                                                                   8264
{% endblock %}                                                           8265
# ==== app/templates/ahp/indicator_edit.html ====                        8266
{% extends "base.html" %}                                                8267
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                 8268
{% block content %}                                                      8269
<div class="card" style="max-width:680px;">                              8270
  <h2>编辑指标 · {{ ind.name }}</h2>                                     8271
  <p class="muted">                                                      8272
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{           8273
    ind.level }} 级，                                                    8274
    由论文指标体系确定，不可修改。                                       8275
  </p>                                                                   8276
  <form method="post">                                                   8277
    {{ form.hidden_tag() }}                                              8278
    <div class="field">                                                  8279
      <label class="field__label" for="name">指标名称</label>            8280
      <input id="name" type="text" name="name"                           8281
    style="width:100%;" required                                         8282
             value="{{ ind.name }}">                                     8283
    </div>                                                               8284
    <div class="field">                                                  8285
      <label class="field__label" for="unit">计量单位</label>            8286
      <input id="unit" type="text" name="unit"                           8287
    style="width:180px;"                                                 8288
             value="{{ ind.unit or '' }}">                               8289
    </div>                                                               8290
    <div class="field">                                                  8291
      <label class="field__label" for="ideal-type">标准化类型</la8292
    bel>                                                                 8293
      <select id="ideal-type" name="ideal_type">                         8294
        {% for code, label in [("benefit","越大越优"),("cost","越8295
    小越优"),("mid","适中型"),("discrete","离散赋值")] %}                8296
          <option value="{{ code }}" {% if ind.ideal_type ==             8297
    code %}selected{% endif %}>{{ label }}</option>                      8298
        {% endfor %}                                                     8299
      </select>                                                          8300
```

```
    </div>                                                                  8301
    <div class="field">                                                     8302
      <label class="field__label" for="ideal-value">理想取值</label>        8303
    bel>                                                                    8304
      <input id="ideal-value" type="text"                                   8305
    name="ideal_value" style="width:100%;"                                  8306
            value="{{ ind.ideal_value or '' }}">                           8307
      <p class="muted" style="margin:4px 0 0;">                             8308
        如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。                  8309
      </p>                                                                  8310
    </div>                                                                  8311
    <fieldset style="border:1px solid                                       8312
    var(--line);border-radius:6px;padding:10px                              8313
    14px;margin:0 0 12px;">                                                 8314
      <legend style="padding:0 6px;font-weight:600;">启用该指标的微8315
    地质单元</legend>                                                       8316
      {% for t in unit_types %}                                             8317
      <label class="checkbox" style="margin-right:18px;">                   8318
        <input type="checkbox" name="enabled_units"                         8319
    value="{{ t }}"                                                         8320
               {% if t in ind.units_list() %}checked{%                      8321
    endif %}>                                                               8322
        <span>{{ t }}</span>                                                8323
      </label>                                                              8324
      {% endfor %}                                                          8325
      <p class="muted" style="margin:8px 0 0;">                             8326
        论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。             8327
      </p>                                                                  8328
    </fieldset>                                                             8329
    <div class="field">                                                     8330
      <label class="checkbox" for="is-constraint">                          8331
        <input id="is-constraint" type="checkbox"                           8332
    name="is_constraint" value="1"                                          8333
               {% if ind.is_constraint %}checked{% endif %}>                8334
        <span>同时作为限制性指标（参与一票否决判定）</span>                 8335
      </label>                                                              8336
    </div>                                                                  8337
    <div class="field">                                                     8338
      <label class="field__label" for="description">取值依据说明</        8339
    label>                                                                  8340
      <textarea id="description" name="description"                         8341
    style="min-height:88px;">{{ ind.description or ''                       8342
    }}</textarea>                                                           8343
    </div>                                                                  8344
    <p class="actions" style="margin-top:16px;">                            8345
      <button type="submit" class="btn">保存</button>                       8346
      <a class="btn btn--ghost" href="/ahp/indicators">返回指标列8347
    表</a>                                                                  8348
    </p>                                                                    8349
  </form>                                                                   8350
```

```
</div>
{% endblock %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家判断矩阵 · 录入与汇总</h2>
  <form method="get" class="actions" style="margin-bottom:14
    px;">
    <div>
      <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
      <select id="unit-type" name="unit_type"
      onchange="this.form.submit()">
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}
        <option value="{{ t }}" {% if unit_type == t
      %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div>
      <label class="field__label" for="level">指标层级</label>
      <select id="level" name="level"
      onchange="this.form.submit()">
        <option value="1" {% if level == 1 %}selected{%
      endif %}>一级指标</option>
        <option value="2" {% if level == 2 %}selected{%
      endif %}>二级指标</option>
      </select>
    </div>
  </form>
  <p class="muted">
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif
    %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
  {% if rows %}
  <table>
    <caption class="muted">已录入 {{ rows|length }}
    位专家的矩阵</caption>
    <thead>
      <tr>
        <th scope="col">专家</th>
        <th scope="col">矩阵规模</th>
        <th scope="col">该矩阵 CR</th>
        <th scope="col">一致性</th>
        <th scope="col">录入时间</th>
```

```
          <th scope="col">操作</th>                                                    8401
        </tr>                                                                          8402
      </thead>                                                                         8403
      <tbody>                                                                          8404
        {% for r in rows %}                                                            8405
        <tr>                                                                           8406
          <td>{{ r.expert_name }}</td>                                                 8407
          <td class="num">{{ n }} × {{ n }}</td>                                       8408
          <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is                           8409
      not none else "—" }}</td>                                                        8410
          <td>                                                                         8411
            {% if r.cr is not none and r.cr < 0.1 %}<span                              8412
      class="tag tag--ok">通过</span>                                                  8413
            {% else %}<span class="tag tag--warn">不通过</span>{%                      8414
      endif %}                                                                         8415
          </td>                                                                        8416
          <td class="mono nowrap">{{ r.created_at.strftime("%Y                         8417
      -%m-%d %H:%M") if r.created_at else "—" }}</td>                                  8418
          <td>                                                                         8419
            <form method="post" action="/ahp/experts/{{ r.id                           8420
      }}/delete"                                                                       8421
                  onsubmit="return confirm('确定删除专家「{{                           8422
      r.expert_name }}」的矩阵吗？');">                                                8423
              {{ form.hidden_tag() }}                                                  8424
              <button type="submit" class="btn btn--sm                                 8425
      btn--danger">删除</button>                                                       8426
            </form>                                                                    8427
          </td>                                                                        8428
        </tr>                                                                          8429
        {% endfor %}                                                                   8430
      </tbody>                                                                         8431
    </table>                                                                           8432
    <p class="actions" style="margin-bottom:10px;">                                    8433
      <a class="btn" href="/ahp/experts/aggregate?unit_type={{                         8434
      unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%                          8435
      endif %}">                                                                       8436
        汇总计算（几何平均）                                                           8437
      </a>                                                                             8438
    </p>                                                                               8439
    {% else %}                                                                         8440
    <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>     8441
    {% endif %}                                                                        8442
    <h3>录入一位专家的判断矩阵</h3>                                                    8443
    <form method="post">                                                               8444
      {{ form.hidden_tag() }}                                                          8445
      <input type="hidden" name="action" value="add">                                  8446
      <input type="hidden" name="unit_type" value="{{                                  8447
      unit_type }}">                                                                   8448
      <input type="hidden" name="level" value="{{ level }}">                           8449
      {% if l1 %}<input type="hidden" name="l1" value="{{ l1                           8450
```

```
      }}">{% endif %}                                                    8451
      <div class="field">                                                8452
        <label class="field__label" for="expert-name">专家姓名</la 8453
      bel>                                                               8454
        <input id="expert-name" type="text" name="expert_name" 8455
               style="width:220px;" placeholder="如：张工">              8456
      </div>                                                             8457
      <table style="width:auto;">                                        8458
        <thead>                                                          8459
          <tr>                                                           8460
            <th></th>                                                    8461
            {% for name in names %}<th scope="col">{{ name       8462
      }}</th>{% endfor %}                                                8463
          </tr>                                                          8464
        </thead>                                                         8465
        <tbody>                                                          8466
          {% for i in range(n) %}                                        8467
          <tr>                                                           8468
            <th scope="row">{{ names[i] }}</th>                          8469
            {% for j in range(n) %}                                      8470
              {% if i == j %}                                            8471
              <td class="muted num">1</td>                               8472
              {% elif i < j %}                                           8473
              <td>                                                       8474
                <label class="muted" style="position:absolute; 8475
      left:-9999px;"                                                     8476
                       for="e-{{ i }}-{{ j }}">{{ names[i] }}  8477
      相对 {{ names[j] }} 的重要性</label>                               8478
                <input id="e-{{ i }}-{{ j }}" type="number"       8479
      step="0.01" min="0.01"                                             8480
                       name="m_{{ i }}_{{ j }}" value="{{          8481
      default[i][j] }}">                                                 8482
              </td>                                                      8483
              {% else %}                                                 8484
              <td class="muted num">{{ default[i][j] }}</td>     8485
              {% endif %}                                                8486
            {% endfor %}                                                 8487
          </tr>                                                          8488
          {% endfor %}                                                   8489
        </tbody>                                                         8490
      </table>                                                           8491
      <p class="actions" style="margin-top:12px;">                       8492
        <button type="submit" class="btn">录入该专家矩阵</button>        8493
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护< 8494
      /a>                                                                8495
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>          8496
      </p>                                                               8497
    </form>                                                              8498
</div>                                                                   8499
{% endblock %}                                                           8500
```

```
# ==== app/templates/ahp/experts_aggregate.html ====
{% extends "base.html" %}
{% block title %}多专家矩阵汇总{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}</h2>
  {% if not result %}
  <p class="muted">
    该范围内尚未录入专家矩阵。
    <a href="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">去录入</a>
  </p>
  {% else %}
  <div class="stat-row">
    <div>
      <div class="stat__label">参与汇总的专家</div>
      <div class="stat__value">{{ result.count }} 位</div>
    </div>
    <div>
      <div class="stat__label">λmax</div>
      <div class="stat__value mono">{{ result.lambda_max }}</div>
    </div>
    <div>
      <div class="stat__label">CI</div>
      <div class="stat__value mono">{{ "%.4f"|format(result.ci) }}</div>
    </div>
    <div>
      <div class="stat__label">CR（要求 &lt; 0.1）</div>
      <div class="stat__value mono">{{ "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </div>
    </div>
  </div>
  <h3>汇总后权重</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
```

```
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <h3>各专家矩阵的一致性</h3>
  <table style="width:auto;">
    <thead>
      <tr><th scope="col">专家</th><th scope="col">该矩阵CR</th><th scope="col">判定</th></tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span class="ok">通过</span>
          {% else %}<span class="warn">不通过</span>{% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% if not result.consistent and result.most_inconsistent %}
  <p style="padding:12px;border-radius:6px;background:var(--danger-soft);
            border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value|round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expected|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <form method="post"
          action="/ahp/experts?unit_type={{ unit_type
```

```
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
            style="display:inline;"
            onsubmit="return confirm('保存后，「{{ unit_type
      }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
        {{ form.hidden_tag() }}
        <input type="hidden" name="action" value="save_agg">
        <button type="submit" class="btn" {% if not
      result.consistent %}disabled{% endif %}>
          保存为生效权重
        </button>
      </form>
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1
      }}{% endif %}">返回录入</a>
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV
    / Excel）</a>
  </p>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  {% for p in patches %}
  <div class="patch-panel">
    <div class="patch-panel__head">
      <h3 class="patch-panel__name">{{ p.name }}</h3>
      <span class="patch-panel__meta">
        所属项目：{{ p.project.name if p.project else "—" }}
      </span>
      <span class="patch-panel__meta">
        微地质单元：{{ p.units.count() }} 个
      </span>
      {% if p.units.count() %}
      <span class="patch-panel__meta patch-panel__tail">
        <a class="link-btn" href="/indicator/import">批量导入本图斑
```

```
数据</a>
    </span>
    {% endif %}
</div>
{% if not p.units.count() %}
<div class="notice notice--warn">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id
}}">去划分微地质单元</a>
    </p>
</div>
{% else %}
<table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
"%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
          {% if u.indicator_data %}
          <span class="tag tag--ok">已录入</span>
          {% else %}
          <span class="tag tag--muted">未录入</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/indicator/{{ u.id
}}">录入 / 编辑</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
```

```
      </table>
      {% endif %}
   </div>
   {% endfor %}
   {% endif %}
</div>
{% endblock %}
# ==== app/templates/indicator/edit.html ====
{% extends "base.html" %}
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}
{% block content %}
<div class="card" style="max-width:820px;">
   <h2>指标数据录入 · {{ unit.unit_type }}</h2>
   <p class="muted">
      输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。
      岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。
   </p>
   <form method="post">
      {{ form.hidden_tag() }}
      {% for group_name, fields, has_hm in groups %}
      <h3>{{ group_name }}</h3>
      {% for fname, cname, unit_ in fields %}
      <div class="field">
         <label class="field__label" for="f-{{ fname }}">
            {{ cname }}{% if unit_ %} ({{ unit_ }}){% endif %}
         </label>
         {% if fname == "aspect" %}
         <select id="f-{{ fname }}" name="aspect">
            <option value="">— 不适用 —</option>
            {% for a in aspect_types %}
            <option value="{{ a }}" {% if current.get("aspect")
      == a %}selected{% endif %}>{{ a }}</option>
            {% endfor %}
         </select>
         {% else %}
         <input id="f-{{ fname }}" type="number" step="any"
      name="{{ fname }}"
               value="{{ current.get(fname) if
      current.get(fname) is not none else '' }}">
         {% endif %}
      </div>
      {% endfor %}
      {% endfor %}
      {% for group_name, fields, has_hm in groups %}
      {% if has_hm %}
      <h3>重金属含量（mg/kg）</h3>
      <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
      <table style="width:auto;">
         <thead>
            <tr>
```

```
                {% for m in heavy_metals %}<th scope="col">{{ m }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number" step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy', {}).get(m) if current.get('heavy', {}).get(m) is not none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
       href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
       href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
```

```
        <label class="field__label" for="patch-id">目标矿山图斑</lab
    el>
        <select id="patch-id" name="patch_id" required>
          <option value="">— 请选择 —</option>
          {% for p in patches %}
          <option value="{{ p.id }}">{{ p.name }} ({{
    p.units.count() }} 个单元）</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="datafile">数据文件</label
    >
        <input id="datafile" type="file" name="datafile"
               accept=".csv,.xlsx,.xls" required>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">开始导入</button>
      </p>
    </form>
    <h3>模板列说明</h3>
    <p class="muted">
      共 {{ headers|length }} 列。留空表示该指标「不适用」。
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
      （按论文注释视为未受污染）。
    </p>
    <p class="mono muted" style="line-height:1.9;">{{
      headers|join(" , ") }}</p>
    <h3>导入规则</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
```

```
      </div>                                                                    8851
      <div>                                                                     8852
        <div class="stat__label">问题行</div>                                   8853
        <div class="stat__value {% if report.errors %}warn{%                    8854
      endif %}">{{ report.errors|length }}</div>                                8855
      </div>                                                                    8856
    </div>                                                                      8857
    {% if report.errors %}                                                      8858
    <h3>未导入的行</h3>                                                         8859
    <ul class="muted" style="padding-left:20px;line-height:1.9                  8860
      ;">                                                                       8861
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}                  8862
    </ul>                                                                       8863
    {% else %}                                                                  8864
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>         8865
    {% endif %}                                                                 8866
    <p class="actions" style="margin-top:16px;">                                8867
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>             8868
    </p>                                                                        8869
</div>                                                                          8870
{% endif %}                                                                     8871
{% endblock %}                                                                  8872
# ==== app/templates/evaluate/index.html ====                                   8873
{% extends "base.html" %}                                                       8874
{% from "_macros.html" import patch_guide %}                                    8875
{% block title %}适宜性定量评价{% endblock %}                                   8876
{% block content %}                                                             8877
<div class="card">                                                              8878
    <h2>适宜性定量评价</h2>                                                     8879
    {{ patch_guide(projects_without_patch) }}                                   8880
    {% if not patches %}                                                        8881
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>      8882
    {% else %}                                                                  8883
    <table>                                                                     8884
      <thead>                                                                   8885
        <tr>                                                                    8886
          <th scope="col">所属项目</th>                                         8887
          <th scope="col">矿山图斑</th>                                         8888
          <th scope="col">前置条件</th>                                         8889
          <th scope="col">最近评价结果</th>                                     8890
          <th scope="col">操作</th>                                             8891
        </tr>                                                                   8892
      </thead>                                                                  8893
      <tbody>                                                                   8894
        {% for p in patches %}                                                  8895
        {% set pr = progress.get(p.id) %}                                       8896
        <tr>                                                                    8897
          <td>{{ p.project.name if p.project else "—" }}</td>                   8898
          <td style="text-align:left;">{{ p.name }}</td>                        8899
          <td style="text-align:left;">                                         8900
```

```
            {% if pr and pr.ready %}                                    8901
            <span class="tag tag--ok">已具备</span>                      8902
            {% elif pr %}                                               8903
            <span class="tag tag--warn">缺：{{                          8904
    pr.blocking[0].label }}</span>                                      8905
            <span class="muted">{{ pr.blocking[0].reason                8906
    }}</span>                                                           8907
            {% endif %}                                                 8908
          </td>                                                         8909
          <td>                                                          8910
            {% if latest.get(p.id) %}                                   8911
            F = <span class="mono num">{{                               8912
    latest[p.id].final_score }}</span>,                                 8913
            <span class="tag tag--ok">{{ latest[p.id].grade             8914
    }}</span>                                                           8915
            {% else %}                                                  8916
            <span class="tag tag--muted">未评价</span>                   8917
            {% endif %}                                                 8918
          </td>                                                         8919
          <td class="nowrap">                                           8920
            <span class="row-actions">                                  8921
              <a class="link-btn" href="/evaluate/{{ p.id               8922
    }}">执行评价</a>                                                    8923
              {% if latest.get(p.id) %}                                 8924
              <a class="link-btn" href="/evaluate/result/{{             8925
    latest[p.id].id }}">查看结果</a>                                    8926
              {% endif %}                                               8927
            </span>                                                     8928
          </td>                                                         8929
        </tr>                                                           8930
        {% endfor %}                                                    8931
      </tbody>                                                          8932
    </table>                                                            8933
    {% endif %}                                                         8934
</div>                                                                  8935
{% endblock %}                                                          8936
# ==== app/templates/evaluate/run.html ====                             8937
{% extends "base.html" %}                                               8938
{% import "_macros.html" as ui %}                                       8939
{% block title %}执行评价 · {{ patch.name }}{% endblock %}              8940
{% block content %}                                                     8941
<div class="card" style="max-width:880px;">                             8942
    <h2>执行适宜性评价 · {{ patch.name }}</h2>                          8943
    {{ ui.workflow_steps(progress) }}                                   8944
    {{ ui.workflow_blocking(progress) }}                                8945
    {% if progress.ready %}                                             8946
    <form method="post">                                                8947
      {{ form.hidden_tag() }}                                           8948
      <p class="actions">                                               8949
        <button type="submit" class="btn">开始评价</button>             8950
```

```
        <a class="btn btn--ghost" href="/evaluate/">返回</a>
      </p>
    </form>
    {% else %}
    <p class="actions">
      <button type="button" class="btn" disabled>开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
    {% endif %}
</div>
<div class="card" style="max-width:880px;">
    <h2>当前流程状态</h2>
    <table>
      <thead>
        <tr><th scope="col">步骤</th><th scope="col">状态</th><th
      scope="col">说明</th></tr>
      </thead>
      <tbody>
        <tr>
          <td>限制性条件判定</td>
          <td>
            {% if progress.constraint.passed %}<span
      class="tag tag--ok">已通过</span>
            {% elif progress.constraint.done %}<span
      class="tag tag--warn">未通过</span>
            {% else %}<span class="tag tag--muted">待完成</span>{
      % endif %}
          </td>
          <td style="text-align:left;">{{
      progress.constraint.detail }}</td>
        </tr>
        <tr>
          <td>划分微地质单元</td>
          <td>
            {% if progress.units.done %}<span class="tag
      tag--ok">已完成</span>
            {% else %}<span class="tag tag--muted">待完成</span>{
      % endif %}
          </td>
          <td style="text-align:left;">{{
      progress.units.detail }}</td>
        </tr>
        <tr>
          <td>录入指标数据</td>
          <td>
            {% if progress.data.ready %}<span class="tag
      tag--ok">已完成</span>
            {% else %}<span class="tag tag--muted">待完成</span>{
      % endif %}
          </td>
```

```
            <td style="text-align:left;">{{                                    9001
        progress.data.detail }}</td>                                            9002
          </tr>                                                                 9003
        </tbody>                                                                9004
      </table>                                                                  9005
</div>                                                                          9006
{% if weight_sources %}                                                         9007
<div class="card" style="max-width:880px;">                                     9008
      <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">           9009
        <thead>                                                                 9010
          <tr><th scope="col">微地质单元</th><th                                9011
        scope="col">权重来源</th></tr>                                          9012
        </thead>                                                                9013
        <tbody>                                                                 9014
          {% for ut, label in weight_sources.items() %}                         9015
          <tr>                                                                  9016
            <td style="text-align:left;">{{ ut }}</td>                         9017
            <td style="text-align:left;">                                       9018
              {% if "自定义" in label %}                                        9019
              <span class="tag tag--warn">{{ label }}</span>                    9020
              {% else %}                                                        9021
              <span class="tag tag--muted">{{ label }}</span>                   9022
              {% endif %}                                                       9023
            </td>                                                               9024
          </tr>                                                                 9025
          {% endfor %}                                                          9026
        </tbody>                                                                9027
      </table>                                                                  9028
</div>                                                                          9029
{% endif %}                                                                     9030
{% endblock %}                                                                  9031
# ==== app/templates/evaluate/result.html ====                                  9032
{% extends "base.html" %}                                                       9033
{% block title %}评价结果{% endblock %}                                         9034
{% block content %}                                                             9035
<div class="card">                                                              9036
      <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>               9037
      <div class="stat-row">                                                    9038
        <div>                                                                   9039
          <div class="stat__label">最终评价因子 F</div>                         9040
          <div class="stat__value mono">{{ r.final_score                        9041
        }}</div>                                                                9042
        </div>                                                                  9043
        <div>                                                                   9044
          <div class="stat__label">适宜性等级</div>                             9045
          <div class="stat__value ok">{{ r.grade }}</div>                       9046
        </div>                                                                  9047
        <div style="flex:1;min-width:240px;">                                   9048
          <div class="stat__label">推荐措施</div>                               9049
          <div style="font-size:15px;">{{ r.measures }}</div>                   9050
```

```
      </div>                                                                    9051
    </div>                                                                      9052
    {% if unit_scores %}                                                        9053
    <h3>各微地质单元得分</h3>                                                   9054
    <table style="max-width:860px;">                                            9055
      <thead>                                                                   9056
        <tr>                                                                    9057
          <th scope="col">微地质单元</th>                                       9058
          <th scope="col">面积占比</th>                                         9059
          <th scope="col">单元评价因子 F'</th>                                  9060
          <th scope="col">加权贡献</th>                                         9061
          <th scope="col">权重来源</th>                                         9062
        </tr>                                                                   9063
      </thead>                                                                  9064
      <tbody>                                                                   9065
        {% for u in unit_scores %}                                              9066
        <tr>                                                                    9067
          <td>{{ u.unit_type }}</td>                                            9068
          <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                     9069
      }}</td>                                                                   9070
          <td class="mono">{{ u.unit_score }}</td>                              9071
          <td class="mono">{{ "%.4f"|format(u.ratio *                           9072
      u.unit_score) }}</td>                                                     9073
          <td class="muted">{{ u.weight_source or "论文表 3                     9074
      默认权重" }}</td>                                                         9075
        </tr>                                                                   9076
        {% endfor %}                                                            9077
      </tbody>                                                                  9078
    </table>                                                                    9079
    {% endif %}                                                                 9080
    <p class="actions" style="margin-top:18px;">                                9081
      <a class="btn btn--ghost" href="/report/{{ r.patch_id                     9082
      }}">查看得分明细</a>                                                      9083
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>              9084
    </p>                                                                        9085
</div>                                                                          9086
{% endblock %}                                                                  9087
# ==== app/templates/report/index.html ====                                     9088
{% extends "base.html" %}                                                       9089
{% from "_macros.html" import patch_guide %}                                    9090
{% block title %}报告输出{% endblock %}                                         9091
{% block content %}                                                             9092
<div class="card">                                                              9093
    <h2>报告与结果输出</h2>                                                     9094
    {{ patch_guide(projects_without_patch) }}                                   9095
    {% if not patches %}                                                        9096
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>      9097
    {% else %}                                                                  9098
    <table>                                                                     9099
      <thead>                                                                   9100
```

```
        <tr>                                                                    9101
          <th scope="col">所属项目</th>                                          9102
          <th scope="col">矿山图斑</th>                                          9103
          <th scope="col">可生成报告</th>                                        9104
          <th scope="col">操作</th>                                              9105
        </tr>                                                                   9106
      </thead>                                                                  9107
      <tbody>                                                                   9108
        {% for p in patches %}                                                  9109
        {% set pr = progress.get(p.id) %}                                       9110
        <tr>                                                                    9111
          <td>{{ p.project.name if p.project else "—" }}</td>                   9112
          <td style="text-align:left;">{{ p.name }}</td>                        9113
          <td style="text-align:left;">                                         9114
            {% if pr and pr.ready %}                                            9115
            <span class="tag tag--ok">可以</span>                               9116
            {% elif pr %}                                                       9117
            <span class="tag tag--warn">暂不可</span>                           9118
            <span class="muted">{{ pr.blocking[0].reason                        9119
      }}</span>                                                                 9120
            {% endif %}                                                         9121
          </td>                                                                 9122
          <td class="nowrap">                                                   9123
            {% if pr and pr.ready %}                                            9124
            <span class="row-actions">                                          9125
              <a class="link-btn" href="/report/{{ p.id                         9126
      }}">查看得分明细</a>                                                     9127
              <a class="link-btn" href="/report/{{ p.id                         9128
      }}/pdf">导出 PDF</a>                                                     9129
              <a class="link-btn" href="/report/{{ p.id                         9130
      }}/word">导出 Word</a>                                                   9131
            </span>                                                             9132
            {% else %}                                                          9133
            <span class="muted">—</span>                                        9134
            {% endif %}                                                         9135
          </td>                                                                 9136
        </tr>                                                                   9137
        {% endfor %}                                                            9138
      </tbody>                                                                  9139
    </table>                                                                    9140
    {% endif %}                                                                 9141
</div>                                                                          9142
{% endblock %}                                                                  9143
# ==== app/templates/report/detail.html ====                                    9144
{% extends "base.html" %}                                                       9145
{% block title %}得分明细 · {{ patch.name }}{% endblock %}                      9146
{% block content %}                                                             9147
<div class="card">                                                              9148
  <h2>得分明细 · {{ patch.name }}</h2>                                          9149
  <div class="stat-row">                                                        9150
```

```
      <div>                                                                    9151
        <div class="stat__label">最终评价因子 F</div>                          9152
        <div class="stat__value mono">{{                                      9153
      "%.4f"|format(result.final_score) }}</div>                               9154
      </div>                                                                   9155
      <div>                                                                    9156
        <div class="stat__label">适宜性等级</div>                              9157
        <div class="stat__value ok">{{ result.grade }}</div>                   9158
      </div>                                                                   9159
      <div style="flex:1;min-width:240px;">                                    9160
        <div class="stat__label">推荐措施</div>                                9161
        <div style="font-size:15px;">{{                                        9162
      result.grade_info.measure }}</div>                                       9163
      </div>                                                                   9164
    </div>                                                                     9165
    <p class="muted" style="margin-top:12px;">                                 9166
      本页与导出报告均取自                                                     9167
      <strong>{% if record %}{{ record.created_at.strftime("%Y                 9168
      -%m-%d %H:%M:%S") }}{% endif %}                                          9169
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else   9170
      "—" }}），                                                               9171
      不会因事后修改指标数据而变化。                                           9172
    </p>                                                                       9173
    <p class="actions" style="margin-top:16px;">                               9174
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF                9175
      报告</a>                                                                 9176
      <a class="btn btn--ghost" href="/report/{{ patch.id                      9177
      }}/word">导出 Word 报告</a>                                              9178
      <a class="btn btn--ghost" href="/report/">返回</a>                       9179
    </p>                                                                       9180
</div>                                                                         9181
<div class="card">                                                             9182
    <h2>结果图表</h2>                                                         9183
    <h3>各微地质单元得分对比</h3>                                             9184
    <div>{{ chart_units|safe }}</div>                                          9185
    <h3>各一级指标平均得分</h3>                                               9186
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  9187
    <div>{{ chart_weight|safe }}</div>                                         9188
</div>                                                                         9189
{% for u in result.unit_results %}                                             9190
<div class="card">                                                             9191
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *                  9192
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>                9193
    {% if u.weight_source %}                                                   9194
    <p class="muted" style="margin-top:-6px;">权重来源：{{                     9195
      u.weight_source }}</p>                                                   9196
    {% endif %}                                                                9197
    <table>                                                                    9198
      <thead>                                                                  9199
        <tr>                                                                   9200
```

```
          <th scope="col">一级指标</th>                                    9201
          <th scope="col">二级指标</th>                                    9202
          <th scope="col">实测值</th>                                      9203
          <th scope="col">标准化值</th>                                    9204
          <th scope="col">权重</th>                                        9205
          <th scope="col">贡献</th>                                        9206
        </tr>                                                              9207
      </thead>                                                             9208
      <tbody>                                                              9209
        {% for d in u.details %}                                           9210
        <tr>                                                               9211
          <td>{{ d.l1 }}</td>                                              9212
          <td style="text-align:left;">{{ d.l2 }}</td>                     9213
          <td class="mono" style="text-align:left;">{{ "—" if              9214
      d.raw is none else d.raw }}</td>                                     9215
          <td class="mono">{{ "—" if d.std_value is none else              9216
      "%.4f"|format(d.std_value) }}</td>                                   9217
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>              9218
          <td class="mono">{{ "—" if d.contribution is none                9219
      else "%.4f"|format(d.contribution) }}</td>                           9220
        </tr>                                                              9221
        {% endfor %}                                                       9222
      </tbody>                                                             9223
    </table>                                                               9224
    <p class="muted" style="margin:0;">                                    9225
      一级指标得分：                                                       9226
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span                 9227
      class="num">{{ "%.4f"|format(v) }}</span>{% if not                   9228
      loop.last %};{% endif %}{% endfor %}                                 9229
    </p>                                                                   9230
</div>                                                                     9231
{% endfor %}                                                               9232
{% endblock %}                                                             9233
# ==== app/templates/report/print.html ====                                9234
<!DOCTYPE html>                                                            9235
<html lang="zh-CN">                                                        9236
<head>                                                                     9237
<meta charset="UTF-8">                                                     9238
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>                   9239
<style>                                                                    9240
  @page { size: A4; margin: 18mm 16mm; }                                   9241
  * { box-sizing: border-box; }                                            9242
  body {                                                                   9243
    margin: 0; color: #23262b; background: #fff;                           9244
    font-family: "Microsoft YaHei", "SimSun", sans-serif;                  9245
    font-size: 10.5pt; line-height: 1.65;                                  9246
  }                                                                        9247
  h1 { font-size: 17pt; text-align: center; margin: 0 0                    9248
    6px; }                                                                 9249
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;                9250
```

```
        border-bottom: 1px solid #dfe3e8; padding-bottom:      9251
    4px; }                                                     9252
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0       9253
    6px; }                                                     9254
  .sub { text-align: center; color: #6b7280; font-size:        9255
    9.5pt; margin-bottom: 18px; }                              9256
  table { width: 100%; border-collapse: collapse; margin:      9257
    8px 0 14px;                                                9258
          font-variant-numeric: tabular-nums; }                9259
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;        9260
    text-align: center; }                                      9261
  th { background: #eef2f7; font-weight: 600; }                9262
  .left { text-align: left; }                                  9263
  .mono { font-family: Consolas, "Courier New", monospace; }9264
  .kv { width: auto; }                                         9265
  .kv th { width: 30%; text-align: left; }                     9266
  .kv td { text-align: left; }                                 9267
  ul { margin: 6px 0 12px; padding-left: 20px; }               9268
  .note { color: #6b7280; font-size: 9.5pt; }                  9269
  .chart { margin: 10px 0 16px; }                              9270
  .sig { margin-top: 28px; font-size: 10pt; }                  9271
  .avoid-break { break-inside: avoid; }                        9272
  svg { max-width: 100%; }                                     9273
</style>                                                       9274
</head>                                                        9275
<body>                                                         9276
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>                9277
  <p class="sub">                                              9278
    {{ patch.name }}                                           9279
    {% if patch.admin_region %} · {{ patch.admin_region        9280
    }}{% endif %}                                              9281
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０ ·          9282
    生成时间 {{ generated_at }}                                9283
  </p>                                                         9284
  <h2>一、图斑基本信息</h2>                                    9285
  <table class="kv">                                           9286
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>        9287
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"    9288
    }}</td></tr>                                               9289
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if       9290
    patch.area_ha is not none else "—" }}</td></tr>            9291
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"      9292
    }}</td></tr>                                               9293
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"     9294
    }}</td></tr>                                               9295
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit9296
    y_missing else "否" }}</td></tr>                           9297
    {% if reference %}                                         9298
    <tr><th>参照生态系统</th>                                  9299
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,  9300
```

```
                多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</th><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning
    }}</td></tr>
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure
    }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
    <thead>
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
    F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
      {% for u in result.unit_results %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.unit_score)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.weighted)
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
    <thead>
      <tr>
        <th>一级指标</th><th>二级指标</th><th>实测值</th>
        <th>标准化值</th><th>权重</th><th>贡献</th>
      </tr>
    </thead>
    <tbody>
```

```
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td class="left">{{ d.l2 }}</td>
          <td class="mono left">{{ "—" if d.raw is none else d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="note">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{ "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{% endfor %}
    </p>
    {% endfor %}
    <h2>六、评价依据说明</h2>
    <ul class="note">
      <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、适中型与离散赋值四类。</li>
      <li>单元评价因子按论文式（9） 由二级指标权重加权求得；整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7 适宜。</li>
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
    </ul>
    <p class="sig">
      评价人（签字）：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿　　日期：＿＿＿＿＿＿＿＿＿＿＿＿
    </p>
  </body>
</html>
# ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
  <h2>搜索结果</h2>
  {% if not q %}
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、标准名称与文号。</p>
  {% else %}
  <p class="muted">
    关键词「<strong>{{ q }}</strong>」共找到
```

```
      <span class="num">{{ total }}</span> 条结果。
    </p>
    {% if not total %}
    <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
    {% endif %}
    {% endif %}
</div>
{% if projects %}
<div class="card">
    <h2>评价项目（{{ projects|length }}）</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">项目名称</th>
          <th scope="col">编号</th>
          <th scope="col">状态</th>
          <th scope="col">图斑数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in projects %}
        <tr>
          <td style="text-align:left;">{{ p.name }}</td>
          <td class="mono">{{ p.code or "—" }}</td>
          <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
          <td class="num">{{ p.patches.count() }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/project/{{ p.id
    }}/patches">矿山图斑</a>
              <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endif %}
{% if patches %}
<div class="card">
    <h2>矿山图斑（{{ patches|length }}）</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">图斑名称</th>
```

```
            <th scope="col">行政区</th>                                          9451
            <th scope="col">矿种</th>                                            9452
            <th scope="col">操作</th>                                            9453
          </tr>                                                                  9454
        </thead>                                                                 9455
        <tbody>                                                                  9456
          {% for m in patches %}                                                 9457
          <tr>                                                                   9458
            <td>{{ m.project.name if m.project else "—" }}</td>                  9459
            <td style="text-align:left;">{{ m.name }}</td>                       9460
            <td>{{ m.admin_region or "—" }}</td>                                 9461
            <td>{{ m.mineral_type or "—" }}</td>                                 9462
            <td class="nowrap">                                                  9463
              <span class="row-actions">                                         9464
                <a class="link-btn" href="/constraint/{{ m.id                    9465
        }}">限制性判定</a>                                                       9466
                <a class="link-btn" href="/unit/{{ m.id                          9467
        }}">微地质单元</a>                                                       9468
                <a class="link-btn" href="/evaluate/{{ m.id                      9469
        }}">执行评价</a>                                                         9470
              </span>                                                            9471
            </td>                                                                9472
          </tr>                                                                  9473
          {% endfor %}                                                           9474
        </tbody>                                                                 9475
      </table>                                                                   9476
    </div>                                                                       9477
    {% endif %}                                                                  9478
    {% if standards %}                                                           9479
    <div class="card">                                                           9480
      <h2>标准与政策文件（{{ standards|length }}）</h2>                          9481
      <table>                                                                    9482
        <thead>                                                                  9483
          <tr>                                                                   9484
            <th scope="col">类别</th>                                            9485
            <th scope="col">省份</th>                                            9486
            <th scope="col">名称</th>                                            9487
            <th scope="col">文号</th>                                            9488
            <th scope="col">操作</th>                                            9489
          </tr>                                                                  9490
        </thead>                                                                 9491
        <tbody>                                                                  9492
          {% for s in standards %}                                               9493
          <tr>                                                                   9494
            <td>{{ s.category }}</td>                                            9495
            <td>{{ s.province }}</td>                                            9496
            <td style="text-align:left;">{{ s.name }}</td>                       9497
            <td class="mono">{{ s.doc_number or "—" }}</td>                      9498
            <td class="nowrap">                                                  9499
              <span class="row-actions">                                         9500
```

```
                <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
                </span>
            </td>
        </tr>
        {% endfor %}
    </tbody>
  </table>
</div>
{% endif %}
{% endblock %}
#  ==== seed_demo.py ====
#  -*- coding: utf-8 -*-
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。
运行：python seed_demo.py                    幂等，只补写缺失的演示数据，
                                                 不会删除用户自建的项目与图斑
      python seed_demo.py --reset                连同标准库与指标定义一并重建
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
#  论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
```

```
        r u l e s = [ ( " 自然恢复适宜性分级 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ] ) ,
d i c t ( c a t e g o r y = " 地方标准 " ,  p r o v i n c e = " 内蒙古 " ,
n a m e = " 历史遗留废弃矿山自然恢复判定规范 " ,
        d o c _ n u m b e r = " D B 1 5 / T  4 2 8 0 — 2 0 2 6 " ,
p u b l i s h _ d a t e = " 2 0 2 6 - 0 1 - 3 0 " ,
        i m p l e m e n t _ d a t e = " 2 0 2 6 - 0 3 - 0 1 " ,
        s u m m a r y = " 调查内容包含参照生态系统调查 " ,
        r u l e s = [ ( " 地质灾害发育程度 " ,  " < = " ,  " 弱发育 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 地形地貌与周边协调 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 有利于植被恢复的表层土壤 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 已形成稳定植被覆盖 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ] ) ,
d i c t ( c a t e g o r y = " 地方标准 " ,  p r o v i n c e = " 江西 " ,
n a m e = " 废弃矿山自然恢复技术规范 " ,
        d o c _ n u m b e r = " D B 3 6 / T  2 2 2 1 — 2 0 2 6 " ,
p u b l i s h _ d a t e = " 2 0 2 6 - 0 2 - 0 2 " ,
        i m p l e m e n t _ d a t e = " 2 0 2 6 - 0 8 - 0 1 " ,
        s u m m a r y = " 对参照生态系统进行了要求；提出自然恢复适宜性指数 " ,
        r u l e s = [ ( " 参照生态系统调查 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 自然恢复适宜性指数 " ,  " > = " ,  " 阈值 " ,  " 充分条件 " ,  N o n e ) ] ) ,
d i c t ( c a t e g o r y = " 地方标准 " ,  p r o v i n c e = " 宁夏 " ,  n a m e = " 矿山自然恢复认定规程 " ,
        d o c _ n u m b e r = N o n e ,  p u b l i s h _ d a t e = " 2 0 2 6 - 0 6 - 1 8 " ,
i m p l e m e n t _ d a t e = N o n e ,
        s u m m a r y = " 分区差异化：三个一级生态修复分区及  2 2  个二级分区 " ,
        r u l e s = [ ( " 无地质灾害风险 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 植被覆盖度趋势 " ,  " = = " ,  " 稳定或正增长 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 与参照系差距 " ,  " < " ,  " 2 0 " ,  " 充分条件 " ,  " % " ) ] ) ,
d i c t ( c a t e g o r y = " 政策文件 " ,  p r o v i n c e = " 安徽 " ,
        n a m e = " 安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知 " ,
        d o c _ n u m b e r = " 皖自然资修函 [ 2 0 2 0 ] 3 4 号 " ,
p u b l i s h _ d a t e = " 2 0 2 0 - 0 3 - 0 3 " ,  i m p l e m e n t _ d a t e = N o n e ,
        s u m m a r y = " 自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好 " ,
        r u l e s = [ ( " 责任主体 " ,  " = = " ,  " 政府 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 无地质灾害 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 无水土污染 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ] ) ,
d i c t ( c a t e g o r y = " 政策文件 " ,  p r o v i n c e = " 山东 " ,
        n a m e = " 山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知 " ,
        d o c _ n u m b e r = " 鲁自然资字 [ 2 0 2 2 ] 1 0 4 号 " ,
p u b l i s h _ d a t e = " 2 0 2 2 - 0 8 - 0 2 " ,  i m p l e m e n t _ d a t e = N o n e ,
        s u m m a r y = " 条件与安徽省相同，另增加自然恢复认定标准 " ,
        r u l e s = [ ( " 无地质灾害 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 无水土污染 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ] ) ,
d i c t ( c a t e g o r y = " 政策文件 " ,  p r o v i n c e = " 湖南 " ,
        n a m e = " 关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知 " ,
        d o c _ n u m b e r = " 湘自资办发 [ 2 0 2 3 ] 5 8 号 " ,
p u b l i s h _ d a t e = " 2 0 2 3 - 0 7 - 2 0 " ,  i m p l e m e n t _ d a t e = N o n e ,
        s u m m a r y = " 自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
> 7 5 %、连续 3 年正向增长 " ,
        r u l e s = [ ( " 植被覆盖度 " ,  " > " ,  " 7 5 " ,  " 必要条件 " ,  " % " ) ,
                  ( " 植被覆盖率趋势 " ,  " = = " ,  " 连续 3 年正向增长 " ,  " 必要条件 " ,  N o n e ) ,
                  ( " 无地质灾害 " ,  " = = " ,  " 是 " ,  " 必要条件 " ,  N o n e ) ,
```

```
                    ("符合国土空间规划", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="新疆",
             name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
             doc_number="新自然资源办函〔２０２４〕２１９号",
        publish_date="２０２４-１２-２０", implement_date=None,
             summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
             rules=[("植被覆盖度", ">=", "周边３０%", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None),
                    ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="广西",
             name="广西壮族自治区历史遗留矿山自然恢复技术指南",
             doc_number="桂自然资办〔２０２４〕２５０号",
        publish_date="２０２４-１２-３０", implement_date=None,
             summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
             rules=[("无地质灾害", "==", "是", "必要条件", None),
                    ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                    ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
             name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
             doc_number="藏自然资发〔２０２５〕１９号",
        publish_date="２０２５-０９-２４", implement_date=None,
             summary="含修复成本高、生态修复效益不显著等适用条件",
             rules=[("责任主体", "==", "政府", "必要条件", None),
                    ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
```

```
                "unit_type": "平台", "ratio": 0.2319,                    9651
                "data": {"soil_ratio": 0.56, "thickness": 30,            9652
    "ph": 8.72,                                                          9653
                         "organic_matter": 10.92,                        9654
                         "heavy_metals": {"Hg": 0.02, "As":              9655
    3.13, "Ni": 12.33,                                                   9656
                                              "Cu": 11.14, "Cd":         9657
    0.09, "Pb": 9.98},                                                   9658
                         "coverage": 58.42, "shannon": 0.72},            9659
            },                                                           9660
    ],                                                                   9661
}                                                                        9662
def clean_case(reset=False):                                             9663
    """清理数据。                                                        9664
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰        9665
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，  9666
    以便把演示环境恢复到初始状态。                                       9667
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目      9668
    会在每次启动时被删除。                                               9669
    """                                                                  9670
    from app.models import EvaluationResult, ScoreDetail,                9671
    ConstraintCheck                                                      9672
    demo_projects = Project.query.filter_by(code=DEMO_CODE).             9673
    all()                                                                9674
    demo_ids = [p.id for p in demo_projects]                             9675
    patch_ids = []                                                       9676
    if demo_ids:                                                         9677
        patch_ids = [p.id for p in MinePatch.query.filter(               9678
            MinePatch.project_id.in_(demo_ids)).all()]                   9679
    if patch_ids:                                                        9680
        unit_ids = [u.id for u in MicroUnit.query.filter(                9681
            MicroUnit.patch_id.in_(patch_ids)).all()]                    9682
        if unit_ids:                                                     9683
            IndicatorData.query.filter(                                  9684
                IndicatorData.unit_id.in_(unit_ids)).delete(             9685
                synchronize_session=False)                               9686
        result_ids = [r.id for r in EvaluationResult.query.f             9687
    ilter(                                                               9688
            EvaluationResult.patch_id.in_(patch_ids)).all()]             9689
        if result_ids:                                                   9690
            ScoreDetail.query.filter(                                    9691
                ScoreDetail.result_id.in_(result_ids)).delet             9692
    e(                                                                   9693
                synchronize_session=False)                               9694
        EvaluationResult.query.filter(                                   9695
            EvaluationResult.patch_id.in_(patch_ids)).delete             9696
    (                                                                    9697
            synchronize_session=False)                                   9698
        ConstraintCheck.query.filter(                                    9699
            ConstraintCheck.patch_id.in_(patch_ids)).delete(             9700
```

```
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
    delete(
            synchronize_session=False)
    if demo_ids:
        Project.query.filter(Project.id.in_(demo_ids)).delet
    e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，
    #  避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).fi
    rst()
        if m is not None:
            return m
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
```

```
    db.create_all()          #  全新环境下先建表，保证本脚本可独立运行
    _purge_orphan_details()
    admin = User.query.filter_by(username="admin").first()
    if not admin:
        from werkzeug.security import generate_password_hash
        admin = User(username="admin",
                     password_hash=generate_password_hash("admin123"),
                     role="admin")
        db.session.add(admin)
        db.session.commit()
    #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
    existing = Project.query.filter_by(code=DEMO_CODE).first()
    if existing and not reset:
        print("演示案例已存在，跳过写入（用户自建项目不受影响）")
        _report()
        _seed_reference_data(reset=False)
        return
    clean_case(reset=reset)
    project = Project(name=CASE["name"], code=DEMO_CODE,
                      description="论文第 5 节典型案例",
                      created_by=admin.id)
    db.session.add(project)
    db.session.flush()
    patch = MinePatch(project_id=project.id,
                      name=CASE["patch_name"],
                      admin_region="河北省唐山市玉田县",
                      area_ha=CASE["area_ha"],
                      mineral_type=CASE["mineral_type"],
                      mining_method="露天开采",
                      stop_years=None,
                      responsible_entity_missing=True)
    db.session.add(patch)
    db.session.flush()
    ref = ReferenceEcosystem(patch_id=patch.id,
                             coverage_benchmark=75.0,
                             shannon_benchmark=0.7)
    db.session.add(ref)
    for u in CASE["units"]:
        unit = MicroUnit(patch_id=patch.id,
                         unit_type=u["unit_type"],
                         ratio=u["ratio"], area=None)
        db.session.add(unit)
        db.session.flush()
        d = u["data"]
        ind = IndicatorData(
            unit_id=unit.id,
```

```
                    soil_ratio=d.get("soil_ratio"),                    9801
                    thickness=d.get("thickness"),                      9802
                    ph=d.get("ph"),                                    9803
                    organic_matter=d.get("organic_matter"),            9804
                    heavy_metals=json.dumps(d["heavy_metals"],         9805
    ensure_ascii=False)                                                9806
                                        if d.get("heavy_metals") else  9807
    None,                                                              9808
                    coverage=d.get("coverage"),                        9809
                    shannon=d.get("shannon"),                          9810
                    slope=d.get("slope"),                              9811
                    aspect=d.get("aspect"),                            9812
                )                                                      9813
                db.session.add(ind)                                    9814
            #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、          9815
            #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。          9816
            db.session.add(ConstraintCheck(                            9817
                patch_id=patch.id,                                     9818
                policy_pass=True, location_pass=True,                  9819
    pollution_pass=True,                                               9820
                geo_pass=True, farmland_pass=True,                     9821
                passed=True, reason="全部通过"))                        9822
            db.session.commit()                                        9823
            print("已写入案例：", CASE["name"])                         9824
            print("   图斑：", CASE["patch_name"])                      9825
            print("   微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台    9826
    23.19%")                                                           9827
            print("   管理员账号：admin / admin123")                    9828
            _seed_reference_data(reset=reset)                          9829
            _report()                                                  9830
def _seed_reference_data(reset=False):                                 9831
    """写入标准库与指标定义。                                           9832
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护          9833
    的标准条目与指标定义。                                              9834
    """                                                                9835
    if reset or Standard.query.count() == 0:                           9836
        seed_standards()                                               9837
    from app.services.indicator_service import                         9838
    seed_indicators                                                    9839
    n = seed_indicators()                                              9840
    print("指标定义：%d 项" % n)                                        9841
def _report():                                                         9842
    total_p = Project.query.count()                                    9843
    total_patch = MinePatch.query.count()                              9844
    print()                                                            9845
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))     9846
    if total_p > total_patch:                                          9847
        print("   提示：项目是归档单位，评价针对的是图斑。")                9848
        print("         新项目需先建立图斑，才会出现在后续各模块中。")      9849
def seed_standards():                                                  9850
```

```
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""
    StandardRule.query.delete()
    Standard.query.delete()
    db.session.commit()
    for s in STANDARDS:
        std = Standard(category=s["category"],
    province=s["province"],
                       name=s["name"],
    doc_number=s["doc_number"],
                       publish_date=s["publish_date"],
                       implement_date=s["implement_date"],
                       summary=s["summary"])
        db.session.add(std)
        db.session.flush()
        for item, op, thr, ctype, note in s["rules"]:
            db.session.add(StandardRule(standard_id=std.id,
    item=item,
                                        operator=op,
    threshold=thr,

    condition_type=ctype, note=note))
    db.session.commit()
    print("已写入标准/政策文件：%d 项，规则 %d 条"
          % (len(STANDARDS), sum(len(s["rules"]) for s in
    STANDARDS)))
if __name__ == "__main__":
    # --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。
    # 无论是否 --reset，都不会删除用户自建的项目与图斑。
    seed(reset="--reset" in sys.argv[1:])
# ==== 测试/test_ahp.py ====
# -*- coding: utf-8 -*-
"""地基自检：验证 AHP 引擎与应用可启动。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import numpy as np
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------
    ----------------
# 1) 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------
    ----------------
print(LINE)
```

```
print("测试 1  完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("  判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------
    ----------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
# ------------------------------------------------------------
    ----------------
# 3） 多专家聚合（几何平均）
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
```

```
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3],专家2 上三角 [3,5,2]")
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------------
    ------------------
# 4) 论文表 3 权重可还原性
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f   CR = %.3e   一致: %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
    ------------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表:" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检:用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
```

```
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
    ------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------
    ------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
```

```
# ------------------------------------------------------------------------
    ----------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------------
    ----------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
```

```
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码:", r.status_code, "(应 200)")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码:", r.status_code, "(应 200)")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
```

```
print("   跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("   含三个一级指标（土壤/植被/地形）:", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3   提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("   结果页含「计算结果」:", "计算结果" in body)
print("   含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("   含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4   不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("   含「不通过」:", "不通过" in body)
print("   含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
```

```
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
```

```
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").first()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡", "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南", "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件：", all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
```

```
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "0",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "1",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
```

```
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
```

```
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
```

```
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录  id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
```

```
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
```

```
        else "[不符合]"))                                                    10551
assert wrapped, "复选框未与文字关联，点击文字不会勾选"                          10552
has_for = 'for="remember"' in html                                           10553
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else         10554
      "[不符合]"))                                                           10555
# 2) 勾选后应下发带过期时间的 remember_token                                 10556
c = app.test_client()                                                        10557
r = c.post("/login", data={"username": "admin", "password":                  10558
      "admin123",                                                            10559
                                "remember": "y"},                            10560
      follow_redirects=False)                                                10561
cookies = r.headers.getlist("Set-Cookie")                                    10562
rt = [h for h in cookies if h.startswith("remember_token=")]                 10563
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else                10564
      "[不符合]"))                                                           10565
assert rt, "未下发 remember_token"                                           10566
exp = re.search(r"Expires=([^;]+)", rt[0])                                   10567
print("  %s 该 cookie 带过期时间：%s"                                       10568
        % ("[符合]" if exp else "[不符合]", exp.group(1) if exp               10569
      else "无（会话 cookie）"))                                             10570
assert exp, "remember_token 无过期时间，关浏览器即失效"                      10571
# 3) 不勾选时不应下发 remember_token                                         10572
c2 = app.test_client()                                                       10573
r2 = c2.post("/login", data={"username": "admin",                            10574
      "password": "admin123"},                                               10575
                 follow_redirects=False)                                     10576
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                        10577
      h.startswith("remember_token=")]                                       10578
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2            10579
      else "[不符合]"))                                                      10580
assert not rt2                                                               10581
# 4) 携带 remember_token 的新会话应自动登录                                  10582
c3 = app.test_client()                                                       10583
c3.set_cookie("remember_token",                                              10584
                rt[0].split(";")[0].split("=", 1)[1],                        10585
      domain="localhost")                                                    10586
r3 = c3.get("/", follow_redirects=False)                                     10587
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"            10588
        % ("[符合]" if r3.status_code == 200 else "[不符合]",                10589
      r3.status_code))                                                       10590
assert r3.status_code == 200                                                 10591
# ------------------------------------------------------------------------   10592
      ------------------                                                     10593
print()                                                                      10594
print(LINE)                                                                  10595
print("二、删除功能")                                                        10596
print(LINE)                                                                  10597
client.post("/login", data={"username": "admin",                             10598
      "password": "admin123"})                                               10599
# 项目：列表页须有删除按钮                                                   10600
```

```
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
html = client.get("/project/%d/patches" %
    patch_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
```

```
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
#  ==== 测试/test_new_features.py ====
#  -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    #  固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
```

```
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "evaluator",
    "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
    "active": "1"})
```

```
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(png), "影像.png"),
                  "image_note": "2025年6月无人机正射影像"},
            content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(b"x"), "bad.txt")},
            content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
```

```
    print("-" * 60)
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
```

```
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
   vals[0] if vals else "")
# ------------------------------------------------------------------
   ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
   "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
   ,坡向\n"
   "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
   87.92,0.60,28.75,阳坡\n"
   "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
   8.42,0.72,,,\n"
   "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
   f-8")), "d.csv")},
           content_type="multipart/form-data",
   follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
   u = MicroUnit.query.filter_by(patch_id=patch_id,
   unit_type="碎石边坡").first()
   ind = IndicatorData.query.filter_by(unit_id=u.id).first(
   )
   ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
   abs(ind.coverage - 87.92) < 1e-6,
       "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
   follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
   ----------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
```

```
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
#  报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
```

```
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
```

```
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------
    ----------------
print(LINE)
```

```
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
```

```
print()                                                                  11101
print("场景　4：判定通过，但指标数据全空")                                  11102
pid = new_patch("4"); add_units(pid); add_constraint(pid,                11103
    passed=True)                                                         11104
r = post_eval(pid)                                                       11105
n = results_of(pid)                                                      11106
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",          11107
   "结果数 %d" % n)                                                      11108
with app.app_context():                                                  11109
    pr = wf.patch_progress(pid)                                          11110
ok(any(b["step"] == "data" for b in pr["blocking"]),                     11111
   "拦原因指向「录入指标数据」")                                          11112
ok(pr["data"]["missing"] == ["碎石边坡"],                                11113
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))          11114
cleanup(pid, "4")                                                        11115
print()                                                                  11116
print("场景　5：单元面积占比合计不为　１００%")                             11117
pid = new_patch("5"); add_constraint(pid, passed=True)                   11118
with app.app_context():                                                  11119
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,          11120
    ratio=0.5)                                                           11121
    db.session.add(u); db.session.commit()                               11122
    db.session.add(IndicatorData(unit_id=u.id,                           11123
    soil_ratio=0.3, thickness=10,                                        11124
                                        ph=8.0,                          11125
    organic_matter=5.0, coverage=80.0,                                   11126
                                        shannon=0.6, slope=20.0,         11127
    aspect="阳坡"))                                                      11128
    db.session.commit()                                                  11129
r = post_eval(pid)                                                       11130
ok(results_of(pid) == 0, "占比不足　１００% → 不产生评价结果")             11131
cleanup(pid, "5")                                                        11132
print()                                                                  11133
print(LINE)                                                              11134
print("二、前置条件齐备时才放行")                                          11135
print(LINE)                                                              11136
pid = new_patch("OK"); add_units(pid); add_constraint(pid,               11137
    passed=True); add_data(pid)                                          11138
with app.app_context():                                                  11139
    pr = wf.patch_progress(pid)                                          11140
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")               11141
r = post_eval(pid)                                                       11142
n = results_of(pid)                                                      11143
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)                    11144
print("  重定向到：", r.headers.get("Location"))                         11145
print()                                                                  11146
print(LINE)                                                              11147
print("三、报告不得先于评价产生")                                          11148
print(LINE)                                                              11149
pid2 = new_patch("R")    # 什么都不做                                    11150
```

```
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
     (r.headers.get("Location") or ""),
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
     rr = c.get("/report/%d%s" % (pid2, suffix),
     follow_redirects=False)
     ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
     %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
     add_data(pid3)
with app.app_context():
     pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
     "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
     pr = wf.report_progress(pid3)
     rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
     rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
```

```
# 导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
```

```
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
              follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
          follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
              data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
```

```
    sys.path.insert(0, ROOT)                                          11301
from app import create_app                                            11302
from app.models import (db, Project, MinePatch, MicroUnit,            11303
    IndicatorData,                                                    11304
                            ConstraintCheck, EvaluationResult,        11305
    WeightScheme,                                                     11306
                            WeightItem)                               11307
from app.services import workflow as wf                               11308
from app.services import weight_service as ws                         11309
LINE = "=" * 72                                                       11310
BASE = ROOT                                                           11311
class TestConfig:                                                     11312
    SECRET_KEY = "test"                                               11313
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                          11314
    os.path.join(BASE, "app", "app.db")                               11315
    SQLALCHEMY_TRACK_MODIFICATIONS = False                            11316
    WTF_CSRF_ENABLED = False                                          11317
    TESTING = True                                                    11318
app = create_app(TestConfig)                                          11319
c = app.test_client()                                                 11320
c.post("/login", data={"username": "admin", "password":               11321
    "admin123"})                                                      11322
_fails = []                                                           11323
def ok(flag, label, extra=""):                                        11324
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",         11325
    label, extra))                                                    11326
    if not flag:                                                      11327
        _fails.append(label)                                          11328
TAG = "W"                                                             11329
UNIT_TYPE = "碎石边坡"                                                 11330
def build_case():                                                     11331
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""          11332
    with app.app_context():                                           11333
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"         11334
    % TAG)                                                            11335
        db.session.add(p); db.session.flush()                         11336
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"      11337
    % TAG)                                                            11338
        db.session.add(patch); db.session.flush()                     11339
        db.session.add(MicroUnit(patch_id=patch.id,                   11340
    unit_type=UNIT_TYPE,                                              11341
                                 area=1.0, ratio=1.0))                11342
        db.session.flush()                                            11343
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi           11344
    st()                                                              11345
        db.session.add(IndicatorData(                                 11346
            unit_id=u.id, soil_ratio=0.34, thickness=10,              11347
    ph=9.24,                                                          11348
            organic_matter=4.89, coverage=87.92,                      11349
    shannon=0.60,                                                     11350
```

```
                slope=28.75, aspect="阳坡"))
            db.session.add(ConstraintCheck(
                patch_id=patch.id, policy_pass=True,
    location_pass=True,
                pollution_pass=True, geo_pass=True,
    farmland_pass=True,
                passed=True, reason="全部通过"))
            db.session.commit()
            return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
```

```
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
```

```
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
```

```
# ==== app/utils/__init__.py ====                    11501
# -*- coding: utf-8 -*-                              11502
# ==== app/views/__init__.py ====                    11503
# -*- coding: utf-8 -*-                              11504
```